D

Data Handling and Statistics Teaching and Learning

Dani Ben-Zvi
Faculty of Education, The University of Haifa, Haifa, Israel

Keywords

Statistics · Data handling · Exploratory data analysis · Teaching and learning statistics · Research on teaching and learning statistics · Statistical reasoning · Statistical literacy · Technological tools in statistics learning

Definition

Over the past several decades, changes in perspective as to what constitute statistics and how statistics should be taught have occurred, which resulted in new content, pedagogy and technology, and extension of teaching to school level. At the same time, statistics education has emerged as a distinct discipline with its own research base, professional publications, and conferences (Ben-Zvi et al. 2018b). There seems to be a large measure of agreement on what content to emphasize in statistics education: exploring data (describing patterns and departures from patterns), sampling and experimentation (planning and conducting a study), anticipating patterns (exploring random phenomena using models, probability and simulation), and statistical inference (estimating population parameters and testing hypotheses) (Scheaffer 2001). Teaching and learning statistics can differ widely across countries due to cultural, pedagogical, and curricular differences and the availability of skilled teachers, resources, and technology.

Changing Views on Teaching Statistics Over the Years

By the 1960s statistics began to make its way from being a subject taught for a narrow group of future scientists into the broader tertiary and school curriculum but still with a heavy reliance on probability. In the 1970s, the reinterpretation of statistics into separate practices comprising exploratory data analysis (EDA) and confirmatory data analysis (CDA, inferential statistics) (Tukey 1977) freed certain kinds of data analysis from ties to probability-based models, so that the analysis of data could begin to acquire status as an independent intellectual activity. The introduction of simple data tools, such as stem and leaf plots and boxplots, paved the way for students at all levels to analyze real data interactively without having to spend hours on the underlying theory, calculations, and complicated procedures. Computers and new pedagogies would later complete the "data revolution" in statistics education.

In the 1990s, there was an increasingly strong call for statistics education to focus more on

S. Lerman (ed.), *Encyclopedia of Mathematics Education*,
https://doi.org/10.1007/978-3-030-15789-0

statistical literacy, reasoning, and thinking. Wild and Pfannkuch (1999) provided an empirically based comprehensive description of the processes involved in the statisticians' practice of data-based inquiry from problem formulation to conclusions. One of the main arguments presented is that traditional approaches to teaching statistics focus on skills, procedures, and computations, which do not lead students to reason or think statistically.

These changes are implicated in a process of democratization that has broadened and diversified the backgrounds and motivations of those who learn statistics at many levels with very diverse interests and goals. There is a growing recognition that the teaching of statistics is an essential part of sound education since the use of data is increasingly common in science, society, media, everyday life, and almost any profession.

A Focus on Statistical Literacy and Reasoning

The goal of teaching statistics is to produce statistically educated students who develop statistical literacy and the ability to reason statistically. Statistical literacy is the ability to interpret, critically evaluate, and communicate about statistical information and messages. Statistically literate behavior is predicated on the joint activation of five interrelated knowledge bases – literacy, statistical, mathematical, context, and critical – together with a cluster of supporting dispositions and enabling beliefs (Gal 2002). Statistical reasoning is the way people reason with the "big statistical ideas" and make sense of statistical information during a data-based activity. Statistical reasoning may involve connecting one concept to another (e.g., center and spread) or may combine ideas about data and chance. Statistical reasoning also means understanding and being able to explain statistical processes and being able to interpret statistical results.

The "big ideas" of statistics that are most important for students to understand and use are data, statistical models and modeling, distribution, center, variability, comparing groups, samples, sampling and sampling distributions, statistical inference, and covariation. Additional important underlying concepts are uncertainty, randomness, evidence strength, significance, and data production (e.g., experiment design). In the past few years, researchers have been developing ideas of *informal* statistical reasoning in students as a way to build their conceptual understanding of the foundations of more formal ideas of statistics (Garfield and Ben-Zvi 2008).

What Does Research Tell Us About Teaching and Learning Statistics?

Research on teaching and learning statistics has been conducted by researchers from different disciplines and focused on students at all levels. Common faulty heuristics, biases, and misconceptions were found in adults when they make judgments and decisions under uncertainty, e.g., the representativeness heuristic, law of small numbers, and gambler's fallacy (Kahneman et al. 1982). Recognizing these persistent errors, researchers have explored ways to help people correctly use statistical reasoning, sometimes using specific methods to overcome or correct these types of problems.

Another line of inquiry has focused on how to develop good statistical reasoning and understanding, as part of instruction in elementary and secondary mathematics classes. These studies revealed many difficulties students have with concepts that were believed to be fairly elementary such as data, distribution, center, and variability. The focus of these studies was to investigate how students begin to understand these ideas and how their reasoning develops when using carefully designed activities assisted by technological tools (Shaughnessy 2007).

A newer line of research is the study of preservice or practicing teachers' knowledge of statistics and probability and how that understanding develops in different contexts. The research related to teachers' statistical pedagogical content knowledge suggests that this knowledge is in many cases weak. Many teachers do not consider themselves well prepared to teach statistics nor face their students' difficulties (Batanero et al. 2011).

The studies that focus on teaching and learning statistics at the college level continue to point out the many difficulties tertiary students have in learning, remembering, and using statistics and point to some modest successes. These studies also serve to illustrate the many practical problems faced by college statistics instructors such as how to incorporate active or collaborative learning in a large class, whether or not to use an online or "hybrid" course, or how to select one type of software tool as more effective than another. While teachers would like research studies to convince them that a particular teaching method or instructional tool leads to significantly improved student outcomes, that kind of evidence is not actually available in the research literature. However, recent classroom research studies suggest some practical implications for teachers. For example, developing a deep understanding of statistics concepts is quite challenging and should not be underestimated; it takes time, a well thought-out learning trajectory, and appropriate technological tools, activities, and discussion questions.

Teaching and Learning

As more and more students study statistics, teachers are faced with many challenges in helping these students succeed in learning and appreciating statistics. The main sources of students' difficulties were identified as: facing statistical ideas and rules that are complex, difficult, and/or counterintuitive, difficulty with the underlying mathematics, the context in many statistical problems may mislead the students, and being uncomfortable with the messiness of data, the different possible interpretations based on different assumptions, and the extensive use of writing and communication skills (Ben-Zvi and Garfield 2004).

The study of statistics should provide students with tools and ideas to use in order to react intelligently to quantitative information in the world around them. Reflecting this need to improve students' ability to reason statistically, teachers of statistics are urged to emphasize statistical reasoning by providing explicit attention to the basic ideas of statistics (such as the need for data, the importance of data production, the omnipresence of variability); focus more on data and concepts, less on theory, and fewer recipes; and foster active learning (Cobb 1992). These recommendations require changes of teaching statistics in *content* (more data analysis, less probability), *pedagogy* (fewer lectures, more active learning), and *technology* (for data analysis and simulations) (Moore 1997).

Statistics at school is usually part of the mathematics curriculum. New K–12 curricular programs set ambitious goals for statistics education, including promoting students' statistical literacy, reasoning, and understanding (e.g., NCTM 2000). These reform curricula weave a strand of *data handling* into the traditional school mathematical strands (number and operations, geometry, algebra). Detailed guidelines for teaching and assessing statistics at different age levels complement these standards. However, school mathematics teachers, which are often not versed in statistics, find it challenging to teach data handling in accordance with these recommendations.

In order to face this challenge and promote statistical reasoning, good instructional practice consists of implementing inquiry or project-based learning environments that stimulate students to construct meaningful knowledge. Ben-Zvi et al. (2018a) suggest several design principles to develop students' statistical reasoning: focus on developing central statistical ideas rather than on presenting set of tools and procedures; use real and motivating data sets to engage students in making and testing conjectures; use classroom activities to support the development of students' reasoning; integrate the use of appropriate technological tools that allow students to test their conjectures, explore and analyze data, and develop their statistical reasoning; promote classroom discourse that includes statistical arguments and sustained exchanges that focus on significant statistical ideas; and use assessment to learn what students know and to monitor the development of their statistical learning, as well as to evaluate instructional plans and progress.

D

Technology has changed the way statisticians work and has therefore been changing what and how statistics is taught. Interactive data visualizations allow for the creation of novel representations of data. It opens up innovative possibilities for students to make sense of data but also place new demands on teachers to assess the validity of the arguments that students are making with these representations and to facilitate conversations in productive ways. Several types of technological tools are currently used in statistics education to help students understand and reason about important statistical ideas. However, using technological tools and how to avoid common pitfalls are challenging open issues (Biehler et al. 2013).

These changes in the learning goals of statistics have led to a corresponding rethinking of how to assess students. It is becoming more common to use alternative assessments such as student projects, reports, and oral presentations than in the past. Much attention has been paid to assess student learning, examine outcomes of courses, align assessment with learning goals, and alternative methods of assessment.

For Further Research

Research in statistics education has made significant progress in understanding students' difficulties in learning statistics and in offering and evaluating a variety of useful instructional strategies, learning environments, and tools (Ben-Zvi et al. 2018b). However, many challenges are still ahead of statistics education, mostly in transforming research results to practice, evaluating new programs, planning and disseminating high-quality assessments, and providing attractive and effective professional development to mathematics teachers (Garfield and Ben-Zvi 2007). The ongoing efforts to reform statistics instruction and content have the potential to both make the learning of statistics more engaging and prepare a generation of future citizens that deeply understand the rationale, perspective, and key ideas of statistics. These are skills and knowledge that are crucial in the current information age of data, big data and data science.

Cross-References

▶ Inquiry-Based Mathematics Education
▶ Mathematical Literacy
▶ Probability Teaching and Learning

References

Batanero C, Burrill G, Reading C (2011) Teaching statistics in school mathematics: challenges for teaching and teacher education (a joint ICMI/IASE study: the 18th ICMI study). Springer, Dordrecht

Ben-Zvi D, Garfield J (2004) The challenge of developing statistical literacy, reasoning, and thinking. Springer, Dordrecht

Ben-Zvi D, Gravemeijer K, Ainley J (2018a) Design of statistics learning environments. In: Ben-Zvi D, Makar K, Garfield J (eds) International handbook of research in statistics education. Springer, Cham, pp 473–502

Ben-Zvi D, Makar K, Garfield J (eds) (2018b) International handbook of research in statistics education, Springer international handbooks of education. Springer, Cham

Biehler R, Ben-Zvi D, Bakker A, Makar K (2013) Technological advances in developing statistical reasoning at the school level. In: Clements MA, Bishop A, Keitel C, Kilpatrick J, Leung F (eds) Third international handbook of mathematics education. Springer, New York, pp 643–690

Cobb GW (1992) Report of the joint ASA/MAA committee on undergraduate statistics. In: The American Statistical Association 1992 proceedings of the section on statistical education. American Statistical Association, Alexandria, pp 281–283

Gal I (2002) Adults' statistical literacy: meaning, components, responsibilities. Int Stat Rev 70:1–25

Garfield J, Ben-Zvi D (2007) How students learn statistics revisited: a current review of research on teaching and learning statistics. Int Stat Rev 75:372–396

Garfield J, Ben-Zvi D (2008) Developing students' statistical reasoning: connecting research and teaching practice. Springer, New York

Kahneman D, Slovic P, Tversky A (1982) Judgment under uncertainty: heuristics and biases. Cambridge University Press, New York

Moore DS (1997) New pedagogy and new content: the case of statistics. Int Stat Rev 65:123–137

NCTM (2000) Principles and standards for school mathematics. National Council of Teachers of Mathematics, Reston

Scheaffer RL (2001) Statistics education: perusing the past, embracing the present, and charting the future. Newsl Sect Stat Educ 7(1). http://www.amstat.org/sections/educ/newsletter/v7n1/Perusing.html

Shaughnessy JM (2007) Research on statistics learning and reasoning. In: Lester FK (ed) The second handbook of

research on mathematics. Information Age Pub, Charlotte, pp 957–1010
Tukey J (1977) Exploratory data analysis. Addison-Wesley, Reading
Wild CJ, Pfannkuch M (1999) Statistical thinking in empirical enquiry. Int Stat Rev 67:223–248

Deaf Children, Special Needs, and Mathematics Learning

Terezinha Nunes
Department of Education, University of Oxford, Oxford, UK

Keywords
Deaf children · Special needs · Mathematics difficulty

Characteristics

The aim of mathematics instruction in primary school is to provide a basis for thinking mathematically about the world. This is as basic a skill as literacy in today's world. Mathematical knowledge is also a means to achieve better employment and to enter higher education. For all these reasons, it is of great importance that deaf children have adequate access to mathematical thinking, but unfortunately most deaf children show a severe delay in mathematics learning. This delay has been persistent over many years. The average score in mathematics achievement tests for deaf children in the age range 8–15 in a study carried out in 1965 showed that they were one standard deviation below the average for hearing children, a result replicated about three decades later. This means that about 50% of the deaf pupils perform similarly to the weakest 15% of the hearing pupils. Later results continue to confirm this weak performance. In the UK, deaf students aged 16–17 years, at the end of compulsory school, were found to have a mathematical age between 10 and 12.5 years. In the USA, the mathematical ability of 80% of the deaf 14-year-olds was described as "below basic" in problem solving and knowledge of mathematical procedures. A recent systematic review confirmed these findings (Gottardis et al. 2011) and analyzed individual differences among deaf children.

This serious and persistent difficulty is not universal among children who are deaf; approximately 15% perform at age appropriate levels. The successful minority indicates that deafness is not a direct cause of difficulty in mathematics learning (see Nunes 2004, for a discussion). This article considers what is involved in learning mathematics in primary school, why deaf children may be at a disadvantage, and how schools can support their learning of mathematics.

D

Learning Mathematics in Primary School

In order to think mathematically, people need to learn to represent quantities, relations, and space using culturally developed and transmitted thinking tools, such as oral and written number systems, graphs, and calculators.

Some researchers argue that numerical concepts have a neurological basis that is independent of language learning, without which learning mathematics is extremely difficult. In view of the pervasiveness of deaf children's mathematical difficulties, it could be hypothesized that they have an inadequate development of such concepts. Basic numerical cognition has been studied in research with young deaf children as well as adults, and the hypothesis has been discarded. Deaf children and adults performed at least as well as their hearing counterparts in such tasks.

The possible consequences of delays in the acquisition of other language-based numerical concepts have also been explored. Two examples are knowledge of counting and understanding of arithmetic operations.

Counting

Deaf children lag behind hearing children in learning to count, independently of whether they are learning to count orally or in sign (Leybaert and Van Cutsem 2002). Consequently, they perform less well than hearing children on school-entry numeracy tests, which typically include tasks that require counting (e.g., “show me 5 blocks”; “tell me which number is bigger”). This delay could be related to the well-established finding that deaf people perform less well than hearing people on serial learning tasks, in which words or gestures must be learned in an exact sequence, just as the number string. However, they perform better if the tasks are presented differently and use spatial cues to organize the information. These findings are provocative rather than conclusive. First, they raise the possibility that deaf children could learn to count more easily if appropriate visual and spatial methods were used for teaching rather than serial learning instruction. Second, serial learning is not an appropriate description of counting skills beyond a certain number (about 20 or 30 in English but this may differ depending on the counting system). Research with hearing and deaf children shows that counting is a structured activity: for example, errors are more likely to occur at the boundaries between decades (e.g., . . .38, 39, 50, 51, 52. . .) than within decades. Therefore, in principle deaf children’s initial disadvantage in counting could be overcome with appropriate teaching methods and with support for mastery of the structure of the system. However, it is possible that their initial struggle with learning to count lowers adults’ expectations about what they can learn in mathematics, resulting in less stimulation on mathematical tasks, and that it also interferes with the children’s own discoveries in the domain of mathematical reasoning.

Early Mathematical Reasoning and Arithmetic Operations

The development of mathematical reasoning starts before school, when children solve practical problems using actions, which they learn to combine with counting. When most children start school (at age 5 or 6), they can already solve simple addition and subtraction problems by putting together or separating objects and counting, and some can also solve multiplication and division problems. By counting, children use explicit numerical representation both for thinking and communicating. When numbers are small and the children can use objects, deaf children do as well as hearing children in solving these problems, but if the numbers go above 10 or 20, most deaf children fall behind. When they are compared with hearing children of the same counting ability, they are just as competent in solving numerical tasks (Leybaert and Van Cutsem 2002), but their disadvantage in counting is reflected in their problem-solving skills when they are compared to same-age hearing peers. Thus, it is possible that, not knowing number words well enough to support their mathematical reasoning, they do not discover how to use counting to solve simple arithmetic problems or important ideas for their later success, such as the inverse relation between addition and subtraction. However, Nunes et al. (2008a, b) have shown that relatively small amounts of teaching can effectively improve young deaf children’s performance in the mathematical reasoning and arithmetic tasks, with which they were struggling before the teaching.

Conclusion

There is little doubt that many deaf children show severe and persistent difficulties in learning mathematics. Evidence suggests that there is no direct connection between deafness and problems with basic number concepts that precede language. However, deaf children lag behind hearing children in learning to count, whether orally or in sign, and at school entry they are behind their hearing counterparts in mathematical knowledge. It is possible that falling behind in counting places deaf children at a disadvantage from the adults’ perspective and that they end up receiving less stimulation to solve mathematical problems early on. It is also possible that their own informal mathematical knowledge is limited by their difficulty in representing quantities explicitly with number words. These findings and conclusions suggest that, if parents and preschool teachers could find visual and spatial ways to teach counting to deaf children, one would see positive changes in the average achievement of deaf children in mathematics in the future.

Cross-References

- ▶ Blind Students, Special Needs, and Mathematics Learning
- ▶ Concept Development in Mathematics Education
- ▶ Inclusive Mathematics Classrooms
- ▶ Language Disorders, Special Needs and Mathematics Learning

References

Gottardis L, Nunes T, Lunt I (2011) A synthesis of research on deaf and hearing children's mathematical achievement. Deaf Educ Int 13:131–150

Leybaert J, Van Cutsem M-N (2002) Counting in sign language. J Exp Child Psychol 81:482–501

Nunes T (2004) Teaching mathematics to deaf children. Whurr, London

Nunes T, Bryant P, Burman D, Bell D, Evans D, Hallett D (2008a) Deaf children's informal knowledge of multiplicative reasoning. J Deaf Stud Deaf Educ 14:260–277

Nunes T, Bryant P, Burman D, Bell D, Evans D, Hallett D et al (2008b) Deaf children's understanding of inverse relations. In: Marschark M, Hauser PC (eds) Deaf cognition. Oxford University Press, Oxford, pp 201–225

Deductive Reasoning in Mathematics Education

Guershon Harel[1] and Keith Weber[2]
[1]Department of Mathematics, University of California, San Diego, La Jolla, CA, USA
[2]Rutgers University, New Brunswick, NJ, USA

Definition

Deductive inference – A deductive inference is a conclusion drawn from premises in which there are rational grounds to believe that the premises necessitate the conclusion. That is, it would be impossible for the premises to be true and the conclusion to be false.

Deductive reasoning – Deductive reasoning is a process when new information is derived from a set of premises via a chain of deductive inferences.

The Importance of Deductive Reasoning

Most mathematics educators hold that proving is a central mathematical practice that should play a prominent role in all mathematical classrooms (Stylianides et al. 2016). Proving is synonymous with deductive reasoning. However, students' conceptions of what constitutes "premise" and "derivation" are not necessarily consistent with mathematician's conceptions of these terms. Further, as we will argue later, a student's conception of deductive reasoning is intricately linked to that student's conceptions of what it means to know and do mathematics (Harel and Sowder 1998). Hence, a teacher or mathematics education researcher must consider students' conceptions of deductive reasoning when teaching proof or conducting research on this topic.

Different Theoretical Perspectives on What a Deductive Inference Is

Although there is a consensus among mathematics educators that proof is important and should be taught, there is significant disagreement as to what a proof is (Balacheff 2008). There is a broad consensus that some types of inferences do not qualify as deductive inferences. Consider the inference: "$1^2 = 1$ is odd, $3^2 = 9$ is odd, $5^2 = 25$ is odd. Since 1^2, 3^2, and 5^2 are odd, any odd number squared is odd." Although the conclusion of this inference is true, mathematics educators would not regard this as a deductive inference because there are not rational grounds for how the premises necessitated the conclusion. The inference appears to be based on the warrant that if three odd numbers have a property, then all odd numbers will have a property, but there are situations where this warrant is false. Thus, there is agreement that some common inferences that students draw are not deductive. However, it is conceptually challenging to define what a deductive *is*.

Below we outline three perspectives and discuss their implications for mathematics education. To frame our discussion, consider the following student justification of the statement "For any triangle, the sum of the lengths of any two sides must be greater of the length of the remaining side" (adapted from a student proof in Williams-Pierce et al. 2017; we lightly revised the text for the sake of clarity):

> Say one side's 5. If the sum of the other sides weren't at least 5 [Student uses each hand to represent a side of an unspecified length with the palm of each hand being situated at the base of a segment of length 5], if you tried to connect them, you think about it, they'd be too short too touch. [Student curves his hands to illustrate the two shorter sides trying, but failing, to close the triangle]. So they would have to be longer than the remaining side [Student repeats the gesture].

Deductive inferences are applications of a decontextualized logical rule: An inference can be characterized as deductive if the inference can be framed as the application of a decontextualized logical rule, such as modus ponens, the law of the excluded middle, or universal instantiation. Mathematics educators using this characterization typically investigate what types of logical structures of implication that students consider to be valid (see Inglis and Attridge 2017, for a comprehensive review). Characterizing deductive inferences as the application of decontextualized logical rules suggests that the teaching of proof should focus on having students acquire these rules. Indeed, common methods for teaching proof prioritize teaching students logical rules. For instance, the two-column proof technique widely used in United States geometry classrooms highlights that each new inference in a proof must be based on a semantic-free consequence of prior rules (see Herbst 2002).

A significant limitation of claiming that deductive inferences are comprised only of the application of decontextualized rules is that this is too *restrictive* for mathematics educators' purposes. For instance, consider the student justification from Williams-Pierce et al. (2017) described above. As the proof is intrinsically kinesthetic, a mathematics educator would not describe the justification as an application of decontextualized rules. It seems doubtful that the student had any such rules in mind when presenting this justification. However, some mathematics educators would sanction proofs using informal representations as acceptable in many mathematics classrooms, especially with younger students (e.g., NCTM 2000).

Deductive inferences include inferences based on transformational reasoning.

Harel and Sowder (1998) defined a transformational inference to be one in which an individual makes by anticipating a logically necessary outcome of applying a goal-oriented operation. To illustrate a transformation inference, consider the student proof from Williams-Pierce et al. (2017). Here, the student performs a goal-oriented operation (swinging two anchored sides of a hypothetical triangle together with the goal of forming a closed figure) and anticipating a result (the two sides would not meet so a closed figure could not be formed). Further, the student viewed the result as logically necessary based on general properties contained in the premises of the inference (the two shorter sides combined were shorter than the length of the longest side; crucially, that the longest side happened to have a length of 5 was irrelevant to the justification). An advantage of considering transformational inferences as deductive inferences that are permissible in a proof is that this construct is permissive enough to allow proofs like the student proof from Williams-Pierce et al. (2017; indeed the authors framed their paper using Harel and Sowder's construct), as well as generic proofs and proofs based on manipulatives and diagrams (see Harel and Sowder 2007, for more details). For illustrations of how focusing students' attention on how anticipating outcomes on goal-oriented operations can foster desirable mathematical justifications, see Harel (2001) and Williams-Pierce et al. (2017).

As a subtle point that further illustrates mathematics educators' disagreement about deductive reasoning and proof, some authors contend that for a justification to qualify as a proof based on transformational inferences, a knowledgeable mathematician ought to be able to translate that justification to a proof in a straightforward manner. It is not clear that the Williams-Pierce et al. (2017) student proof cited above satisfies this criterion.

Deductive inferences are based on socially sanctioned rules for reasoning. Some commentators believe that the permissibility of an inference is primarily a social phenomenon. That is, it is permissible to apply a theorem or technique if the mathematical community believes the theorem or technique works (i.e., will never yield a false inference). Some mathematics educators have characterized proof similarly – as an argument that convinces a particular community at a particular time (e.g., Balacheff 1987). Within this perspective, asking whether the student justification in Williams-Pierce et al. is valid is not a meaningful question; one would need to know the norms of the student's mathematical community before evaluating the student contribution.

Mathematics educators who privilege the social aspect of deductive mathematical reasoning encourage students to negotiate the rules for what constitutes an acceptable inference and freely apply standards that become "taken-as-shared" (Cobb et al. 1992) by the classroom community (e.g., Alibert and Thomas 1991; for the detailed negotiation of basic logical rules by a pair of students, see Dawkins and Cook 2017). The teacher's role in these environments is to create situations that encourage classroom communities to come to accept inference rules that are regarded as normatively correct by the broader community. One limitation of this perspective is that it offers limited guidance for what norms for deduction and proof *should be encouraged* in a classroom community. Specifying that these norms should be compatible with mathematical practice is a useful heuristic, but mathematics educators do not agree on how mathematicians practice their craft and mathematicians themselves disagree on the legitimacy of some types of inference (e.g., Weber et al. 2014).

The Role of Axioms in Deductive Reasoning

The deductive mode of thought was conceived of by the Greeks more than 20 centuries ago and is still the dominant mode of drawing mathematical conclusions today. However, there is an important difference between how the Greek mathematicians and contemporary mathematicians practice their craft. Much of Greek mathematics was concerned with planar geometry where the Greeks viewed their axioms describing a single idealized physical reality. Since Hilbert's *Grundlagen*, contemporary mathematicians view axiom systems as being a freestanding system of relationships between undefined terms. Whereas the Greeks' geometry was restricted to a single interpretation (i.e., its presumed description of humans' spatial experiences), Hilbert's geometry could be satisfied by any points, lines, and planes that satisfied the relationships described in Hilbert's axioms, including the many non-Euclidean geometries that are studied today.

The difference between Greek axiomatic systems and modern axiomatic systems is profound: although both the Greeks and modern mathematicians privileged deductive reasoning, what deductive reasoning meant to the two groups of mathematicians differed. The Greeks believed they were describing a unique mathematical reality, which is one reason that Euclid would often draw inferences that were not direct logical consequences of his axioms but were rooted in humans' intuitive physical experience. In modern mathematics, deductive inferences are applicable to any model that satisfies the axiomatic system in which one is working.

The shift between Greek axiomatics and modern axiomatics can shed light on many shifts that students find difficult in their mathematics education, such as the shift from descriptive to stipulated definitions as students progress to university mathematics. Understanding the historical social, cultural, and intellectual needs that led mathematicians from Greek axiomatics to modern axiomatics may provide useful guidelines for instructional interventions that can foster students' views on deduction and proof to shift similarly.

Purposes of Deductive Reasoning and Proof

Bestowing psychological certainty. A common purpose of proof in mathematical practice and mathematics classrooms is to remove all doubt that a statement is true (Harel and Sowder 1998). Instruction highlighting the relationship between psychological certainty and deductive reasoning often involves having students realize that other types of inference (e.g., generalizing from a small number of examples) are not sufficient to know mathematical claims with certainty (e.g., Brown 2014), to develop students' intellectual needs to know mathematical claims with certainty (e.g., Harel 1998), and to appreciate how deductive reasoning is sufficiently general to provide this certainty.

Bestowing logical certainty or providing a priori *knowledge*. Some scholars have remarked that there are situations in which proofs cannot provide psychological certainty, not even to mathematicians, because the proof is sufficiently complex that the reader cannot be certain the proof does not contain a mistake. Doyle et al. (2014) remarked that although proofs are used to place theorems beyond doubt, it is not "psychological doubt" that the proof can remove: "No proof can defeat all lingering psychological doubts. For example, serious doubts about one's logical ability or memory will not be allayed by rehearsing a gap-free absolutely correct proof." What proof can offer instead is conditional guarantee; a proof cannot guarantee that a theorem is true, but if the theorem is false, this must be due to a faulty premise or a logical error in the proof. Such a conditional guarantee cannot be provided by other types of inference.

A related purpose of proof is to demonstrate that theorems are a priori consequences of how terms are defined. Considering this purpose, Mamona-Downs and Downs (2010) argued that, "the point [of proof] is not so much about conviction, but how we can clarify the bases of the reasoning employed" (p. 2338). Dawkins and Weber (2017) argued that mathematicians' value of a priori knowledge can account for proof-related behaviors in mathematics classrooms, such as a teacher's insistence that one prove a statement whose truth-value is obvious (e.g., a geometry statement that has been verified with a dynamic geometry software package) and the fact that students should avoid appealing to personal experience when writing a proof.

Providing explanation. In mathematicians' practice, proofs are valued because they explain why mathematical theorems are true (e.g., de Villiers 1990). A number of mathematics educators argued that proof should play a similar role in mathematics classrooms. See Harel (2001) for a classroom intervention in which students came to value proofs because of their explanatory power. A significant weakness in this area is that there is no agreement on what it means for a proof to be explanatory for a student (for a recent discussion, see Stylianides et al. 2016).

Illustrating methods and promoting discovery. A primary reason that mathematicians read their colleagues' work is to identify methods that they could use to solve problems that they are working on. Some mathematics educators, notably Hanna and Barbreau (2008), have contended that proof can play a similar role in mathematics classrooms as well: students can learn new problem-solving methods by studying the proofs their teachers present.

Systematization. One role of proof is to ensure that a definition or axiom system faithfully reflects the concept that it was designed to categorize (de Villiers 1990). Mathematicians do so by demonstrating widely believed statements about a theory are logical consequences of the theory's axioms and definitions. For instance, when logicians demonstrate that addition commutes using Peano's axioms, the point is not to persuade the reader that addition is commutative, but rather to illustrate to the reader that Peano's axioms can account for this fact. For a discussion of how systematization can help students understand concepts and their definitions, see Durrand-Gurrier (2016).

Students' Proving Abilities, Conceptions, and Perceptions

There is a large body of literature that students at all levels have difficulty writing proofs (e.g., Healy and Hoyles 2000; for a review, see Stylianides et al. 2017). In many cases, these difficulties can be attributed to students holding unproductive conceptions of deductive reasoning. Here we discuss three areas of research on these conceptions.

What Types of Evidence Provide Students with Mathematical Certainty?

Students often obtain certainty from evidence that would not be persuasive to a mathematician. Harel and Sowder's (1998, 2007) proof schemes framework is the most prevalent theoretical perspective to investigate students' views on the relationship between types of mathematical evidence and certainty. Within the proof schemes framework, *ascertaining* is a process that an individual uses to gain personal certainty in mathematical statements. *Persuading* is a process that an individual uses to provide others with certainty in a mathematical statement. An individual's *proof scheme* is what constitutes ascertaining and persuading for that individual.

Harel and Sowder (1998, 2007) presented a comprehensive taxonomy of proof schemes held by students. The following proof schemes have garnered extensive attention in the research literature. An *authoritarian proof scheme* is a scheme in which an individual gains certainty in a statement because it was sanctioned by an authoritative source such as a teacher or textbook. An *empirical inductive proof scheme* is a scheme in which an individual gains certainty in a universal statement by verifying that the statement holds with a small number of examples (e.g., verifying that the square of every odd integer is odd by verifying the claim for $n = 1, 3$, and 5). A *ritual proof scheme* is a scheme in which an individual believes that a proof needs to have a certain appearance determined by an authority (such as the use algebraic symbols) to bestow certainty. A *deductive transformational proof scheme* is a scheme in which an individual believes that certainty in a statement can be obtained by demonstrating that the statement can be deduced from accepted premises through a chain of transformational inferences (as defined above).

There is extensive research demonstrating that many students and elementary teachers hold the undesirable authoritarian, empirical inductive, or ritual proof schemes (e.g., see Harel and Sowder 2007, and Stylianides et al. 2017, for comprehensive reviews). Likewise, there is evidence that many students and elementary teachers do not hold deductive transformational proof schemes, viewing proofs merely as defeasible evidence in support of a statement (e.g., Chazan 1993). It is perhaps for this reason that students will continue to seek evidence in support of a conjecture, even if they have read a proof of the conjecture and verified that the proof was correct (Fischbein and Kedem 1982). These findings can explain many nonnormative student behaviors, including their propensity to "prove by example," their focus on the form of a proof rather than its content, and their lack of appreciation of the proofs that they encounter in their studies (Harel and Sowder 1998).

Do Students Justify with the Aim of Obtaining Certainty?

A relatively unexplored question in mathematics education is what type of conviction that students are trying to obtain from the justifications that they produce. Bieda and Lepak (2014) reported that when middle school students were shown two justifications to support a number theoretic statement, many preferred an empirical justification (i.e., a verification of a universal statement with several examples) to a deductive justification. However, these students were often aware that empirical justifications could not guarantee truth and the deductive justifications potentially could. Rather, the students favored the transparency and the concreteness of the empirical justification, while adding that their ideal would be have both justifications since they served different epistemic functions. If students do not respect obtaining certainty as an ideal in mathematics, they may still hold inductive empirical proof schemes, even if they are aware of the limitations of empirical justifications (see, for instance, Healy and Hoyles 2000).

What Do Students and Teachers Think the Role of Proof Is?

Many elementary and secondary teachers believe the only function of proof is to verify that a mathematical claim is true. They do not appreciate that proofs can be used to further other pedagogical goals, such as providing explanation and facilitating communication (e.g., Knuth 2002). Students also hold similar perspectives on proof and do not see proof as serving any other purpose than verification (e.g., Healy and Hoyles 2000). As we noted earlier, some students do not even see proof as providing conclusive verification. Some students view proof as merely defeasible evidence (e.g., Chazan 1993) or as a tool that one can use to verify a general solution in particular cases (Vinner 1983). Further, Herbst and Brach (2006) suggested that for many high school geometry students, proof does not serve any epistemic purpose in knowledge generation or verification at all. Rather, these authors suggest that many students view proving tasks as a means to receive credit by demonstrating their logical ability to their teachers. Needless to say, students who hold such views are unlikely to engage in proving tasks in a manner that can promote their mathematical growth and understanding.

In summary, these three subsections highlight that students' conceptions of deductive reasoning are intricately related to their understanding of what it means to know and do mathematics. Students who do not appreciate the ideal of knowing mathematical statements with certainty are unlikely to seek proofs to guarantee that theorems are true or to be bothered that an empirical argument that they produce is insufficient to guarantee truth. Students who think that certainty is obtained from an authority are unlikely to try to verify claims for themselves. Students who believe that they only engage in deductive reasoning to receive external credit from their teacher are unlikely to think about the meaning of the justifications that they produce. For further discussion, see Harel and Sowder (1998).

Open Research Questions with Regard to Deductive Reasoning and Proof

Here we briefly describe three active areas of research investigating important strands of research with regard to proof.

What Factors Prevent Deductive Reasoning from Playing a Prominent Role in K-12 Mathematics Classrooms?

Using the large international database of mathematics lessons from the TIMSS study, Hiebert et al. (2003) analyzed the problems presented to students during their lessons. While problems requiring deductive reasoning were fairly common in Japanese classrooms, such problems were largely absent from classrooms in other countries. In United States classrooms, problems requiring deductive reasoning were virtually nonexistent. These findings are consistent with the results of subsequent smaller studies (e.g., Bieda 2010). Researchers have highlighted many reasons that deductive reasoning plays a tangential role in many classrooms, including teachers lacking the pedagogical knowledge to teach proof effectively (e.g., Knuth 2002) and a lack of meaningful proving opportunities in textbooks (e.g., Otten et al. 2014). Bieda (2010) found that even knowledgeable middle school teachers who used curricula that supported the teaching of proof were reluctant to have proof play a role in their class. Bieda attributed these teachers' reluctance to be at least in part due to their beliefs about proof. These teachers viewed proof as being accessible only to the brightest students and not essential for understanding the middle school content. Consequently, they viewed proof as more of an enrichment activity or a challenge problem to give to their brightest students, rather than as something central to learning mathematics.

How Can We Have Proof Serve the Functions of Providing Insight, Promoting Discovery, and Illustrating Methods?

The mathematician Yuri Manin famously said that "good proofs are proof that make us wiser" (quoted in Aigner and Schmidt 1998). As we noted earlier, mathematics educators agree and desire that the proofs presented in classrooms should make students wiser by offering explanations and illustrating methods. What is needed is research into how this can be done effectively. As noted above, one area of research involves operationalizing what it means for a proof to be explanatory and illustrate methods. While there are examples of proofs that fulfill this purpose (e.g., Hanna 2018; Hanna and Barbeau 2008), there are not general guidelines for producing such proofs or determining if a new proof is explanatory. Further, there is increasing evidence that what can be learned from a particular proof depends not only on the proof itself but also on how the student engages with the proof (e.g., Lew et al. 2016). For instance, Hodds et al. (2014) demonstrated students understanding of a proof (including their ability to summarize the proof and apply the methods used in a proof to a new problem) significantly increased if they provided self-explanations while reading the proof. An alternative perspective is that the explanation contained in a proof might be contained in the activity of producing the proof, rather than in the product that is produced (c.f., Harel 2001; Stylianides et al. 2016).

What Strategies and Heuristics Can Help Students Learn to Prove?

Weber et al. (in press) demonstrated that students may provide empirical justifications for their claims not because these students thought such justifications were adequate or desirable, but rather because these students lacked the cognitive skills to produce deductive justifications. Similar suggestive findings were reported by Stylianides and Stylianides (2009). These results highlight that even if students hold desirable conceptions about deductive reasoning and proof, they still might not be able to produce proofs. These findings introduce the questions of what strategies and heuristics can help students produce proofs? How do successful students come to learn these strategies? To what extent can they be taught? Contemporary work in this area has largely centered on how mathematicians and successful students use informal representations of mathematical concepts in their reasoning, such as examples (e.g., Sandefur et al. 2013), and gestures (e.g., Williams-Pierce et al. 2017). However, researchers are only beginning to highlight the competencies that are needed to reason about these representations effectively. The critical issues of how students come to acquire these competencies and whether these competencies can be taught remain important open research questions.

References

Aigner M, Schmidt VA (1998) Interview with Yuri I. Manin: Good proofs are proofs that make us wiser. The Berlin Intelligencer, pp 16–19

Alibert D, Thomas M (1991) Research on mathematical proof. In: Tall D (ed) Advanced mathematical thinking. Kluwer, Dordrecht, pp 215–230

Balacheff N (1987) Processus de preuves et situations de validation. Educ Stud Math 18:147–176

Balacheff N (2008) The role of the researcher's epistemology in mathematics education: an essay on the case of proof. ZDM 40(3):501–512

Bieda KN (2010) Enacting proof-related tasks in middle school mathematics: challenges and opportunities. J Res Math Educ 41:351–382

Bieda KN, Lepak J (2014) Are you convinced? Middle-grade students' evaluations of mathematical arguments. Sch Sci Math 114(4):166–177

Brown S (2014) On skepticism and its role in the development of proof in the classroom. Educ Stud Math 86:311–335

Chazan D (1993) High school geometry students' justification for their views of empirical evidence and mathematical proof. Educ Stud Math 24:359–387

Cobb P, Wood T, Yackel E, McNeal B (1992) Characteristics of classroom mathematics traditions: an interactional analysis. Am Educ Res J 29(3):573–604

Dawkins PC, Cook JP (2017) Guiding reinvention of conventional tools of mathematical logic: students' reasoning about mathematical disjunctions. Educ Stud Math 94(3):241–256

Dawkins P, Weber K (2017) Values and norms of proof for mathematicians and students. Educ Stud Math 95:123–142

de Villiers M (1990) The role and function of proof in mathematics. Pythagoras 24:17–24

Doyle T, Kutler L, Miller R, Schueller A (2014) Proofs without words and beyond. Convergence. Retrieved from: https://www.maa.org/press/periodicals/convergence/proofs-without-words-and-beyond. Last downloaded 16 Jan 2018

Durand-Guerrier V (2016) Conceptualizations of the continuum, an educational challenge for undergraduate students. Int J Res Undergrad Math Educ 2:338–361

Fischbein E, Kedem I (1982) Proof and certitude in the development of mathematical thinking. In: Vermandel A (ed) Proceedings of the sixth international conference for the psychology of mathematics education. PME, Antwerp, pp 128–131

Hanna G (2018) Reflections on proof as explanation. In A. Stylianides & G. Harel (Eds.) Advanced in mathematics education research on proof and proving. (pp. 3–18). Springer: Dordrect

Hanna G, Barbeau E (2008) Proofs as bearers of mathematical knowledge. ZDM – Int J Math Educ 40:345–353

Harel G (1998) Two dual assertions: the first on learning and the second on teaching (or vice versa). Am Math Mon 105:497–507

Harel G (2001) The development of mathematical induction as a proof scheme: a model for DNR-based instruction. In: Campbell S, Zaskis R (eds) Learning and teaching number theory: research in cognition and instruction. Kluwer, Dordrecht, pp 185–212

Harel G, Sowder L (1998) Students' proof schemes: results from exploratory studies. In: Schoenfeld AH, Kaput J,

D

Dubinsky E (eds) Research in collegiate mathematics education III. American Mathematical Society, Providence, pp 234–283
Harel G, Sowder L (2007) Toward comprehensive perspectives on the learning and teaching of proof. In: Lester FK (ed) Second handbook of research on mathematics teaching and learning. Information Age, Greenwich, pp 805–842
Healy L, Hoyles C (2000) Proof conceptions in algebra. J Res Math Educ 31:396–428
Herbst P (2002) Establishing a custom of proving in American school geometry: evolution of the two-column proof in the early twentieth century. Educ Stud Math 49:283–312
Herbst P, Brach C (2006) Proving and doing proofs in high school geometry classes: what is it that is going on for students? Cogn Instr 24(1):73–122
Hiebert J, Gallimore R, Garnier H, Givvin KB, Hollingsworth H, Jacobs J, Chui AM, Wearne D, Smith M, Kersting N, Manaster A, Tseng E, Etterbeek W, Manaster C, Gonzales P, Stigler J (2003) Teaching mathematics in seven countries: results from the TIMSS 1999 video study. NCES 2003-013. U. S. Department of Education, Institute of Education Sciences, Washington, DC
Hodds M, Alcock L, Inglis M (2014) Self-explanation training improves proof comprehension. J Res Math Educ 45:62–101
Inglis M, Attridge N (2017) Does mathematical study develop logical thinking? Testing the theory of formal discipline. World Scientific, ISBN: 978-1-78634-068-9. http://www.worldscientific.com/worldscibooks/10.1142/q0020
Knuth EJ (2002) Secondary school mathematics teachers' conceptions of proof. J Res Math Educ 33:379–405
Lew K, Fukawa-Connelly T, Mejia-Ramos JP, Weber K (2016) Lectures in advanced mathematics: why students might not understand what the professor is trying to convey. J Res Math Educ 47:162–198
Mamona-Downs J, Downs M (2010) Necessary realignments from mental argumentation to proof presentation. In: Durrand-Gurrier V, Soury-Lavergne S, Arzarello F (eds) Proceedings of CERME 6. Institute National de Recherche Pedagogique, Lyon, pp 2337–2345
National Council of Teacher of Mathematics (2000) Principles and standards for school mathematics. Author, Reston
Otten S, Males LM, Gilbertson NJ (2014) The introduction of proof in secondary geometry textbooks. Int J Educ Res 64:107–118
Sandefur J, Mason J, Stylianides GJ, Watson A (2013) Generating and using examples in the proving process. Educ Stud Math 83(3):323–340
Stylianides AJ, Stylianides GJ (2009) Proof constructions and evaluations. Educ Stud Math 72(2):237–253
Stylianides GJ, Sandefur J, Watson A (2016) Conditions for proving by mathematical induction to be explanatory. J Math Behav 43:20–34
Stylianides G, Stylianides A, Weber K (2017) Research on the teaching and learning of proof: taking stock and moving forward. In: Cai J (ed) Compendium for research in mathematics education. National Council of Teachers of Mathematics, Reston
Vinner S (1983) The notion of proof: some aspects of students' view at the senior high level. In: Hershkowitz R (ed) Proceedings of the seventh international conference for the psychology of mathematics education. Weizmann Institute of Science, Rehovot, pp 289–294
Weber K, Inglis M, Mejía-Ramos JP (2014) How mathematicians obtain conviction: implications for mathematics instruction and research on epistemic cognition. Educ Psychol 49:36–58
Weber K, Lew K, Mejia-Ramos JP. Using expectancy value theory to account for students' mathematical justifications. To appear in Cognition and Instruction (in press)
Williams-Pierce C, Pier EL, Walkington C, Boncoddo R, Clinton V, Alibali MW, Nathan MJ (2017) What we say and how we do: action, gesture, and language in proving. J Res Math Educ 48(3):248–260

22q11.2 Deletion Syndrome, Special Needs, and Mathematics Learning

Sophie Brigstocke
Department of Psychology, University of York, York, UK

Keywords

Genetic disorder · Mathematics difficulties · Cognitive impairment

Characteristics

Chromosome 22q11.2 deletion syndrome (22q) is the most common genetic deletion syndrome with an estimated prevalence of between one in 3000 and 6000 births (e.g., Kobrynski and Sullivan 2007). It has only been detectable with 100% accuracy since 1992 using techniques such as the FISH test (fluorescence in situ hybridization). Prior to identification of a single associated deletion, the syndrome had been given a number of different labels according to the primary medical condition, for example, velocardiofacial syndrome, DiGeorge syndrome, Cayler syndrome, Shprintzen syndrome, and Catch 22.

The majority of individuals with 22q experience some degree of learning difficulty and generally show a marked imbalance in performance across different subtests within IQ batteries. Verbal IQ scores are usually significantly higher than performance IQ scores (e.g., Moss et al. 1999; Wang et al. 2007). The majority of children will receive some form of support at school although some individuals experience no difficulties at all. Indeed a very wide level of individual differences in attainment in individuals with 22q is noted in all studies to date.

There is consistent evidence that mathematics skills are weaker than literacy skills in the majority children with 22q. This profile is unusual as children with mathematics difficulties are often reported to have comorbid reading difficulties. Typically, performance on standardized tests of reading and spelling is within the normal range, but performance on mathematical reasoning and arithmetic tasks is at least one standard deviation below age norms in children with 22q. Children with 22q specifically selected to have full scale IQ of at least 70 also demonstrate this profile, thereby suggesting that it is associated with 22q per se rather than low general ability.

There are very few studies examining number skills in detail in children with 22q. De Smedt et al. (2006, 2007a, b) tested children, selected to have an IQ of more than 70, on a series of computerized tests assessing performance in number reading and writing, number comparison, counting, and single and multi-digit arithmetic. A mathematical word-solving task was also included and reading ability was measured. Children were individually matched with typically developing children from the same class at school for gender, age, and parental education level. Consistent with their hypotheses, De Smedt et al. (2007a, b) report group differences on multi-digit operations involving a carry, word-solving problems, and speed in judging the relative value of two digits. There was no difference in reading, number reading and writing, single digit addition, or verbal and dot counting accuracy. The difficulties with multi-digit operations are unsurprising given the visuospatial requirements of operations such as borrowing and carrying. Previous researches suggest that multi-digit arithmetic is an area of particular difficulty in children with visuospatial learning disability as well as arithmetic difficulties (Venneri et al. 2003). More research is needed to further uncover the nature of the mathematical difficulties experienced by children with 22q and to aim to uncover best practice methods for teaching number skills in 22q as so far, certainly in the UK, no consensus has been reached.

Cross-References

▶ Autism, Special Needs, and Mathematics Learning
▶ Deaf Children, Special Needs, and Mathematics Learning
▶ Down Syndrome, Special Needs, and Mathematics Learning
▶ Inclusive Mathematics Classrooms
▶ Language Disorders, Special Needs and Mathematics Learning
▶ Learning Difficulties, Special Needs, and Mathematics Learning

References

De Smedt B, Swillen A, Devriendt K, Fryns J, Verschaffel L, Ghesquiere P (2006) Mathematical disabilities in young primary school children with velo-cardio-facial syndrome. Genet Couns 17:259–280

De Smedt B, Devriendt K, Fryns JP, Vogels A, Gewillig M, Swillen A (2007a) Intellectual abilities in a large sample of children with velo-cardio-facial syndrome: an update. J Intellect Disabil Res 51:666–670

De Smedt B, Swillen A, Devriendt K, Fryns J, Verschaffel L, Ghesquiere P (2007b) Mathematical disabilities in children with velo-cardio-facial syndrome. Neuropsychologia 45:885–895

Kobrynski L, Sullivan K (2007) Velocardiofacial syndrome, DiGeorge syndrome: the chromosome 22q11.2 deletion syndromes. Lancet 370(9596):1443–1452

Moss E, Batshaw M, Solot C, Gerdes M, McDonald-McGinn D, Driscoll D, Emanuel B, Zackai E, Wang

P (1999) Psychoeducational profile of the 22q11.2 microdeletion: a complex pattern. J Pediatr 134(2): 193–198

Venneri A, Cornoldi C, Garuti M (2003) Arithmetic difficulties in children with visuospatial learning disability (VLD). Child Neuropsychol 9(3):175–183

Wang P, Woodin F, Kreps-Falk R, Moss E (2007) Research on behavioral phenotypes: velocardiofacial syndrome (deletion 22q11.2). Dev Med Child Neurol 42:422–427

Design Research in Mathematics Education

Malcolm Swan
Centre for Research in Mathematics Education, Jubilee Campus, School of Education, University of Nottingham, Nottingham, UK

Keywords
Engineering research · Design experiments · Design research

Definition

Design-based research is a formative approach to research, in which a product or process (or "tool") is envisaged, designed, developed, and refined through cycles of enactment, observation, analysis, and redesign, with systematic feedback from end users. In education, such tools might, for example, include innovative teaching methods, materials, professional development programs, and/or assessment tasks. Educational theory is used to inform the design and refinement of the tools and is itself refined during the research process. Its goals are to create innovative tools for others to use, describe, and explain how these tools function, account for the range of implementations that occur, and develop principles and theories that may guide future designs. Ultimately, the goal is *transformative*; we seek to create new teaching and learning possibilities and study their impact on teachers, children, and other end users.

Malcolm Swan: deceased.

The Origins and Need for Design Research

Educational research may broadly be categorized into three groups: the *humanities* approach, scholarly study that generates fresh insights through critical commentary, the *scientific* approach that analyzes phenomena empirically to better understand how the world works, and the *engineering* approach that not only seeks to understand the status quo but also attempts to use existing knowledge to systematically develop "high-quality solutions to practical problems" (Burkhardt and Schoenfeld 2003). Design research falls into this "engineering" category and, as such, seeks to provide the tools and processes that enable the end users of mathematics education (teachers and students, administrators, and politicians) to tackle practical problems in authentic settings.

Design research is an unsettled construct and the field is in its youth. It is only at the beginning of the last two decades that we see design research as an emerging paradigm for the study of learning through the systematic design of teaching strategies and tools. The beginnings of this movement, at least in the USA, are usually attributed to Brown (1992) and Collins (1992), though in a sense, it was an idea waiting to be named (Schoenfeld 2004). In Europe there have long been traditions of principled design-based research under other guises, such as curriculum development and didactical engineering (e.g., Bell 1993; Brousseau 1997; Wittmann 1995).

Prior to the 1990s, much educational and psychological research had relied heavily on quasi-experimental studies that had been developed successfully in other fields such as agriculture. These involved experimental and control treatments to evaluate whether or not particular variables were associated with particular outcomes. In mathematics education, for example, one might design a novel approach to teaching a particular area of content, assign students to an experimental or control group, and assess their performance on some defined measures, using pre- and post-testing. Though sounding straightforward, this practice proved highly problematic (Schoenfeld 2004): the goals of education are more complex than the mastery of specific skills; the control of variables in naturalistic settings is often impossible, undesirable, and sometimes even unethical; and much of the theory is "emergent," only becoming apparent as one engages in the research.

In the early 1990s, a number of researchers began to question the limitations of traditional experimental psychology as a paradigm for educational research. Brown's paper on "design experiments" was seminal (Brown 1992). Brown recounts how her own research moved away from laboratory settings toward naturalistic ones in which she attempted to transform classrooms from "worksites under the management of teachers into communities of learning." She vividly recounts her own struggles in reconceptualizing her focus and methodology, deconstructing methodological criticisms against it (such as the Hawthorne effect). Interestingly Brown still saw the need for lab-based research, both to precede and stimulate work in naturalistic settings and also for the closer study of phenomena that had arisen in those settings. At about the same time, Collins (1992, pp. 290–293) began to argue for a design science in education, distinguishing *analytic* sciences (such as physics or biology) as where research is conducted in order to *explain* phenomena from *design* sciences (such as aeronautics or acoustics) where the goal is to determine how designed artifacts (such as airplanes or concert halls) behave under different conditions. He argued strongly for the need of the latter in education. In mathematics education, such designed artifacts might include, for example, new teaching methods, materials, professional development programs, assessment tasks, or any combination of these.

Since that time, "design research" has become more widespread and respectable in education. However, it must be said that not all so-called "design research" studies satisfy the definition described above. Some, for example, do not satisfy the requirement that the designs should be theory-based and develop theory, while others do not move beyond the early stages and test their designs in the hands of others not involved in the development process.

Characterizing Design-Based Research

There have been many attempts to characterize design-based research (Barab and Squire 2004; Bereiter 2002; Cobb et al. 2003; DBRC 2003, p. 5; Kelly 2003; Lesh and Sriraman 2010; Swan 2006, 2011; van den Akker et al. 2006). While design research is still in its infancy and its characterization is far from settled, most researchers do seem to agree that design-based research is:

Creative and Visionary

The researcher identifies a problem in a defined context and, drawing on prior research, envisions a tool that might help end users to tackle it. A draft design is developed, possibly with the assistance of end users. For example, the researcher identifies a particular student learning need and uses research to design a series of lessons. The ultimate aim is to produce an effective design, an account of the theory and principles underpinning the design, and an analysis of the range of ways in which the design functions in the hands of a typical sample of the target population of teachers and students.

Ecologically Valid

The researcher studies and refines the design in authentic settings, such as classrooms. This precludes the prior manipulation of variables in the study. It is important, therefore, to distinguish those aspects of the design that are being studied from those that are extraneous.

Interventionist and Iterative

The role of the researcher evolves as the research proceeds. During early iterations, the design is usually sketchy, and the researcher needs to intervene to make it work. With teaching materials, for example, this phase may be conducted with small samples of students. Later, as the design evolves, the researcher holds back, in order to see how the design functions in the hands of end users. Early iterations are often conducted in a few favorable contexts. Early drafts of teaching materials, for example, may be tested in carefully chosen classrooms with confident teachers, in order to gain insights into what is possible with faithful implementation. Later iterations aim to study how the design functions in a wider range of authentic contexts, with teachers who have not been involved in the design process. Under these conditions, "design mutations" invariably occur. Rather than viewing these as negative, interfering factors, the designs and theories evolve to explain these mutations. With each cycle of the process, the sample size is increased and becomes more typical of the target population. From time to time, a particular issue may arise that the researcher wants to study closely. In such a case, it is possible to go back to the small-scale study of that isolated issue.

Theory-Driven

The outputs of design research include developing theories about learning, interventions, and tools. Rather than focusing on learning outcomes, using pre- and posttests, the research seeks to understand *how* designs function under different conditions and in different classroom contexts. The theories that evolve in this way are *local* and *humble* in scope and should not be judged by their claims to "truth" but rather their claims to be useful (Cobb et al. 2003). Theory in design research usually focuses on an explanation of how and why a particular design feature works in a particular way. It is both specific and generative in that it can be used to predict ways in which future designs will function if they embody this feature.

Some Issues and Challenges

Firstly, design research done well requires great skill on the part of researchers. Indeed, the combination of skills required is not usually found in individuals but in teams. A design research team will typically involve people with knowledge of the literature (researchers), an understanding of pedagogy (teachers), creative "care and flair" (designers), and facility with "delivering" the design (publishers' IT technicians).

Secondly, design research often takes a great deal longer than other forms of research. There is often a significant "entry fee" in terms of time and energy taken up with producing a prototype before any study of it can begin. This is particularly true if the design involves creating new software. Then, each cycle of design, implementation, analysis, and redesign can occupy weeks, if not months.

Thirdly, design research is data rich. A mixture of qualitative and quantitative methods is used to develop a rich description of the way the design works as well as the kinds of learning outcomes that may be expected. This often results in a proliferation of data. Brown, for example, found that she "had no room to store all the data, let alone time to score it" (Brown 1992, p. 152). Data may include lesson observations, videos of the designs in use, and questionnaires and interviews with users. In early iterations, observation plays a dominant role. Later, however, more indirect means are also needed as the sample size grows. Reliability may be improved through the use of triangulation from multiple data sources and repetition of analyses across cycles of implementation and through the use of standardized measures.

Fourthly, design research requires discipline. It is all too tempting to turn a "good idea" into a draft design and then ask someone to try it out to "see what happens." Good design-based research is more than formative evaluation; however, it is theory-driven. In preparation for a design-based research study, one must try to articulate the theory and draw clear lines of connection between this and the design itself. This may be done by eliciting "principles" to direct the design. The research involves putting these principles in "harm's way" (Cobb et al. 2003). Then, the focus of the research needs to be articulated. For early iterations this may be on the potential impact of the faithful use of the design, while on later iterations, we may be more interested in refining the design by studying end users' interpretations and mutations.

Finally, writing up design research is problematic. Most designs are too extensive to be described and analyzed in traditional journal articles that emphasize methods and results over tools. Recently e-journals have begun to appear that allow for a much clearer articulation of design-based research. These, for example, allow extensive extracts of teaching and professional development materials to be displayed, along with videos of the designs in use (see, e.g., http://www.educationaldesigner.org).

Cross-References

- ▶ Curriculum Resources and Textbooks in Mathematics Education
- ▶ Didactic Engineering in Mathematics Education
- ▶ Didactic Situations in Mathematics Education
- ▶ Mathematics Curriculum Evaluation

References

Barab S, Squire K (2004) Design-based research: putting a stake in the ground. J Learn Sci 13(1):1–14

Bell A (1993) Principles for the design of teaching. Educ Stud Math 24(1):5–34

Bereiter C (2002) Design research for sustained innovation. Cogn Stud Bull Jpn Cogn Sci Soc 9(3):321–327

Brousseau G (1997) Theory of didactical situations in mathematics (trans: Balacheff N, Cooper M, Sutherland R, Warfield V), vol 19. Kluwer, Dordrecht

Brown AL (1992) Design experiments: theoretical and methodological challenges in creating complex interventions in classroom settings. J Learn Sci 2(2):141–178

Burkhardt H, Schoenfeld A (2003) Improving educational research: toward a more useful, more influential and better-funded enterprise. Educ Res 32(9):3–14

Cobb P, Confrey J, diSessa A, Lehrer R, Schauble L (2003) Design experiments in educational research. Educ Res 32(1):9

Collins A (1992) Towards a design science in education. In: Scanlon E, O'Shea T (eds) New directions in educational technology. Springer, New York, pp 15–22

DBRC (2003) Design-based research: an emerging paradigm for educational inquiry. Educ Res 32(1):5–8

Kelly A (2003) Theme issue: the role of design in educational research. Educ Res 32(1):3–4

Lesh R, Sriraman B (2010) Reconceptualizing mathematics education as a design science. In: Sriraman B, English L (eds) Theories of mathematics education: seeking new frontiers. Springer, Berlin, pp 123–146

Schoenfeld A (2004) Design experiments. In: Elmore PB, Camilli G, Green J (eds) Complementary methods for research in education. American Educational Research Association, Washington

Swan M (2006) Collaborative learning in mathematics: a challenge to our beliefs and practices. National Institute for Advanced and Continuing Education (NIACE) for the National Research and Development Centre for Adult Literacy and Numeracy (NRDC), London

Swan M (2011) Towards the creative teaching of mathematics. In: Thomson P, Sefton-Green J (eds) Researching creative learning: methods and issues. Routledge, Abingdon, pp 54–67

van den Akker J, Graveemeijer K, McKenney S, Nieveen N (eds) (2006) Educational design research. Routledge, London/New York

Wittmann E (1995) Mathematics education as a design science. Educ Stud Math 29(4):355–374

Dialogic Teaching and Learning in Mathematics Education

Ole Skovsmose
Department of Learning and Philosophy, Aalborg University, Aalborg, DK, Denmark

D

Keywords

Dialogue · Literacy · Mathemacy · Investigation · Inquiry · Inquiry cooperation model · Critical mathematics education

Definition

Dialogic teaching and learning refers to certain qualities in the interaction between teachers and students and among students. The qualities concern possibilities for the students' involvement in the educational process, for establishing enquiry processes, and for developing critical competencies.

Characteristics

Sources of Inspiration

There are different sources of inspiration for bringing dialogue into the mathematics classroom, and let me just refer to two rather different.

The notion of dialogue plays a particular role in the pedagogy of Paulo Freire. He sees dialogue as crucial for developing *literacy*, which refers to a capacity in reading and writing the world: reading it, in the sense that one can interpret sociopolitical phenomena, and writing it, in the sense that one becomes able to make changes.

With explicit reference to mathematics, the crucial role of dialogue can be argued with allusion to Imre Lakatos' presentation in *Proof and Refutations* (Lakatos 1976). Here Lakatos shows that a process of mathematical discovery is of dialogic nature, characterized by proofs and refutations.

Critical mathematics education and social constructivism have developed dialogic teaching and learning through a range of examples and studies. It has been emphasized that dialogue is principal for establishing critical perspectives on mathematics and for a shared construction of mathematical notions and ideas. In fact dialogic teaching and dialogic learning represents two aspects of the same process.

Marilyn Frankenstein (1983) has emphasized the importance of Freire's ideas (Freire 1972) for developing critical mathematics education, and Paul Ernest (1998) has opened the broader perspective of social constructivism, also acknowledging the importance of Lakatos work.

The Inquiry Cooperation Model

The notion of dialogue appears to be completely open. As a consequence, it becomes important to try to characterize what a dialogue could mean. The *inquiry cooperation model* as presented in Alrø and Skovsmose (2002) provides such a specification with particular references to mathematics.

This model characterizes different dialogic acts: *Getting in contact* refers to the act of tuning in at each other. *Locating* and *identifying* refer to forms of grasping perspectives, ideas, and arguments of the other. *Advocating* means providing arguments for a certain point of view – although not necessary one's own. *Thinking aloud* means making public details of one's thinking, for instance, through gestures and diagrams. *Reformulating* refers to particular attempts in grasping other ideas by rethinking, rephrasing, and reworking them. *Challenging* means questioning certain ideas, which is an important way of sharpening mathematical arguments. *Evaluating* refers to reflexive questioning, like: What insight might we have reached? What new questions have we encountered?

Dialogic teaching and learning can be characterized as a process rich of such dialogic acts.

New Qualities in Teaching and Learning

The idea of dialogic teaching and learning is to promote an education with new qualities. Let me refer to just a few having to do with the students' interest, making investigations, and developing a mathemacy.

Students' Interest. It has been emphasized that dialogic teaching and learning includes a sensitivity to the students' perspectives and possible interests for learning. This sensitivity has not only to do with the dialogic act of "getting in contact" but with all the acts represented by the inquiry cooperation model. A principal point of dialogic teaching is to invite students into the learning process as active learners.

Making Investigations. Dialogic teaching and learning can be characterized in terms of investigative approaches, where both teacher and students participate in the same inquiry process. Barbara Jaworski (2006) makes a particular emphasis on establishing communities of inquiry, and in any such communities, dialogue plays a defining role. Landscapes of investigations (Skovsmose 2011) might also provide environments that facilitate dialogic teaching and learning.

Similar to literacy, *mathemacy* refers not only to a capacity in dealing with mathematical notions and ideas but also to a capacity in interpreting sociopolitical phenomena and acting in a mathematized society. Thus, mathemacy combines a capacity in reading and writing *mathematics* with a capacity in reading and writing *the world* (see Gutstein 2006). Dialogue teaching and learning is in hectic development, both in theory and in practice. A range of new studies and new classroom initiatives are being developed. In particular, the very notion of dialogue is in need of further development; see, for instance, Alrø and Johnsen-Høines (2012) and Hunter et al. (2018).

Cross-References

▶ Critical Mathematics Education
▶ Mathematization as Social Process

References

Alrø H, Johnsen-Høines M (2012) Inquiry – without posing questions? Math Enthus 9(3):253–270

Alrø H, Skovsmose O (2002) Dialogue and learning in mathematics education: intention, reflection, critique. Kluwer, Dordrecht

Ernest P (1998) Social constructivism as a philosophy of mathematics. State University of New York Press, Albany

Frankenstein M (1983) Critical mathematics education: an application of Paulo Freire's epistemology. J Educ 164:315–339

Freire P (1972) Pedagogy of the oppressed. Penguin Books, Harmondsworth
Gutstein E (2006) Reading and writing the world with mathematics: toward a pedagogy for social justice. Routledge, New York
Hunter R, Cevil M, Herbel-Eisenmann B, Planas N, Wagner D (eds) (2018) Mathematical discourse that breaks barriers and create space for marginalized learners. Sense, Rotterdam
Jaworski B (2006) Theory and practice in mathematics teaching development: critical inquiry as a mode of learning in teaching. J Math Teach Educ 9(2):187–211
Lakatos I (1976) Proofs and refutations: the logic of mathematical discovery. Cambridge University Press, Cambridge
Skovsmose O (2011) An invitation to critical mathematics education. Sense, Rotterdam

Didactic Contract in Mathematics Education

Guy Brousseau[1], Bernard Sarrazy[2,3] and Jarmila Novotná[4]
[1]Institut Universitaire de Formation des maîtres d'Aquitaine, Mathématiques, Laboratoire Cultures Education Societes (LACES), Anthropologie et diffusion des savoir, University of Bordeaux – France, Bordeaux Cedex, France
[2]Departement Sciences de l'Education, Laboratoire Cultures Education Societes (LACES), EA 4140, Anthropologie et diffusion des savoir, Univ. Bordeaux – France, Bordeaux Cedex, France
[3]Département des sciences de l'éducation, Faculté des Sciences de l'Homme, Université Bordeaux Segalen, Bordeaux Cedex, France
[4]Faculty of Education, Charles University, Praha, Czech Republic

Keywords
Didactical situations · Mathematical situations · Didactical contract · Didactique · Milieu · Devolution · Institutionalization

Introduction

Teachers manage *didactical situations* that create and exploit *mathematical situations* where practices are exercised and students' mathematical knowledge is developed. The study of the *didactical contract* concerns the compatibility on this precise subject of the aspirations and requirements of the students, the teachers, the parents, and the society.

Definition

A "*didactical contract*" is an interpretation of the commitments, the expectations, the beliefs, the means, the results, and the penalties envisaged by one of the protagonists of a *didactical situation* (student, teacher, parents, society) for him- or herself and for each of the others, *à propos of the mathematical knowledge being taught* (Brousseau and Otte 1989; Brousseau 1997). The objective of these interpretations is to account for the actions and reactions of the partners in a didactical situation.

The didactical contract can be broken down into two parts: a contract of devolution – the teacher organizes the mathematical activity (see ► "Didactic Situations in Mathematics Education") of the student who in response commits him- or herself to it – and a contract of institutionalization – the students propose their results and the teacher vouches for the part of their results that conforms to reference knowledge.

Customary practices (Balacheff 1988), whether explicit or tacit, leave the hope that divergences are accidental and reducible and that there exist real contracts, whether or not they can be made explicit, that are compatible and satisfactory. This is not so, owing to various paradoxes that became apparent in the course of teaching in a way that is based on mathematical situations. This gave rise to many questions, among them are as follows:

How could students commit themselves to the subject of knowledge that they have not yet learned?

What are the respective roles of what is inexpressible, of what is said, of what is not said or cannot be said to the other in the teaching relationship?

Does there exist knowledge that ought not to be made explicit before being learned?

The study of these questions was the origin of the theory of didactical situations.

Characteristics

Background: Illustrative Examples

These questions arose in the course of research at the COREM (Center for Observation and Research on Mathematics Education, entity formed of a laboratory and a school establishment by the IREM of the University of Bordeaux (1973–1999)) on the possibility of assigning to *mathematical situations* the job of managing what the teacher cannot say or the student cannot yet understand from a text, and in the clinical observation of students failing selectively in mathematics:

(a) *The Case of Gaël.* Gaël (8 years old) always responded in the manner of a very young child. It was not a developmental delay, but rather a posture. By replacing some lessons with "games" in which he could take a chance and see the effects of his decisions and by getting him to make bets – without too much risk – on whether his answers were right, the experimenters saw his attitude changes radically and his difficulties disappear. A new "didactical contract" with him had been constructed (Brousseau and Warfield 1998).

(b) *The Age of the Captain.* Researchers at the Institute for Research on the Teaching of Mathematics (IREM of Grenoble) offered students at age 8 the following problem: "On a boat there are 26 sheep and 10 goats. How old is the captain?" 76 of the 97 students answered, "36 years old."

This experiment produced a scandal. Some accused the teachers of stupefying their students; others reproached the researchers for "laying stupid traps for the children." In a letter to the experimenters, G. Brousseau indicated to them that it was a matter of an "effect of the contract" for which neither the students nor the teachers were responsible. So the researchers asked the students: "What do you think of this problem?" The students responded: "It is stupid!" The researchers ask: "Then why did you answer it?" The students answered: "Because the teacher asked for it!" The researchers ask: "And if the captain was 50 years old?" The students made a response: "The teacher didn't give the right numbers." A similar experiment done with established teachers produced the same behavior: for various reasons (such as the hope of an explanation that the teacher wanted to hear) the subjects produce the answer least incompatible with their knowledge, even when they see very well that it is false: the obligation of answering is stronger than that of answering correctly. Despite these explanations, for years the initial observation elicited strong criticisms of the work of the teachers (Sarrazy 1996).

Didactical and Ethical Responsibility

The teacher has the responsibility of supporting the collective and individual activity of the students, of attesting in the end to the truth of the mathematics that has been done, of confirming it or giving proofs, of organizing it in the standard way, of identifying errors that have been or might be made and passing judgment on them (without passing judgment on their authors), and of providing the students with a moderate amount of individual help (as with the natural learning of a language.) Occasional individual help conforms to the collective process of mathematical communities. If the teacher finds himself acceding to an institutional function, he may be subject to obligations of equity and of means for which the responsibility is shared with the institution. Decisions made about the teacher and the students based on individual and isolated results are a dangerous absurdity. Experts, parents, and society share the responsibility for the effects of such decisions.

Paradoxes of the Didactical Contract

The teacher wants to teach what she knows to a student who does not know it. This has many consequences, among them are as follows:

(a) Custom can determine pedagogical and psychological relationships, but not those proper to new knowledge, because new knowledge is a specific unexpected adventure that consists of a modification and an augmentation of old knowledge and of its implications. Thus, it cannot be known in advance by the student: the teacher can only commit himself to general procedures, and for her part the student cannot commit herself to a project of which she does not know the main part.
(b) Paradox of devolution: the knowledge and will of the teacher need to become those of the student, but what the student knows or does by the will of the teacher is not done or decided by his own judgment. The didactical contract can only succeed by being broken: the student takes the risk of taking on a responsibility from which he already releases the teacher (a paradox similar to that of Husserl).
(c) Paradox of the said and unsaid (consequence of the preceding): it is in what the teacher does not say that the student finds what she can say herself.
(d) Paradox of the actor: the teacher must pretend to discover with his students knowledge that is well known to him. The lesson is a stage production.
(e) The paradox of uncertainty: knowledge manifests itself and is learned by the reduction of uncertainty that it brings to a given situation. Without uncertainty or with too much uncertainty, there is neither adaptation nor learning. The result is that the optimal progression of normal individual or collective learning is accompanied by a normal optimal rate of errors. Artificially reducing it damages both individual and collective learning. It is useful to arrange things so that it is not always the same students who are condemned to supply the necessary errors.
(f) As in the case of learning, excessive or premature adaptation of complex knowledge to conditions that are too particular leads it to be replaced by a simplified and specific knowledge. This can then constitute an *epistemological* or *didactical obstacle* to its later adaptation to new conditions. (For example, division of natural numbers is associated with a meaning, sharing, which becomes an obstacle to understanding it in the case where a decimal number needs to be divided by a larger decimal number, e.g., 0.3/0.8.)
(g) The paradox of rhetoric and mathematics. To construct the students' mathematical knowledge and its logical organization, the teacher uses various rhetorical means, designed to capture their attention. The culture, pedagogical procedures, and even mathematical discourse (commentaries on mathematics) overflow with metaphors, analogies, metonyms, substitutions, word pictures, etc. The mathematical concepts are often constructed against these procedures (e.g., "correlation is not causation"). The teacher should thus at the same time as an educator teach the culture with its historical mistakes and as a specialist cause the rejection of the parts that science has disqualified.

These paradoxes can only be unraveled by specific situations and processes carefully planned out in the light of well-shared knowledge of mathematical and scientific *didactique* (Brousseau and Otte 1989; Brousseau 2005).

Observations of Reactions of Teachers to Difficulties

These observations and the experimental and theoretical studies of the *didactical contract* make it possible to understand and predict the cumulative effects of teachers' decisions.

The contract manifests itself essentially in its ruptures. These are revealed by the reactions of the students or by the interventions of the teachers, and they can be classified as follows:

(a) *Abandonment*. The teacher does not react to an error made by the students (e.g., because it would be too complicated to explain it), or she repeats the question identically or she gives the complete solution.
(b) The progressive *reduction* or manipulation of the students' uncertainty, using a great variety of means:

Bringing in mathematical, technical, or methodological information

Decomposition of the problem into intermediate questions (decomposition of the objectives)

Use of various extra-mathematical rhetorical means: analogies, metaphors, metonyms, or mnemonic minders (the "Topaze effect")

(c) *Critical commentary on the errors*, the question, the knowledge, or the material.

(d) *A trial of the student* and its consequences: penalties, discrimination, and individualization.

In case of failure, the contract obligates the teacher to try again. The new attempt either replaces the preceding one or criticizes and corrects it, making of it a new teaching object (a meta-process).

For each of these types of response, there are conditions under which it is the most appropriate response; *thus there is no universal response.*

For example, Novotná and Hošpesová (2007) identify and classify the behaviors whose systematic repetition generates Topaze effects:

1. Explicitly, the teacher
   (a) Gives the steps of the solution and transforms it into the execution of a sequence of tasks
   (b) Asks questions in a sequence that mandates the procedures of the solution
   (c) Gives warnings about a possible error
   (d) Enumerates previous experiences or knowledge, pointing out analogies with problems that have previously been resolved or are obvious or well known.
2. Implicitly, he
   (a) Reformulates students' propositions or his own
   (b) Uses "guide" words
   (c) Pronounces the first syllable of words
   (d) Poses new questions that orient the student towards the solution
   (e) Shows doubt about dubious initiatives

Their research confirms that the resulting Topaze effects go unnoticed but have a high cost. The students, apparently active, become dependent on this aid and lose their confidence in themselves. An error is understood to be a transgression of the didactical contract and proof itself, badly supported, becomes something to be learned rather than understood.

By using jointly the notions of *milieu*, of situation, and of the didactical contract, Perrin-Glorian and Hersant (2003) were able to show in numerous examples on the one hand what the student and the *milieu* are in charge of and thus the occasions for learning that are their responsibility, and on the other hand the help brought in by the teacher.

Predicting and Explaining Certain Long-Term Effects

The uncontrolled recursive resumption of the same type of response leads to drifting and inevitable failures. For example, for the students studying the procedure for solving problems by the same pedagogical methods, studying theorems is just as costly, less sure, and less useful.

As another example, a sequence of meta-slippages contributed to the failure of the reform of "modern math": the foundations of mathematics were interpreted by "naïve" set theory, which was itself formalized into algebra. This was metaphorically represented by "graphs," which were finally interpreted in vernacular language. Each representation betrayed the preceding one slightly and supported new conventions, and the slippages were ultimately uncontrollable. In the absence of didactical situations and proven epistemological processes, varying the types of response seems to be the best strategy.

Enforcing requirements based on the results of individuals leads to a mincing up of the objectives, to the abandonment of high-level objectives, and to addressing the objectives by painful behaviorist methods. These slow the learning and lead to an individualization that slows it yet further. Each of these tends to destroy the role of provisional knowledge and to augment mechanically the time for teaching and learning without positive impact on the results.

Specifying the means of teaching a subject involves precise and specific protocols for performances that are known and accepted by the population. Specifying required results for the teachers as for the students has absolutely no scientific basis. Its disastrous effects, predicted since 1978, have been observable for 40 years.

The mean rate of success is a “regulated” variable of the system. Otherwise stated, the global progress of *all* the students is less rapid if one requires at every stage a 100% rate of success. The conception of mathematical activity as an adventure and a collective practice makes it possible to mitigate the effects of difference in rhythms of learning.

It seems that today the requirements of the different partners of teaching towards one another are less and less compatible with each other, perhaps because of the variety of possibilities, of offers, and of perspectives provided by numerous ill-coordinated sciences.

The experiments on teaching rational and decimal numbers (Brousseau 1997) or statistics and probability (Brousseau et al. 2002) prove that it is possible to organize efficient and communicable processes with the help of didactical contracts based on the nature of the knowledge to be acquired.

Extensions

Sarrazy (1996, 1997) studied the pitfalls of these meta-didactical slippages and more particularly those that are consequences of a teaching that aims at making the contractual expectations explicit, frequently taking the form of the teaching of metacognitive or heuristic procedures – or even of algorithms for solving problems. Complementing the work engaged in by Schubauer-Léoni (1986) in a psychosocial approach to the didactical contract, Sarrazy radicalized the paradox of the consubstantial rule (A rule does not contain in itself its conditions for use) of the contract at the intersection of the theory of situations and Wittgensteinian anthropology (Wittgenstein 1953). Contrary to the psychological or linguistic interpretations of the contract (such as that of “the age of the captain”), he showed how these slippages lead to a veritable demathematization of teaching by a displacement of the goals of the contract. These works also made it possible to establish the primacy of the role of situations and that of school cultures (Sarrazy 2002; Clanché and Sarrazy 2002; Novotná and Sarrazy 2011) and family habits conceived as *backgrounds* (Searle 1979) of the didactical contract. These backgrounds make it possible to explain the differences *in sensitivity to the didactical contract*, that is, the objective differences of the various positions of the students with regard to the implicit elements of the contract and thus of their spontaneous (and not necessarily conscious or thought-out) “representations” of the division of responsibilities in the contract (e.g., some of the students answer that the captain is 36 years old, others refrain from giving an answer, still others finally say that they do not know, and some of them authorize themselves to declare that this problem is absurd). These results reaffirm the importance of the Theory of Situations and notably the explicative power of the contract, but also underline the interest of considering the pedagogical ideologies of the teachers and the cultures of the students in the interpretation of contractual phenomena. These works together lead into a perspective of study baptized “anthropo-didactique,” situating the phenomena of the didactical contract in the double perspective mentioned above. This theoretical current has made it possible to reinterpret in a fertile way a certain number of phenomena of teaching (*lato sensu*), as much on the micro-didactical level as the macro-didactical, and of their interactions, such as school inequities (Sarrazy 2002), school difficulties (Clanché and Sarrazy 2002; Sarrazy and Novotná 2005) heterogeneities, didactical time and didactical visibility (Chopin 2011), student teacher interactions, and the effects of the *genre*. These themes have traditionally been studied by connected disciplines (psychology, sociology, anthropology, etc.) but independently of the didactical dimensions which in fact are necessarily involved in these phenomena. This approach thus realized the study of what Brousseau designated in 1991 “didactical conversions”: “The causes of phenomena of a non-didactical nature can only influence didactical phenomena by the intermediary of elements having their origin in didactical theory.” This "reinterpretation" of a non-didactical phenomenon in didactical terms is a didactical conversion (Brousseau and Centeno 1991, p. 186).

Cross-References

▶ Didactic Situations in Mathematics Education

References

Balacheff N (1988) Contract and custom: two registers of didactical interactions (trans: Herbst P). Math Educat Univ Georgia Athens 9(2):23–29

Brousseau G (1997) Theory of didactical situations in mathematics (eds and trans: Balacheff N, Cooper M, Sutherland R, Warfield V). Kluwer, Dordrecht/Boston/London

Brousseau G (2005) The study of the didactical conditions of school learning in mathematics. In: Hoffmann JL, Falk S (eds) Activity and sign grounding mathematics education. Springer, New York, pp 159–168

Brousseau G, Centeno J (1991) Rôle de la mémoire didactique de l'enseignant. Rech Didact Math 11(2–3):167–210

Brousseau G, Otte M (1989) The fragility of knowledge. In: Bishop A, Mellin-Olsen S, Van Dormolen J (eds) Mathematical knowledge: its growth through teaching. Kluwer, Dordrecht, pp 13–38

Brousseau G, Warfield V (1998) The case of Gaël. J Math Behav 18(1):1–46

Brousseau G, Davis RB, Werner T (1986) Observing students at work. In: Christiansen B, Howson AG, Otte M (eds) Perspective on mathematics education, vol II. D. Reidel, Dordrecht, pp 205–241

Brousseau G, Brousseau N, Warfield V (2002) An experiment on the teaching of statistics and probability. J Math Behav 20:363–441

Chopin M-P (2011) Le temps de l'enseignement L'avancée du savoir et la gestion des hétérogénéités dans la classe. Presses Universitaires de Rennes, Rennes

Clanché P, Sarrazy B (2002) Contribution à l'étude des rapports entre l'expérience quotidienne 'réelle' et l'expérience quotidienne scolaire dans l'enseignement des mathématiques: Cas de l'enseignement d'une structure additive dans un cours préparatoire kanak. Rech Didact Math 22(1):7–30

Novotná J, Hošpesová A (2007) What is the price of Topaze? In: Woo J-H, Lew H-C, Park K-S, Seo D-Y (eds) Proceedings of 31st conference of the international group for the psychology of mathematics education, vol 4. PME, Seoul, pp 25–32

Novotná J, Sarrazy B (2011) Didactical variability in teacher education. In: Zaslavsky O, Sullivan P (eds) Constructing knowledge for teaching secondary mathematics: tasks to enhance prospective and practicing teacher learning. Springer, New York, pp 103–116, Section 4.1: Teachers' learning to attend to and enhance students' problem solving skills and strategies

Perrin-Glorian MJ, Hersant M (2003) Milieu et contrat didactique, outils pour l'analyse de séquences ordinaires. Rech Didact Math 23(2):217–276

Sarrazy B (1996) La sensibilité au contrat didactique: rôle des arrière-plans dans la résolution de problèmes d'arithmétique au cycle trois. Thèse de doctorat en Sciences de l'éducation, Bordeaux 2

Sarrazy B (1997) Sens et situations: une mise en question de l'enseignement des stratégies métacognitives en mathématiques. Rech Didact Math 17(2):135–166

Sarrazy B (2002) Effects of variability on responsiveness to the didactic contract in problem-solving among pupils of 9–10 years. Eur J Psychol Educ 17(4):321–341

Sarrazy B, Novotná J (2005) Didactical contract: theoretical frame for the analysis of phenomena of teaching mathematics. In: Novotná J (ed) Proceedings SEMT 05. Univerzita Karlova v Praze, Pedagogická fakulta, Prague, pp 33–45

Schubauer-Leoni M-L (1986) Le contrat didactique: un cadre interprétatif pour comprendre les savoirs manifestés par les élèves en Mathématiques. J Eur Psychol Éduc 1(Spécial 2):139–153

Searle J (1979) Expression and meaning: studies in the theory of speech acts. Cambridge University Press, Cambridge, MA

Wittgenstein L (1953) Philosophical investigations. Blackwell, Oxford

Didactic Engineering in Mathematics Education

Michèle Artigue
Laboratoire de Didactique André Revuz (EA4434), Université Paris-Diderot, Paris, France

Keywords

Didactic engineering · Theory of didactic situations · A priori and a posteriori analysis · Research methodology · Classroom design · Development activities

Definition

In mathematics education, there exists a tradition of research giving a central role to the design of teaching sessions and their experimentation in classrooms. Didactical engineering, which emerged in the early 1980s and continuously developed since that time, is an important form taken by this tradition. In the educational community, it mainly denotes today a research methodology based on the controlled design and experimentation of teaching sequences and adopting an internal mode of validation based on the comparison between the a priori and a posteriori analyses of these. However, since its emergence, the expression didactical engineering has also been used for denoting development activities, referring to the design of educational resources based on research results or constructions and to the work of didactical engineers.

History

From its emergence as an academic field of study, mathematics education has been associated with the design and experimentation of innovative teaching practices, in terms of both mathematical content and pedagogy. The importance to be attached to design was early stressed by researchers as Brousseau and Wittman, for instance, who very early considered that mathematics education was a genuine field of research that should develop its own frameworks and practices and not just a field of application for other sciences such as mathematics and psychology.

The idea of didactical engineering (DE), which emerged in French didactics in the early 1980s, contributed to firmly establish the place of design in mathematics education research. Foundational texts regarding DE such as Chevallard (1982) make clear that the ambition of didactic research of understanding and improving the functioning of didactic systems where the teaching and learning of mathematics takes place cannot be achieved without considering these systems in their concrete functioning, paying the necessary attention to the different constraints and forces acting on them. Controlled realizations in classrooms should thus be given a prominent role in research methodologies for identifying, producing, and reproducing didactic phenomena, for testing didactic constructions. As a research methodology, DE emerged with this ambition, relying on the conceptual tools provided by the Theory of Didactic Situations (TDS), and conversely contributing to its consolidation and evolution (Brousseau 1997). It quickly became a well-defined and privileged methodology in the French didactic community, accompanying the development of research from elementary school up to university level as evidenced in the synthesis proposed at the 1989 Summer School of Didactics of Mathematics (Artigue 1990, 1992).

From the 1990s, DE migrated outside its original habitat, being extended to the design of teacher preparation and professional development sessions, used by didacticians from other disciplines, for instance, physical sciences or sports, and also by researchers in mathematics education in different countries. Simultaneously, the progressive shift of research attention towards teachers increased the use of methodologies based on naturalistic observations of classrooms, leading to theoretical developments and results that, in turn, affected DE. Moreover, design-based research perspectives emerged in other contexts, independently of DE (Design-Based Research Collaborative 2003). These evolutions and the resulting challenges are analyzed in Margolinas et al. (2011).

DE as a Research Methodology

As a research methodology, DE is classically structured into four different phases: preliminary analyses; design and a priori analysis; realization, observation, and data collection; a posteriori analysis and validation (Artigue 1990, 2014).

Preliminary analyses usually include three main dimensions: an epistemological analysis of the mathematical content at stake, an analysis of the institutional conditions and constraints that the DE will face, and an analysis of what educational research has to offer for supporting the design.

In the second phase, design and a priori analysis, research hypotheses are engaged in the process. Design requires a number of choices, from global to local. They determine *didactic variables*, which condition the interactions between students and knowledge, between students and between students and teachers, thus the opportunities that students have to learn. In line with TDS, in design, particular importance is attached:

To the search for *fundamental situations*, i.e., mathematical situations encapsulating the epistemological essence of the concepts

To the characteristics of the *milieu* with which the students will interact in order to maximize the potential it offers for autonomous action and productive feedback

To the organization of *devolution* and *institutionalization* processes by which the teacher, on the one hand, makes students accept the mathematical responsibility of solving the tasks and, on the other hand, connects the knowledge they produce to the scholarly knowledge aimed at

The a priori analysis makes clear these choices and their relation to the research hypotheses. Conjectures are made regarding the possible dynamic of the situation, students' interaction with the *milieu*, students' strategies, their evolution and their outcomes, about teacher's necessary input and role. Such conjectures regard not individuals but a *generic and epistemic student* entering the mathematical situation with some supposed knowledge background and accepting to enter the mathematical game proposed to her. The actual realization will involve students with their personal specificities and history, but the goal of the a priori analysis is not to anticipate all these personal behaviors; it is to build a reference with which classroom realizations will be contrasted in the a posteriori analysis.

During the phase of realization, data are collected for a posteriori analysis. The nature of these data depends on the precise goals of the DE, the hypotheses tested, and the conjectures made in the a priori analysis. The realization can lead to some adaptation of the design in *itinere*, especially when the DE is of substantial size. These adaptations are documented and taken into account in the a posteriori analysis.

A posteriori analysis is organized in terms of contrast with the a priori analysis. Up to what point the data collected during the realization support the a priori analysis? What are the significant convergences and divergences and how to interpret them? The hypotheses underlying the design are put to the test in this contrast. There are always differences between the reference provided by the a priori analysis and the contingence analyzed in the a posteriori analysis. The validation of the hypotheses underlying the design does not thus impose perfect match between the two analyses. Moreover, the validation of the research hypotheses may require the collection of complementary data to those collected during the classroom, especially for appreciating the learning outcomes of the process. Statistical tools can be used, but what is essential is that validation is internal, not in terms of external comparison between control and experimental groups.

These are the characteristics of DE as research methodology when associated with the conception of a sequence of classroom sessions having a precise mathematical aim. However, as shown in Margolinas et al. (2011), this methodology has been extended to other contexts such as teacher education, more open activities such as project work or modeling activities, and even mathematical activities carried out in informal settings. In these last cases, the content of preliminary and a priori analyses must be adapted; what the design ambitions to control in terms of learning trajectories and the reference provided by the a priori analysis cannot exactly have the same nature.

Realizations

The first exemplars of DE research regarded primary school. Paradigmatic examples are the long-term designs produced by Brousseau, on the one hand, and by Douady, in the other hand, for extending the field of numbers from whole numbers to rational numbers and decimals (Brousseau et al. 2014; Douady 1986). The two constructions were different, but they proved both to be successful in the experimental settings where they were tested, and they significantly contributed to the state of the art regarding the learning and teaching of numbers. Beyond that, they had theoretical implications. The development of the *tool-object dialectics* and the identification of the learning potential offered by the organization of *games between mathematical settings* by Douady are intrinsically linked to her DE for the extension of the number field; the idea of *obsolescence of didactic situations* emerged from the attempts made at reproducing Brousseau's DE year after year. These are only two examples among the many we could mention. DEs were progressively developed at all levels of schooling, covering a diversity of mathematical domains and addressing a diversity of research issues. At university level, for instance, paradigmatic examples remain the construction developed by Artigue and Rogalski for the study of differential equations, combining qualitative, algebraic, and numerical approaches to this topic (Artigue 1993) and that developed by Legrand for the teaching of Riemann integral within the theoretical framework of the scientific debate (Legrand 2001). Both were experimented with first year students and showed their resistance to students' diversity. Constraints met at more advanced levels of schooling contributed to the deepening of the reflection on an optimized organization of the sharing of mathematical responsibilities between students and teacher in DE and to the softening of the conditions and structures often imposed to design at more elementary levels. DE was also enriched by its use in other domains than mathematics and by researchers trained in other cultural traditions. A good example of it is provided by its use in sports, already mentioned, and by the elaboration of DE combining the theoretical support of TDS and that of semiotic approaches (cf. for instance, (Falcade et al. 2007; Maschietto 2008) using such combination for studying the educational potential of digital technologies). More globally, ICT has always been a privileged domain for DE, for exploring and testing the potential of new technologies, and for supporting technological development as well as theoretical advances in that area. Another interesting example is the use of DE within the socio-epistemological framework in mathematics education (Farfán 1997; Cantoral and Farfán 2003).

Challenges and Perspectives

DE developed as a research methodology, but DE from the beginning had also the ambition of providing a model for productive interaction between fundamental research and action on didactic systems. DEs produced by research were natural candidates for supporting such a productive interaction. Quite soon, researchers however experienced the fact that the DEs they had developed and successfully tested in experimental settings did not resist to the usual dissemination processes. This problem partly motivated the shift of interest towards teachers' representations and practices. Addressing it requires to clearly differentiate research DE (RDE) and development DE (DDE), acknowledging that these cannot obey the same levels of control. In Margolinas et al. (2011), this issue is especially addressed by Perrin-Glorian through the idea of DE of second generation, in which the progressive loss of control that the elaboration of a DDE requires is co-organized in collaboration with teachers and illustrated by an example. Such a strategy implies a renewed conception of dissemination of research results, in line with the current evolution of vision of relationships between researchers and teachers that also impacts DE (see for instance the concept of Cooperative DE associated with the Theory of Joint Action in Didactic (Joffredo-Lebrun et al. 2018).

Another challenge is the issue of relationships between the tradition of DE described above and the different forms of design which are developing in mathematics education under the umbrella of design-based research, reflecting the increased interest for design in the field, or the vision of design introduced in the Anthropological Theory of Didactics (ATD) in the last decade in terms of Activities of Study and Research (ASR) and Courses of Study and Research (CSR) (Chevallard 2006), and then Study and Research Paths (Barquero and Bosch 2015). Despite de fact that ATD and TSD emerged in the same culture, the visions of design they propose today present substantial differences. Establishing productive connections between the two approaches without losing the coherence proper to each of them is a problem also addressed in

D

Margolinas et al. (2011) and further discussed in (Artigue 2014; Barquero and Bosch 2015).

Cross-References

- ▶ Anthropological Theory of the Didactic (ATD)
- ▶ Cooperative Didactic Engineering
- ▶ Design Research in Mathematics Education
- ▶ Didactic Situations in Mathematics Education

References

Artigue M (1990) Ingénierie didactique. Recherches en Didactique des Mathématiques 9/3:281–308. (English translation: Artigue M (1992) Didactical engineering. In: Douady R, Mercier A (eds) Recherches en Didactique des Mathématiques, Selected papers, pp 41–70). La Pensée Sauvage Editions, Grenoble

Artigue M (1993) Didactic engineering as a framework for the conception of teaching products. In: Biehler R et al (eds) Mathematics didactics as a scientific discipline. Kluwer, Dordrecht

Artigue M (2014) Perspectives on design research: the case of didactical engineering. In: Bikner-Ahsbahs A, Knipping C, Presmeg N (eds), Approaches to qualitative research in mathematics education. Springer, New York, pp. 467–496

Barquero B, Bosch M (2015) Didactic engineering as a research methodology: from fundamental situations to study band research paths. In: Watson A, Ohtani M (eds) Task design in mathematics education. Springer, New York, pp 249–272

Brousseau G (1997) Theory of didactical situations. Kluwer, Dordrecht

Brousseau G, Brousseau N, Warfield V (2014) Teaching fractions through situations: a fundamental experiment. Springer. https://doi.org/10.1007/978-94-007-2715-1

Cantoral R, Farfán R (2003) Mathematics education: a vision of its evolution. Educ Stud Math 53(3):255–270

Chevallard Y (1982) Sur l'ingénierie didactique. Preprint. IREM d'Aix Marseille. Accessible at http://yves.chevallard.free.fr

Chevallard Y (2006) Steps towards a new epistemology in mathematics education. In: Bosch M (ed) Proceedings of the IVth congress of the European society for research in mathematics education (CERME 4). Universitat Ramon Llull Editions, Barcelona, pp 22–30

Design-Based Research Collaborative (2003) Design-based research: an emerging paradigm for educational enquiry. Educ Res 32(1):5–8

Douady R (1986) Jeux de cadres et dialectique outil-objet. Recherches en Didactique des Mathématiques 7(2):5–32

Falcade R, Laborde C, Mariotti MA (2007) Approaching functions: Cabri tools as instruments of semiotic mediation. Educ Stud Math 66(3):317–334

Farfán R (1997) Ingeniería Didactica y Matemática Educativa. Un estudio de la variación y el cambio. Grupo Editorial Iberoamérica, México

Joffredo-Lebrun S, Morellato M, Sensevy G, Quilio S (2018) Cooperative engineering as a joint action. Eur Educ Res J. https://doi.org/10.1177/1474904117690006

Legrand M (2001) Scientific debate in mathematics courses. In: Holton D (ed) The teaching and learning of mathematics at University level: an ICMI study. Kluwer, Dordrecht, pp 127–136

Margolinas C, Abboud-Blanchard M, Bueno-Ravel L, Douek M, Fluckiger A, Gilel P, Vandebrouck F, Wozniak F (eds) (2011) En amont et en aval des ingénieries didactiques. XVe école d'été de didactique des mathématiques. La Pensée Sauvage Editions, Grenoble

Maschietto M (2008) Graphic calculators and micro-straightness: analysis of a didactical engineering. Int J Comput Math Learn 13(3):207–230

Didactic Situations in Mathematics Education

Guy Brousseau[1] and Virginia Warfield[2]

[1]Institut Universitaire de Formation des maîtres d'Aquitaine, Mathématiques, Laboratoire Cultures Education Societes (LACES), Anthropologie et diffusion des savoir, University of Bordeaux – France, Bordeaux Cedex, France

[2]Department of Mathematics, University of Washington, Seattle, WA, USA

Keywords

Didactical situation · A-didactical situation · Mathematical situation · Acculturation · Didactique

Didactical Situation

A *didactical situation* in mathematics is a project organized so as to cause one or some students to *appropriate* some piece of mathematical reference knowledge. (The organizer and the student may be individuals, a population, institutions, and so on.)

Components

Every didactical process is a sequence of situations, each pertaining to one of the following three types:

A "**situation of devolution**" in which the teacher sets the students up:
- to accept boldly and confidently the challenge of an engaging and instructive mathematical situation whose instructions he gives in advance: conditions, rules, goal, and above all the criterion for success
- and to do it without his help, on their own responsibility (Brousseau 1997, pp. 230–235)

A "**mathematical situation**" that supports the students in autonomous mathematical activities, both individual and collective, that represent those in use by mathematicians. Rather than looking to gain credit for themselves, the students are engaged in:
- Producing "new" statements and discussing their validity
- Making decisions, formulating hypotheses, predicting and judging their consequences, attempting to communicate information, producing and organizing models, arguments and proofs, etc., adequate for certain precise projects
- and evaluating and correcting by themselves the consequences of their choices

It is thus not the students who are in question, but some conjectures and some knowledge (Brousseau 1997, pp. 230–235).

A "**situation of institutionalization**" in which the teacher:
- Takes note of the progress of the mathematical situation, of the questions and answers that have been obtained or studied from it, and of those that have emerged, and places them within the perspective of the curriculum
- Distinguishes among the pieces of knowledge (*connaissances*) that have appeared those that have revealed themselves to be false and those that are correct, and among the latter those that will serve as references, presenting in that case the canonical way of formulating them
- And draws conclusions for the organization of further sequences (exercises, problems, etc.) (Brousseau 1997, pp. 235–243).

Teaching Methods

Teaching methods can be distinguished first by the interpretation, the role, and the importance assigned to each of the components. Here are two very different examples of this:

Example 1: In certain methods, devolution consists of a prerequisite teaching of new knowledge (a lecture), followed by examples and exercises, and followed by the presentation of problems whose autonomous solution by the students constitutes the mathematical situation. Institutionalization consists of correction, evaluation, and the conclusions that the teacher draws from them. Sometimes the mathematical situation is considered only as a means of verifying the individual learning produced by the lecture.

Example 2: In other methods, devolution is reduced to the organization, presentation, and staging of an individual or collective *mathematical situation* aimed at provoking activities and processes like those of mathematicians: a search for solutions or proofs but also production of questions, hypotheses or conjectures, reformulations, definitions and study of objects, sorting, debates, challenges, etc. Learning is the means and the product of this activity. Institutionalization then consists of identifying and organizing, among the correct pieces of knowledge produced by the students, those consistent with common usage and with *accepted mathematical knowledge*, and among those the ones that are sufficiently "acquired" by all of the students so that the teacher and students can refer to them with each other in future mathematical situations. The "lecture" consists of a conclusion and of putting things in order. Exercises are a means of training available to the students (Margolinas et al. 2005; Illustrative examples in Warfield 2007).

Origin and Necessity of the Concept of "Didactical Mathematical Situation"

The Reform of the Foundations (1907–1980)

The term "didactical situation" appeared in the 1960s with the meaning "mathematical situation for teaching."

The new mathematical concepts on which teaching was to be rebuilt were communicated by formalized texts in a symbolic language unintelligible to students and/or by reformulations, metaphorical representations, and ambiguous commentaries. On the other hand, they referred necessarily to examples taken from the classical mathematics that they were reorganizing. The "fundamental" concepts were thereby postponed to the end of the studies.

The challenge was thus to imagine conditions, situations, that could induce in the students the geneses of fundamental mathematical concepts, in a form and by processes comparable to those put into operation by mathematicians *before* the final presentation of their results, in the process mathematical development. This idea found justification in the work of the period: the acquisition of language does not follow the classic formulation of its grammar, and Piaget identified certain mathematical structures in the genesis of logical thought in children.

Conceiving of similar geneses, and especially imagining conditions capable of inducing them, could only arise from the competence of the mathematical community. It did so through a gigantic effort of its researchers and of its teachers, realizing as it did so the aspirations of pedagogues like Dewey, Montessori, or Freinet. But diffusing these conceptions more widely, *against* the traditional culture of teaching, posed yet more redoubtable problems, which have not at this point been surmounted.

Learning Mathematics by Doing It Reverses the Classic Pedagogical Order

The teaching of mathematics is based on a text or some texts that express it in a canonical way (i.e., in the order: definitions, properties and theorems, and finally proofs). The classical conception consists of teaching using the texts first, so that a student could never argue that he or she is being required to use a piece of knowledge that was not first revealed and taught. Teaching pieces of knowledge before needing to use them gives the appearance of being a "rational" method, but it introduces a disassociation (learn with metamethods that have no relationship with the object and its use), an inversion (learn terms before understanding them and doing anything with them), and finally teleological requirements: the student is blamed in the course of learning for not having first learned what is in fact the goal of the teaching that is going on. This epistemological error greatly limits the field of application, the age of learning, and the degree of success of the classical method.

Conversely, direct *acculturation* to specific mathematical practices that can produce these texts brings their learning closer to that of vernacular language or natural thought. Everything then rests on the power of the situations to induce in the children the "process of mathematization."

It would be absurd and detrimental to want to exclude some method or to uniformly recommend it over some other. The conditions to which each is best adapted must be scientifically studied and their advantages combined. For example, situations of cooperative discovery and collective adventures create homogeneity and motivation and make it possible to acquire the classical practices by use. Exercises can help in doing well and rapidly what is worthwhile and has been understood (Brousseau 1992).

The Project of a Mathematical Science: *Didactique*

The organization of these mathematical situations and their succession obey various reasons: mathematical, epistemological, rational, empirical, ideological, etc. Their scientific study combines:

1. The (anthropological) observation and the analysis (semiological) of the practices and conceptions of the teachers and of the students
2. The conception, realization, and experimental study of original mathematical situations appropriate to each of the pieces of mathematical knowledge aimed for (Didactic engineering in mathematics education)
3. The inventory of possible choices, their modeling in the form of situations, the experimental and theoretical study of their conditions and of

their properties, and the creation of appropriate instruments of analysis (theory of didactical situations)

The conception of these situations requires prior and specific mathematical study of the *knowledge to be taught*, along with that of its historical genesis, of its epistemological properties, and of its possible didactical geneses and their properties. But the scientific confrontation of these speculations with actual teaching is fundamental.

The theory of situations, its concepts, and its research methods is one of the most ambitious among the numerous scientific approaches to the phenomenon of *didactique.*

But well before being able to offer teachers, in the name of mathematicians, an aid, or some ready-to-implement solutions for teaching mathematics, *didactique* must describe, understand, and explain in a scientific manner mathematical activity and its possible *didactical transpositions*.

Didactique plays a role in the reorganization and transformation of mathematical knowledge. Its results are thus first addressed to the community of mathematicians, to whom falls – for good reasons – the responsibility towards society of the reference in teaching materials to the established knowledge of its specialty. *Didactique* of mathematics requires specific concepts and methods of study. It thus joins logic, computer science, epistemology, history of mathematics, and so on as one of the mathematical sciences. It takes charge of the knowledge of everything that is specific to the discovery, the diffusion, or the appropriation of each piece of mathematical knowledge, new or not, that results from the adventures specific to it. It extends, enriches, and puts to the test the general contributions of classical social sciences, which are indispensible but insufficient for clarifying all the facets of this teaching.

Mathematical Situations

Definition

Every mathematical concept is the solution of at least one specific system of mathematical conditions, which itself can be interpreted by at least one situation, for example, a game, whose solution (decision, message, argument) is one of the typical manifestations of the concept. A situation is composed of a milieu and a project. The duration of the life of a mathematical situation (the time of studying it) can vary from a few seconds to several centuries for humanity or several months for teaching.

D

Examples

Example 1: Children 4–5 years old. From a collection of thirty or so familiar objects, 5 or 6 are hidden in a box by a child in the morning. In the afternoon, she is supposed to enumerate them to another child, who confirms the presence or absence of the objects she names. The solution of this game is the creation, enumeration, and use of lists. Knowing neither how to read nor how to write, the children represent the objects in their own way (pictograms) to distinguish them, first individually and then collectively. The lists of symbols represent sets; belonging or not, conjunctions and disjunctions of characters are used, corrected, understood, and formulated in vernacular language (Pérès 1984; Digneau 1980).

Example 2: Children 10–11 years old. To be certain of the number of white marbles contained in a firmly closed opaque bottle with a known number of marbles, some white and some black, students invent hypothesis testing and the measure of events (33 short sessions) (Brousseau et al. 2002).

A great many researchers have imagined and studied various types of situations destined for all sorts of notions, for all levels of school and even university. See, for example, Bessot (2000), Laborde and Perrin-Glorian (2005), Bloch (2003).

Types of Mathematical Knowledge, Reference Knowledge (*Savoirs*)

Classical methods forbid the teacher from tolerating without immediate correction, the manifestation of anything contrary to written established mathematics. A genuine mathematical activity necessarily gives rise to all sorts of knowledge. Some is knowledge sought for – these are the references, recognized as correct, true and known: they are professed and expected. But there also necessarily appear pieces of knowledge that are ill made, ill formed, incomplete, doubtful, false, or even inexpressible. They are "knowledge" in the sense of "the trace of an encounter." Their presence, whether or not firmly nailed down, is indispensible to thought. For example, a theorem that the student knows very well (savoir), but about whose usefulness in a situation is unsure, functions provisionally as a simple piece of nonestablished knowledge (connaissance).

The teacher cannot intervene in this flow of activities without blocking its functioning and must therefore delegate the responsibility for exercising a pragmatic penalty to the initiatives of the students that result from their knowledge. He entrusts it to a *milieu* that is clearly stripped of teleological or pedagogical intentions [its reactions depend neither on the intended goal nor on the individuals].

The *milieu* of a situation is what the students exercise their actions on and what gives them objective responses. The teacher thus entrusts to the *milieu* the job of showing the students' errors by their effects, without using an argument of authority or revealing any intentions. The milieu may comprise informative texts; material objects; other students, cooperating or concurrent; and so on. To this must be added the established knowledge of the student as well as her memories of relevant previous events, and objective conditions, that may not be known to the student but that intervene in her choices and in the effects of her decisions. The *cognitive variables* of the situation are those whose value has an influence on the issue of the situation or on the knowledge developed. These variables are didactical if their value can be chosen by the teacher (the sex of the students may influence the progress of a situation, but it is not a didactical variable). The *milieu* can be interpreted metaphorically by games that present some states that are permissible and some that are excluded, rules of action, and issues of which one would be the goal sought (Warfield 2007).

Examples of *Milieux*

1. *Cabri geometry* permits the student to realize, in the context of geometrical objects and transformations, which of her projects are constructible, that is, compatible with the axioms (Laborde and Laborde 1995). The projects lead the students to gain knowledge of, formulate, and test what the *milieu* permits them to glimpse.
2. Analysis of a situation. The reader will find an example of the analysis of a didactical situation (the Race to 20), of its *milieu*, of the strategies used by students, of the theorems in action that support them, and of the didactical methods that make it possible to lead them to a complete proof and then to extend it so as to have them reinvent an algorithm: the search for the remainder of a Euclidean division, in Brousseau (1997, pp. 3–18). This work also includes numerous other examples.

The project is an objective, a final state of the *milieu*, the response to a question, or even a pretext for exploration. It is what explains, justifies, or condemns after the fact the choices that have been chosen or ventured by the subject.

The resolution is the occasion to put to trial not the student, but a way of knowing.

Remarks: The milieu of a situation is not a natural milieu and does not turn mathematics into a sort of experimental science. The project is essential, and its goal is to establish the consistency of certain statements.

Different branches of mathematics developed in different *milieux*: geometry in the knowledge of space, probability in the statistics of games, algebra in arithmetic, arithmetic in the measurement of amounts, etc.

In elementary teaching, knowledge of these *milieux* is neither spontaneous nor contained in their mathematical interpretation. For example, the knowledge that is useful for finding one's way around a big city merits specific work that cannot be reduced to some geometry.

Types of Mathematical Situations Characteristic of Activities, of Pieces of Knowledge, and of Pieces of Mathematical Learning

The mathematical knowledge of a student manifests itself in her interactions with a milieu, as a means of attaining or maintaining a desired state. These interactions are grouped in four types of situations which are, in the order of didactical necessity, inverse to the ordinary chronological order:

1. Situation of *reference*: A person (student or teacher) refers the person asking to a piece of mathematical knowledge (a proof, a theorem, a definition, etc.) that belongs to their common repertoire (Perrin-Glorian 1993).
2. Situation of *argumentation* (of *proof*): A proposer communicates to an opponent an argument, an element of proof. He makes use for that of their common repertoire which his message tends to augment. The argument makes reference to a *milieu* and a (mathematical) project in common that gives it its meaning and its value. The two speakers have the same information, in particular, on the *milieu*, the same rights of refutation, and the same interest in arriving at a consistent agreement (for an action on the *milieu*).
3. Situation of *information (communication)*: The transmitter and receiver cooperate on an action on the *milieu*, in whose success they are interested and which depends on their joint action. Neither of the two has at the same time all of the information and all of the necessary means of action. They exchange messages in order to realize a common mathematical project.
4. Situation of *action*: A subject intervenes on the *milieu* to modify it with a determined aim. She observes the effect of her actions and attempts to anticipate them by constructing pieces of knowledge, conscious and explainable or not. This situation encompasses all of the others, but it extends beyond them by stimulating the existence of inexpressible and possibly even unconscious models of action.

Each of these types of situation creates *distinct typical motivations* (modify a *milieu*, communicate some information, debate the validity of a declaration, establish a reference) that mobilize and expand the *corresponding repertoires* (implicit models of action, semiological or linguistic repertoires, logical repertoires, mathematics or metamathematics, established knowledge and theory) which are themselves acquired according to specific different *modes of learning or acculturation.*

The *actual situations* are, every one of them, specific to a precise piece of knowledge.

This is the level which must be appealed to in order to judge the relevance of the contributions of other scientific domains (pedagogy, psychology, sociology, etc.).

D

The Processes

Different modes of composition and articulation of these elementary situations make it possible to create composite situations and sequences of situations that form processes:

1. *Process of mathematization*: A sequence of autonomous mathematical situations that are introduced by didactical interventions of the teacher and that work together towards the construction of the same complex knowledge (e.g., rational and decimal numbers (Brousseau et al. 2004, 2007, 2008, 2009)).
2. *Genetic situation*: It introduces and without other external intervention generates the sequence of situations that lead to the acquisition of a concept (e.g., how many white marbles [article cited]).

The didactical work of the teacher then consists of maintaining the intensity and the relevance of the exchanges and implementing their progress and their conclusion. Examples of process: on areas, Perrin-Glorian (1992), and on geometry, Salin and Berthelot (1998).

Some of the Results of Research on Didactical Situation

The notion of didactical situation was used in many research projects. It gave rise to numerous reflections and, with modifications, was expanded in more work than it is possible to summarize or cite here:

1. One of its first results was to establish that adaptation to certain conditions tends to render it more difficult to adapt to others and thus creates the phenomenon of didactical obstacles, then to show that the history of mathematics presents phenomena similar to the epistemological obstacles detected by G. Bachelard, and finally, to take advantage of this phenomenon in teaching by use of situations presenting "jumps in informational complexity" (▶ "Epistemological Obstacles in Mathematics Education")
2. Research on situations had the goal of furnishing alternatives to the classical conceptions that showed their limitations in the face of the influx of knowledge to be taught and of the fundamental reorganizations necessitated by that influx. This research showed the importance of the role of the *unsayable* in mathematical situations and of the *unsaid* in the didactical relationship.

Rather than imagining teaching and producing learning of the *texts* that resulted from real mathematical activity by universal, that is nonspecific, nonmathematical teaching methods, it appeared that it would be preferable to have the students themselves produce this knowledge and these texts, thanks to specific mathematical activities that best stimulated the real activity of mathematicians.

The many didactical situations realized showed that this project was realizable. Experiments proved it. Curricula were conceived, experimented with, and reproduced for all the branches of mathematics and for all the basic levels of teaching (3–12 years old) in an establishment conceived for the purpose (the COREM).

Currently they cannot be developed because of the complexity of knowledge necessary for the teachers to conduct them and for the public to accept them.

This research produced counterexamples to most of the "universal principles," explicit or implicit, of classical didactics, for behaviorist methods as well as radical constructivism. It showed, for example, that in the classical conception, errors can have no status other than that of being far from some norm. They are interpreted as a failure of the student and/or the teacher that involves their responsibility and ultimately their guilt for a failure of their will. This absurd process generates very bad working conditions for the students and for the teachers.

Among many other results, The classical conception led to seeking out individualization of teaching, but this individualization did not improve the results, because mathematical knowledge is produced by the cooperation of numerous individuals operating in the same community, and no isolated brain can produce the exact form that history has given it. For a large portion of the students, the real use of communication and mathematical debates is indispensible.

The concept of situation has been the object and has been illustrated in a great deal of research of different types:

1. *Empirical*, so as to identify the observables of a given teaching episode and analyze them a priori and a posteriori
2. *Experimental*, to conceive of either a precise teaching project (engineering) or a teaching design (of cognitive psychology, of sociology, of *didactique*, etc.)
3. *Theoretical*, to study their properties (economic, ergonomic, etc.) on appropriate models, possibly mathematics (automata, games, various systems), or conceive of modes or specific indices for these studies (implicative statistical analysis for the study of dependencies) (Artigue and Perrin-Glorian 1991)

The results of these studies were used in many research projects more particularly centered on students, teachers, or school knowledge and the didactical transposition (Mercier et al. 2000).

Research Perspectives

1. The study of the optimal conditions for articulation of mathematical situations and of institutionalization is a necessity. Pieces of "knowledge" proposed in mathematical

situations, whether erroneous or valid, must evolve sufficiently rapidly to arrive at established knowledge. Making these pieces of provisional knowledge the object of classical teaching, on the pretext that they were produced by the students themselves, is a major error. On the contrary, the reorganization of spontaneous knowledge around established knowledge with a complement of information (a lecture) is a mathematical necessity that offers an indispensible time gain. *Didactique* is a science of dynamic equilibrium of situations.

2. What are the relationships between the teaching of mathematics (*microdidactique)* and the explicit or latent mathematical or didactical conceptions held by the various social, economic, cultural, and scientific components of a society (*macrodidactique*)?
3. What are the factors in the failure of the reform of modern mathematics?

Cross-References

► Didactic Contract in Mathematics Education
► Didactic Engineering in Mathematics Education
► Epistemological Obstacles in Mathematics Education

References

Artigue M, Perrin-Glorian MJ (1991) Didactic engineering, research and development tool: some theoretical problems linked to this duality. Learn Math 11(1):13–17

Balacheff N (1999) Contract and custom: two registers of didactical interactions. Math Educ 9(2):23–29

Bloch I (2003) Teaching functions in a graphic milieu: what forms of knowledge enable students to conjecture and prove? Educ Stud Math 52(1):3–28

Brousseau G (1992) Didactique: what it can do for the teacher, Research in Didactique of Mathematics (selected papers). In: Douady R, Alain M (eds) La Pensée Sauvage Grenoble, pp 7–40

Brousseau G (1997) Theory of didactical situations in mathematics, Didactique des mathématiques 1970–1990 (eds and trans: Cooper M, Balacheff N, Sutherland R, Warfield V). Kluwer, Dordrecht

Brousseau G, Brousseau N, Warfield V (2002) An experiment on the teaching of statistics and probability. J Math Behav 20: 363–441

Brousseau G, Brousseau N, Warfield V (2004) Rationals and decimals as required in the school curriculum. Part 1: rationals as measurement. J Math Behav 23(1):1–20

Brousseau G, Brousseau N, Warfield V (2007) Rationals and decimals as required in the school curriculum. Part 2: rationals as measurement. J Math Behav 26(4):281–300

Brousseau G, Brousseau N, Warfield V (2008) Rationalsand decimals as required in the school curriculum. Part 3: rationals and decimals as linear functions. J Math Behav 27, 153–176

Brousseau G, Brousseau N, Warfield V (2009) Rationals and decimals as required in the school curriculum. Part 4: problem solving composed mapping and division. J Math Behav 28:79–118

Digneau JM (1980) Création d'un code à l'école maternelle, IREM de Bordeaux,173pp. http://guy-brousseau.com/wp-content/uploads/2010/08/THESE-JM-DIGNEAU-1980.pdf

Laborde C, Laborde JM (1995) The case of Cabri-geometre: learning geometry in a computer based environment. In: Watson D, Tinsley D (eds) Integrating information technology into education. International Federation for Information Processing. Chapman and Hall, London, pp 95–106. ISBN0–412–62250-5

Laborde C, Perrin-Glorian MJ (2005) Introduction to the special issue teaching situations as object of research: empirical studies within theoretical perspectives. Educ Stud Math Educ 59:1–12

Margolinas C, Coulange L, Bessot A (2005) What can the teacher learn in the classroom? Educ Stud Math Educ 59(1):205–234

Mercier A, Sensevy G, Schubauer-Leoni M-L (2000) How social interactions within a class depend on the teacher’s assessment of the various pupil’s mathematical capabilities, a case study. Z Didakt Math Int Rev Math Educ 32(5):126–130

Pérès J (1984) Utilisation d’une théorie des situations en vue de l’identification des phénomènes didactiques au cours d’une activité d’apprentissage scolaire. Construction d’un code à l’école maternelle IREM de Bordeaux, 279p. http://guy-brousseau.com/wp-content/uploads/2010/08/THESE-J-PERES.pdf

Perrin-Glorian MJ (1992) Aires de surfaces planes et nombres décimaux. Questions didactiques liées aux élèves en difficulté aux niveaux CM-6ème. Thèse d'état Université Paris 7

Perrin-Glorian MJ (1993) Questions didactiques soulevées à partir de l'enseignement des mathématiques dans des “classes faibles”. Rech Didact Math 13(1.2):5–118

Salin MH, Berthelot R (1998) The role of pupils’ spatial knowledge in the elementary teaching of geometry. In: Mammana C, Villani V (eds) Perspectives on the teaching of geometry for the 21st century, ouvrage collectif édité par. Kluwer, pp 71–78

Warfield VM (2007) Invitation to didactique. Xlibris corporation 1-888-795-4274Xlibris.com

D

Didactic Transposition in Mathematics Education

Yves Chevallard[1,2] and Marianna Bosch[3]
[1]Aix-Marseille Université, Marseille, France
[2]Apprentissage Didactique, Evaluation, Formation, UMR ADEF – Unité Mixte de Recherche, Marseille, France
[3]IQS School of Management, Universitat Ramon Llull, Barcelona, Spain

Keywords

Anthropological theory of the didactic · Scholarly knowledge · Knowledge to be taught · Institutional transposition · Noosphere · Ecology of knowledge · Reference epistemological models

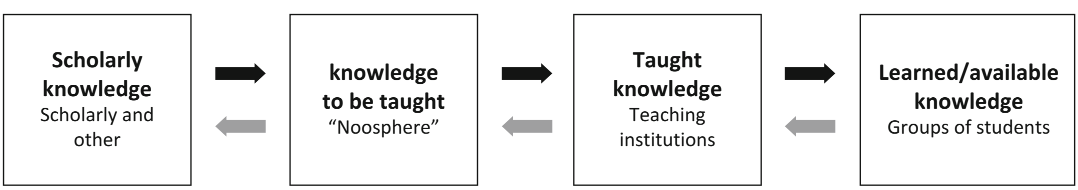
Definition

The process of didactic transposition refers to the transformations an object or a body of knowledge undergoes from the moment it is produced, put into use, selected, and designed to be taught until it is actually taught in a given educational institution. The notion was introduced in the field of didactics of mathematics by Yves Chevallard (1985, 1992b). It highlights the fact that what is taught at school is originated in other institutions, constructed in concrete practices, and organized in particular sets of objects. In the case of mathematics or any other subject, the *taught knowledge*, the concrete practices and bodies of knowledge proposed to be learned at school, originates from what is called the *scholarly knowledge*, generally produced at universities and other scholarly institutions, also integrating elements taken from a variety of related social practices. When one wishes to "transpose" a body of knowledge from its original habitat to school, specific work should be carried out to rebuild an appropriate environment with activities aimed at making this knowledge "teachable," meaningful, and useful.

Different actors participate in this *transpositive work* (see Fig. 1): producers of knowledge, teachers, curriculum designers, etc. They belong to what is called the *noosphere*, the sphere of those who "think" about teaching, an intermediary between the teaching system and society. Its main role is to negotiate and cope with the demands made by society on the teaching system while preserving the illusion of "authenticity" of the knowledge taught at school, thus possibly denying the existence of the process of didactic transposition itself. It must appear that *taught knowledge* is not an invention of school. Although it cannot be a reproduction of *scholarly knowledge*, it should look like preserving its main elements. For instance, the body of knowledge taught at school under the label of "geometry" (or "mechanics," "music," etc.) has to appear as genuine. It is thus important to understand the choices made in the designation of the *knowledge to be taught* and the construction of the *taught knowledge* to analyze what is transposed and why and what mechanisms explain its final organization and to understand what aspects are omitted and will therefore not be diffused.

Scope

Besides mathematics, research on didactic transposition processes has been carried out in many other educational fields, such as the natural sciences, philosophy, music, language, technology, and physical education. These investigations have spread faster in the French- and Spanish-speaking communities



Didactic Transposition in Mathematics Education, Fig. 1 Diagram of the process of didactic transposition

(Arsac 1992; Arsac et al. 1994; Bosch and Gascón 2006) than in the English-speaking ones, although some prominent figures soon contributed to develop the first transpositive analyses (Kang and Kilpatrick 1992). The notion of didactic transposition has been generalized to *institutional transposition* (Chevallard 1989, 1992a; Artaud 1995) when knowledge is transposed from one social institution to another. Because of social needs, bodies of knowledge originated and developed in different "places" or institutions of society need to "live" in other institutions where they should be transposed. They have to be transformed, deconstructed, and reconstructed in order to adapt to their new institutional setting. For instance, the mathematical objects used by economists, geographers, or musicians need to be integrated in other practices commonly ignored by the mathematicians who produced them. It is clear from the history of science that institutional transpositions – including didactic ones – do not necessarily produce degraded versions of the initial bodies of knowledge. Sometimes the transpositive work *improves* the organization of knowledge and makes it more understandable, structured, and accurate to the point that the knowledge originally transposed is itself bettered. The organization of knowledge in fields and disciplines as it exists today is the fruit of complex and changing historical interactional processes of institutional and didactic transpositions that are not well known yet.

An Emancipatory Tool

In a field of research, new notions are not only introduced to describe reality but to provide new ways of questioning and new possibilities to modify it. The notion of didactic transposition is conceived, first of all, as an analytical instrument to avoid the "illusion of transparency" concerning educational phenomena and, more particularly, the nature of the knowledge involved, that is, to emancipate research from the viewpoint of the scholarly and the teaching institutions about the knowledge involved in educational processes.

Any *taught* field or discipline is the product of an intricate process the singularity of which should never be underrated. As a consequence, one should not take for granted the current, observable organization of a field or discipline taught at school, as if it were the only possible one. Instead one should see it against the (fuzzy) set of organizations that *could* have existed, some of which may someday turn into reality. Considering the "scholarly knowledge" as part of the object of study of research in didactics is part of this emancipatory movement of detachment. Although school teaching has to be legitimized by external entities that guarantee the pertinence and epistemological relevance of the knowledge taught (in a complex process of negotiations which includes crises and disagreements), researchers do not have to consider these institutional perspectives as the true or correct viewpoints or as the wrong ones; they just need to know them and integrate them in the analysis of educational phenomena.

In some cases, the "scholarly legitimation" of school knowledge can be questioned by the noosphere, on behalf of its cultural relevance: "Is this the geometry citizens need?" Such a conflict situation can change significantly the conditions of teaching and learning, by allowing a self-referential, epistemologically confined teaching. Moreover, there are certain teaching processes in which the scholarly body of knowledge is created afterwards because of the need to teach a given content that has to be organized, labeled, and recognized as something relevant (an illustrative example is the case of accounting and its corresponding body of knowledge, accountancy). It is also possible that something that is not even commonly recognized as a proper body of knowledge may appear as "scholarly knowledge" for the role it assumes in a given educational process. For instance, in the teaching of sports, the scholarly knowledge, albeit not academically tailored, includes that of high-level sport players, even if they are a far cry from what we normally consider "scholars" to be!

Enlargement of the Object of Study

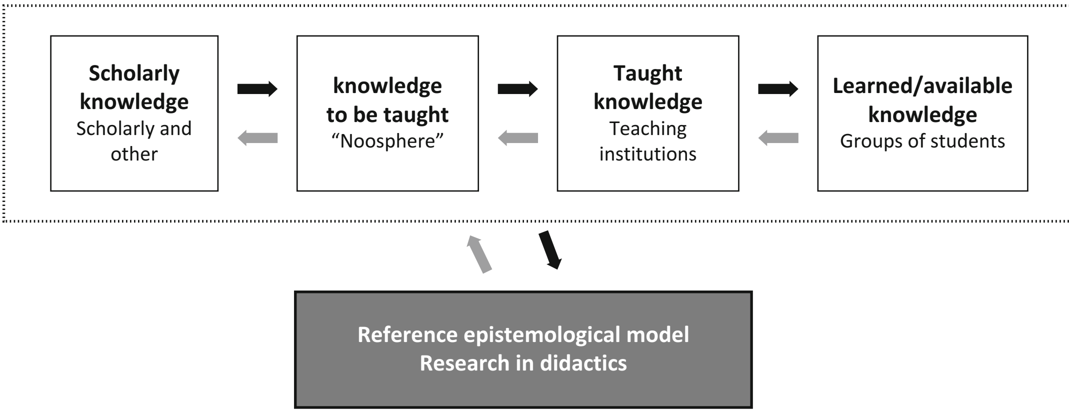
The second consequence of the detachment process introduced by the notion of didactic transposition is the evolution of the object of study of didactics as a research discipline. Besides studying students' learning processes and how to improve them through new teaching strategies, the notion of didactic transposition points at the object of the learning and teaching itself, the "subject matter," as well as its possible different ways of living – its diverse *ecologies* – in the institutions involved in the transposition process.

Let us take an example on negative numbers. Regarding the transpositive process, the first issue is to consider what the *taught knowledge* is made of (what concrete activities that are proposed to the students, their organization, the domain or block of contents they belong to, etc.) and how official guidelines and *noospherian* discourses present and justify these choices (the *knowledge to be taught*). Today, at most schools, negative numbers are officially related to the measure of quantities with opposite directions and introduced in the context of real-life situations. Where does this school organization come from? It results from different scholar ("new mathematics") or social ("back-to-basics") pressures, canalized by the noosphere, that cannot be presented here but that delimit the kind of mathematical practices our students learn (or fail to learn) about this body of knowledge. If we look at scholarly knowledge, the environment is different: negative numbers are defined as an extension of the set of natural numbers **N** and form the ring of integers **Z**, without any specific discussion (http://www.encyclopediaofmath.org/index.php/Integer). This has not always been the case: it is very well known that until the mid-nineteenth century, the possibility of "quantities less than zero" was still denied by many scholars. Their final acceptation was strongly related to the needs of algebraic work, which explains why, for a long time, integers were called "algebraic numbers." It also explains why the introduction of negative numbers was considered one of the main differences between arithmetic and algebra. This relationship to elementary algebraic work has now completely disappeared from the scholar's and school's conception of negative numbers, despite the fact that some practices of calculation – for instance, those involving the product of integers – acquire their full sense when interpreted in this context.

Various other analyses have brought similar results regarding how the transposition process has affect other different mathematical contents (school algebra, linear algebra, limits of functions, proportionality, geometry, irrational numbers, functions, arithmetic, statistics, proof, modeling, etc.): more generally speaking, there is no such thing as an eternal, context-free notion or technique, the matter taught being always shaped by institutional forces that may vary from place to place and time to time. These investigations underline the institutional relativity of knowledge and show to what extend most of the phenomena related to the teaching and learning of mathematics are strongly affected by constraints coming from the different steps of the didactic transposition process. Consequently, the empirical unit of analysis of research in didactics becomes clearly enlarged, far beyond the relationships between teachers and students and their individual characteristics.

The Need for Researchers' Own Epistemological Models

Taking into consideration *transpositive phenomena* means moving away from the classroom and being provided with notions and elements to describe the bodies of knowledge and practices involved in the different institutions at different moments of time. To do so, the epistemological emancipation from scholarly and school institutions requires researchers to create their own perspective on the different kinds of knowledge intervening in the didactic transposition process, including their own way of describing knowledge and cognitive practices, their own epistemology. In a sense, there is no privileged reference system from which to observe the phenomena occurring in the different institutions involved in the teaching process: the scholarly one, the noosphere, the school, and the classroom. Researchers should build their own *reference epistemological models* (Barbé et al. 2005) concerning the bodies of knowledge involved in the reality they wish to approach (see Fig. 2). The term "model" is used to emphasize the fact that any perspective provided by researchers (what mathematics is, what algebra is, what measuring is, what negative numbers are, etc.) always constitutes a methodological proposal for the analysis; as such, it should constantly be questioned and submitted to empirical confrontation.



Didactic Transposition in Mathematics Education, Fig. 2 The external position of researchers

From Didactic Transposition to the Anthropological Approach

When knowledge is considered a changing reality embodied in human practices taking place in social institutions, one cannot think about teaching and learning in individualistic terms. The evolution of the research perspective towards a systematic epistemological analysis of knowledge activities explicitly appears at the foundation of the anthropological theory of the didactic (Chevallard 1992a, 2007; Winslow 2011). It is approached through the study of the conditions enabling and the constraints hindering the production, development, and diffusion of knowledge and, more generally, of any kind of human activity in social institutions.

Cross-References

▶ Curriculum Resources and Textbooks in Mathematics Education
▶ Didactic Engineering in Mathematics Education
▶ Didactic Situations in Mathematics Education

References

Arsac G (1992) The evolution of a theory in didactics: the example of didactic transposition. In: Douady R, Mercier A (eds) Research in Didactique of mathematics. Selected papers. La Pensée sauvage, Grenoble, pp 107–130

Arsac G, Chevallard Y, Martinand JL, Tiberghien A, Balacheff N (1994) La transposition didactique à l'épreuve. La Pensée sauvage, Grenoble

Artaud M (1995) La mathématisation en économie comme problème didactique. La communauté des producteurs de sciences économiques: une communauté d'étude. In: Margolinas C (ed) Les débats de didactique des mathématiques. La Pensée sauvage, Grenoble, pp 113–129

Barbé J, Bosch M, Espinoza L, Gascón J (2005) Didactic restrictions on the teacher's practice. The case of limits of functions in Spanish High Schools. Educ Stud Math 59:235–268

Bosch M, Gascón J (2006) Twenty-five years of the didactic transposition. ICMI Bull 58:51–64

Chevallard Y (1985) La Transposition Didactique. Du savoir savant au savoir enseigné, 2nd edn 1991, La Pensée sauvage, Grenoble [Spanish translation: Chevallard Y (1997) La transposición didáctica. Del saber sabio al saber enseñado. AIQUE, Buenos Aires]

Chevallard Y (1989) On didactic transposition theory: some introductory notes. In: Proceedings of the international symposium on selected domains of research and development in mathematics education, Bratislava, pp 51–62. http://yves.chevallard.free.fr/spip/spip/IMG/pdf/On_Didactic_Transposition_Theory.pdf. Accessed 25 Oct 2012

Chevallard Y (1992a) Fundamental concepts in didactics: perspectives provided by an anthropological approach. In: Douady R, Mercier A (eds) Research in Didactique of mathematics. Selected papers. La Pensée sauvage, Grenoble, pp 131–167

Chevallard Y (1992b) A theoretical approach to curricula. J Math 13(2/3):215–230. http://yves.chevallard.free.fr/spip/spip/IMG/pdf/A_Theoretical_Approach_to_Curricula.pdf. Accessed 25 Oct 2012

Chevallard Y (2007) Readjusting didactics to a changing epistemology. Eur Educ Res J 6(2):131–134. http://www.wwwords.co.uk/pdf/freetoview.asp?j=eerj&vol=6&issue=2&year=2007&article=4_Chevallard_EERJ_6_2_web. Accessed 25 Oct 2012

Kang W, Kilpatrick J (1992) Didactic transposition in mathematics textbooks. Learn Math 12(1):2–7
Winslow C (2011) Anthropological theory of didactic phenomena: some examples and principles of its use in the study of mathematics education. In: Bosch M et al (eds) Un panorama de la TAD. An overview of ATD. CRM Documents, vol 10. Centre de Recerca Matemàtica, Barcelona, pp 533–551. http://www.recercat.net/bitstream/handle/2072/200617/Documents10.pdf?sequence=1. Accessed 25 Oct 2012

Didactical Phenomenology (Freudenthal)

Marja Van den Heuvel-Panhuizen
Freudenthal Institute for Science and Mathematics Education, Faculty of Science and Faculty of Social and Behavioural Sciences, Utrecht University, Utrecht, The Netherlands

Keywords

Phenomena in reality · Mathematical thought objects · Didactics · Realistic mathematics education · Analyses of subject matter

What Is Meant by Didactical Phenomenology?

The term *didactical phenomenology* was coined by Hans Freudenthal. Although his initial ideas for it date from the late 1940s, he likely first used the term in a German article in 1974. A few years later, the term appeared in English in his book *Weeding and Sowing – Preface to a Science of Mathematical Education* (Freudenthal 1978). Understanding the term requires comprehending Freudenthal's notion of a *phenomenology* of mathematics, which refers to describing mathematical concepts, structures, or ideas, as thought objects (*nooumena*) in their relation to the phenomena (*phainomena*) of the physical, social, and mental world that can be organized by these thought objects.

The term *didactical* is used by Freudenthal in the European continental tradition referring to the way we teach students and the organization of teaching processes. This definition of didactics goes back to Comenius' (1592–1670) *Didactica Magna* (Great Didactics) that contains a well-founded view on what and how students should be taught. As such, this meaning of didactics contrasts with the Anglo-Saxon tradition in which it merely has a superficial meaning involving a set of instructional tricks.

Combining the two terms into *didactical phenomenology* implies considering the *phenomenology* of mathematics from a *didactical* perspective.

Merit of a Didactical Phenomenology for Mathematics Education

In Freudenthal's words (1983, p. ix), a didactical phenomenology of mathematics can "show the teacher the places where the learner might step into the learning process of mankind." In other words, a didactical phenomenology informs us on how to teach mathematics, including how mathematical thought objects can help organizing and structuring phenomena in reality, which phenomena may contribute to the development of particular mathematical concepts, how students can come in contact with these phenomena, how these phenomena beg to be organized by the mathematics intended to be taught, and how students can be brought to higher levels of understanding. As such, Freudenthal's didactical phenomenologies are landmarks for developing teaching outlines.

Relation with Realistic Mathematics Education

By disclosing the sources of mathematics in reality, a didactical phenomenology is strongly related to Realistic Mathematics Education (RME), the domain-specific instruction theory for mathematics, which has been developed in the Netherlands and in which Freudenthal was heavily involved (Freudenthal 1991). In RME, rich, realistic situations have a prominent position in the learning process. These situations serve as sources for initiating the development of mathematical concepts, tools, and procedures. What situations can serve as contexts for this development is revealed by a didactical phenomenology. By tracing phenomena in reality that can elicit mathematical thoughts, the students are given access to the sources of mathematics in everyday experiences. Building on these sources offers them an orientation basis they experience as real and opens the possibility of personal engagement and solving problems in a way they find meaningful. This attachment of meaning to mathematical constructs students have to develop touches on a main principle of RME.

Examples of Didactical Phenomenology

In *Weeding and Sowing*, Freudenthal exemplified his idea of a didactical phenomenology by providing an analysis of the topic of ratio and proportion. Furthermore, he announced to deal comprehensively with didactical phenomenology in a following book. That book was *Didactical phenomenology of mathematical structures* (Freudenthal 1983). In this book, he gave more examples of didactical phenomenologies, including those of length, natural numbers, fractions, geometry and topology, negative numbers and directed magnitudes, algebraic language, and functions.

Remarkably, these examples did not just deal with connecting mathematical thought objects to phenomena in reality to find starting points for learning mathematics. In fact, these examples were profoundly scrutinized analyses of subject matter in which the key concepts of a particular mathematical topic were disclosed together with contexts which have a model character and with significant landmarks in students' learning pathways.

The Method

Unfortunately, in *Didactical phenomenology of mathematical structures*, Freudenthal did not elaborate much on how to establish these didactical phenomenologies. Although the book contains a short chapter titled *The method*, this did not reveal how to generate such phenomenologies. Nevertheless, a corner of the veil was lifted when Freudenthal (1983, p. 29) considered the material he needed to write this book:

> I have profited from my knowledge of mathematics, its application, and its history. I know how mathematical ideas have come or could have come into being. From an analysis of textbooks I know how didacticians judge that they can support the development of such ideas in the minds of learners. Finally, by observing learning processes I have succeeded in understanding a bit about the actual process of constitution of mathematical structures and the attainment of mathematical concepts.

This statement and the provided examples show how a didactical phenomenology results from a number of analyses, each taking a different perspective: didactical, phenomenological, epistemological, and historical-cultural.

Mathematics-Related Analyses Constituting the Didactics of Mathematics

These analyses have in common that they all take mathematics as their starting point. Didactical analyses examine the nature of the mathematical content as a basis for teaching this content. By identifying the determining aspects of mathematical concepts and their relationships, knowledge is gathered about didactical models that can help students to understand these concepts. Phenomenological analyses disclose possible manifestations of these mathematical concepts in reality and can suggest contexts for students to meet these concepts. Epistemological analyses focus on students' learning processes and can uncover how the mathematical understanding of students in a classroom interaction may shift. Finally, in historical-cultural analyses, we may encounter current and past approaches to teaching mathematics through which we can gain a better understanding of learning mathematics and how education can contribute to it.

These analyses are all included in Freudenthal's didactical phenomenology and surpass its narrow literal meaning, which would certainly have his approval, as in *Weeding and Sowing* Freudenthal (1978, p. 116) already stated: "[T]he name does not matter; nor is that activity [didactical phenomenology] an invention of mine; more or less consciously it has been practiced by didacticians of mathematics for a long time" (Freudenthal 1978, p. 116). Indeed, the name is not essential, but these analyses are. In Freudenthal's view, they form the heart of researching and developing mathematics education.

Cross-References

► Realistic Mathematics Education

References

Freudenthal H (1978) Weeding and sowing. Preface to a science of mathematical education. Reidel, Dordrecht
Freudenthal H (1983) Didactical phenomenology of mathematical structures. Reidel, Dordrecht
Freudenthal H (1991) Revisiting mathematics education. China lectures. Kluwer, Dordrecht

Differential Equations Teaching and Learning

Oh Nam Kwon
College of Education, Department of Mathematics Education, Seoul National University, Seoul, Korea

Keywords
Differential equations · Students' understanding and difficulties · Inquiry-oriented teaching · Tertiary education

Definition

Differential equations (DEs) are ubiquitous in applied mathematics and form a major component of undergraduate mathematics curricula at most universities. They are also part of undergraduate and graduate mathematics courses for students in a wide variety of disciplines: chemistry, life sciences, and economics. DEs provide models for many real-life situations and thus allow the formulation of phenomena from other disciplines. Research studies in mathematics education have been concerned about the processes of teaching and learning of the concept of ordinary differential equations and related themes such as directional fields, solution methods, equilibrium solutions, or those solutions that involve systems of ordinary differential equations.

Differential Equations Curriculum

Although analytic techniques for finding closed-form solutions to DEs have long been a mainstay of the traditional introductory DEs course, in practice, when modeling a physical or realistic problem with a DE, solutions are usually inexpressible in closed form. Numerical and graphical methods, on the other hand, are applicable to all differential equations. Numerical methods are most easily employed with the aid of technology and, in many cases, provide reliable approximate solutions. With graphical (or qualitative) methods, one obtains overall information about solutions to DEs by viewing the solutions geometrically and by analyzing the DE itself. These changes reflect the evolution of the subject from an algebraic setting to a numerical and geometric setting characteristic of analytic approaches to and interest in dynamical systems. Drawing on a dynamical systems point of view, DEs can be treated as mechanisms that describe how functions evolve and change over time. Interpreting and characterizing the behavior and structure of these solution functions are important goals, with central ideas including the long-term behavior of solutions, the number and nature of equilibrium solutions, and the effect of varying parameters on the solution space.

Learning

A number of studies document student difficulties with DEs, within mathematics itself, as well as related to their use in other disciplines. One of the learning difficulties met by students is from misunderstanding of basic underlying concepts such as rate, rate of change, and derivative (Rowland and Jovanoski 2004). See also ▶ "Calculus Teaching and Learning." There is also the fact that, for the first time, students meet equations whose solutions are not numbers but functions. For example, Rasmussen (2001) reported that the concept of a solution of a differential equation might be difficult for students to comprehend because they are used to identifying solutions of equations with numerical values and not with functions. The shift away from conceptualizing solutions as numbers is nontrivial for students. Habre (2000) noted that the idea of solving DEs has remained purely algebraic in the minds of all students and few students succeeded in moving back and forth between the visual and the algebraic aspects of DEs. Similarly, students tend to privilege algebraic approaches over graphical approaches even in courses that value and emphasize graphical and qualitative analysis (Arslan 2010). Scholars have also noted that for many students, the connection between the differential equation and its solutions is largely procedural, rather than conceptual. These findings reveal that students are successful in algebraic solutions but not in conceptualizing DEs and the concepts of DE solution. In overcoming students' learning difficulties, research makes it clear that learning is possible if adequate strategies are developed. For example, Keene (2007) found that students reasoned successfully about both qualitative and analytic solutions to differential equations in a reform-oriented approach. She developed the notion of dynamic reasoning as a way to frame solutions and has had students learn to use time as a dynamic parameter that coordinates with other quantities in order to understand and solve problems. In examining student understanding of solutions and solution methods for a system of DEs, Dana-Picard and Kidron (2008) investigated the cognitive processes of two students in a computer algebra system (CAS) environment and found that the order in which graphical and symbolic representations were brought to bear in their sense-making was markedly different. Central to the students' mathematical work was the interplay between numerical, graphical, and algebraic work that promoted a dynamical view of the graphs of dy/dt versus y and corresponding y versus t solution graphs. The inquiry-oriented approach to DEs (IO-DE) of Rasmussen and his colleagues (2006) provides an example of a curriculum based on Realistic Mathematics Education (see ▶ "Realistic Mathematics Education"). In addition to curricular products, their work has led to empirical results about student understanding of specific concepts (for example, DE solution, direction field, bifurcation diagram) and theoretically grounded local instructional theories for DEs. These studies show that the way in which the mathematical contents are structured and presented to students plays an important role in the students' development of conceptual learning.

Teaching

Lectures are still the most dominant form of teaching practices in traditional DE courses. Researchers note that when traditional teaching styles, such as transmission, are used, there is a significant gap between what the lecturer teaches and what the student actually learns. Rasmussen and Kwon (2007) suggest that an inquiry-oriented approach to DE courses could guide students in learning new mathematical concepts through inquiry, by engaging in mathematical discussions, posing and following up conjectures, explaining and justifying their thinking, and solving novel problems. Their inquiry-oriented instruction is founded on three cornerstones: (1) deeper student engagement in the mathematics, (2) peer-to-peer collaboration, and (3) instructor inquiry into student thinking. Although such classrooms may include small group work, this instruction can also apply to large classrooms in which the instructor uses individual response systems (see ▶ "Inquiry-Based Mathematics Education").

Literature from quantitative studies has proved that an inquiry-oriented DE approach enhances desirable student learning outcomes as compared to traditional DE approaches. Rasmussen et al. (2006) showed that IO-DE group students performed better on conceptually oriented items than comparison groups. In a similar context, Kwon et al. (2005) conducted a follow-up study 1 year after instruction for a subset of the students on the retention effects of conceptual and procedural knowledge. Researchers concluded that IO-DE enables students to emphasize a variety of strategies both simultaneously and with equal importance, and because of this, students retained multiple ways to approach problems and performed better even after 1 year. Other evaluation studies also proved the positive outcome of the IO-DE approach on students' conceptual understanding, problem solving, retention, and justification. Another form of inquiry-oriented instruction is the notion of "Study and Research Path" (SRP) developed in the frame of the Anthropological Theory of the Didactic (see ▶ "Anthropological Theory of the Didactic (ATD)"). Barquero et al. (2016) proposed SRP as a bridge between inquiry and transmission in the context of modelling the evolution of populations through sequences and DEs. As a whole, the research on inquiry-oriented instruction speaks its clear benefit for learners.

One of the studies examining effective instructional practices in teaching differential equations focused on the instructor-student interaction patterns that facilitated students' reinvention of the ideas of bifurcation. Combining the theoretical constructs of classroom practice, Rasmussen et al. (2009) identify three broad "brokering moves" that facilitated the emergence and reinvention of a bifurcation diagram. These brokering moves forge connections between the different small groups, the classroom community as a whole, and the norms and practices of the broader mathematical community. In other work examining instructor-student interactions that contributed to significant student progress in creating, interpreting, and using phase portraits, Kwon et al. (2008) detail the following four functions of instructor revoicing: (1) as a binder, (2) as a springboard, (3) for ownership, and (4) as a means for socialization. In a related analysis of this same DE classroom, Rasmussen et al. (2009) developed a framework that coordinates student argumentation (analyzed according to the terms established by Toulmin 1969) and instructor discursive moves that enable and constrain progress in student argumentation. This work offers some first steps in coordinating student and instructor activity.

For Further Research

Research in DE teaching and learning has made progress in countering students' difficulties in learning DEs and in offering a variety of instructional strategies and learning environments. There has been an increasing trend toward inquiry-oriented instruction. The increase in inquiry-oriented instruction has resulted in greater opportunities to, as well as a greater need to, better understand the ways in which educational contexts, discourse, instructional materials, and technological tools enable and constrain student participation in enculturation into disciplinary practices. The studies on inquiry-oriented instruction highlight the need for research exploring different avenues that mathematicians can use to further refine and develop their pedagogical skills and pedagogical content knowledge for their teaching of DEs. Many challenges still lie ahead for DE teaching and learning, mostly in transforming research to practice and focusing on the connections between DEs and other STEM courses in undergraduate education. Collaborative research across content disciplines has the potential to inform the field as how to best serve students in undergraduate STEM education.

Cross-References

▶ Anthropological Theory of the Didactic (ATD)
▶ Calculus Teaching and Learning
▶ Inquiry-Based Mathematics Education
▶ Realistic Mathematics Education

References

Arslan S (2010) Do students really understand what an ordinary differential equations is? Int J Sci Math Educ 41(7):873–888

Barquero B, Serrano L, Ruiz-Munzon N (2016) A bridge between inquiry and transmission: the study and research paths at university level. In: First conference of international network for didactic research in University Mathematics, March 2016, Montpellier

Dana-Picard T, Kidron I (2008) Exploring the phase space of a system of differential equations: different mathematical registers. Int J Sci Math Educ 6(4):695–717

Habre S (2000) Exploring students' strategies to solve ordinary differential equations in a reformed setting. J Math Behav 18(4):455–472
Keene KA (2007) A characterization of dynamic reasoning: reasoning with time as parameter. J Math Behav 26(3):230–246
Kwon ON, Rasmussen C, Allen K (2005) Students' retention of mathematical knowledge and skills in differential equations. Sch Sci Math 105(5):1–13
Kwon ON, Ju MK, Rasmussen C et al (2008) Utilization of revoicing based on learners thinking in an inquiry-oriented differential equations. Class SNU J Educ Res 17:111–134
Rasmussen C (2001) New directions in differential equations: a case of students' understanding and difficulties. J Math Behav 20(1):55–87
Rasmussen C, Kwon ON (2007) An inquiry-oriented approach to undergraduate mathematics. J Math Behav 26(3):189–194
Rasmussen C, Kwon ON, Allen K et al (2006) Capitalizing on advances in mathematics and K-12 mathematics education in undergraduate mathematics: an inquiry-oriented approach to differential equations. Asia Pac Educ Rev 7(1):85–93
Rasmussen C, Zandieh M, Wawro M (2009) How do you know which way the arrows go? The emergence and brokering of a classroom mathematics practice. In: Roth W (ed) Mathematical representations at the interface of the body and culture. international perspectives on mathematics education, cognition, equity and society. IAP, Charlotte, pp 171–218
Rowland D, Jovanoski Z (2004) Student interpretations of the terms in first-order ordinary differential equations in modelling contexts. Int J Math Educ Sci Technol 35(4):503–516

Discourse Analytic Approaches in Mathematics Education

Candia Morgan
Institute of Education, University of London, London, UK

Keywords
Discourse · Discourse analysis · Foucault · Identity · Language · Linguistics · Methodology · Social practice

Introduction

The first challenge in addressing this topic is the multiplicity of ways in which the term *discourse* is used and defined – or not defined – within mathematics education (see Ryve 2011). It is frequently found, especially in discussions within the context of curriculum reform, simply to signify student engagement in talk in the classroom. Without denying the value of the development of such engagement, the approaches to discourse and discourse analysis considered in this article all take rather more complex and theoretically shaped views of the nature of discourse – views that influence the focus of research and the analytic methods. An important component of the ways these approaches conceive of discourse is a concern with the relationship between language (and other modes of communication), the social context in which it is used, and the meanings that are produced in this context (Howarth 2000). It is this concern and the fundamental assumption that studying the way language is used can provide insight into the activity or practice (mathematics or mathematics education) in which it is used that leads researchers to adopt discourse analytic approaches. Of course, a very high proportion of the data used in studies across many branches of mathematics education research is primarily linguistic or textual: interviews, written responses to questionnaires, classroom transcripts, written texts produced by students, etc. Increasingly it has also been recognized by researchers using a wide range of approaches that the language produced by students or other research subjects is not a transparent medium through which it is easy to decipher an underlying truth. What distinguishes research that adopts a discourse analytic approach is the assumption that the language is itself an inextricable part (or, for some researchers, even the whole) of the object of study. This assumption is shared with another analytic approach, conversation analysis, and some discourse analysts make use of methods developed in conversation analysis. However, whereas discourse analysis is generally interested in characterizing the practices within which language plays a role, conversation analysis focuses primarily on how linguistic interactions themselves are organized to achieve social actions (see Wooffitt 2005, for an introduction to the two approaches from a conversation analytic perspective).

D

Gee (1996) makes a useful distinction between *discourse*, defined as instances of communication, and *Discourses*, the conjunctions of ways of speaking, subject positions, values, etc. that characterize and structure particular social practices. The notion of *Discourses* has its origin in the thinking of the French philosopher Foucault (e.g., Foucault 1972) whose work includes studies of the construction of "regimes of truth" about notions such as madness or sexuality. Though not all discourse analytic research in mathematics education comes from this tradition, it can generally be characterized as tending either towards analysis of *discourse*, focusing on communication events and the local social practices within which they arise, or towards analysis of *Discourse*, taking larger scale social practices and structures as the object of research. Of course, some approaches move between the two, generating interpretation of specific communication events by applying knowledge of wider social practices and structures or building a picture of a significant social practice through analysis of local communication events. Discourse analytic approaches thus vary in two dimensions: the extent to which they make use of detailed linguistic analysis and the extent of their focus on social practices, structures, and institutions.

The adoption and development of discourse analytic approaches in mathematics education research largely coincided with what Lerman termed the "social turn" (Lerman 2000). Increased recognition of the importance of studying and taking account of the social nature of mathematics education practices as well as of individual cognition demanded the development of theoretical ways of conceiving of social practices and methodological approaches to studying them. Discourse analytic approaches provided one way of addressing this demand. This development within the field of mathematics education reflected a much wider development of theories of discourse and discourse analytic methods within social science and the humanities. As researchers have begun to draw on theories and methods originating outside the field of mathematics education, they have faced the challenge of ensuring that both theory and methods take account of the specialized nature of mathematical communication and practices and that they have the power to illuminate issues of interest to mathematics education. Facing this challenge is a continuing project; notable contributions have come from within mathematics education (e.g., Morgan 1998; Sfard 2008) and from linguistics (e.g., O'Halloran 2005).

With a few exceptions, notably the work of Walkerdine (1988) who used analyses of *Discourses*, including Discourses of gender and of child-centered education, in order to understand how differences between various social groups are constructed in mathematics education practices, early interest in discourse analytic approaches, such as that represented in the Special Issue of *Educational Studies in Mathematics* edited by Kieran et al. (2001), was dominated by analysis of communication events (*discourse*), focusing on understanding classroom interaction and the development of mathematical thinking in interaction. At a time when the majority of research in mathematics education focused on the mathematical thinking of individuals, this application of discourse analysis may be seen as an incremental manifestation of the "social turn," addressing the same interest in mathematical thinking but reconceptualizing it as a phenomenon that is evident (and, for some researchers, produced) in social interaction. More recently, the issues addressed by the mathematics education research community have expanded, incorporating a wider conceptualization of mathematics and mathematics education as social practices. Thus more research has addressed, *inter alia*, identity, power relationships, and social justice – issues that lend themselves to study using approaches that focus on *Discourses*. Some of this research has adopted approaches that may be characterized as structuralist, drawing on sociological accounts of social structures such as the work of Basil Bernstein (e.g., Bernstein 2000) to describe and interpret discursive phenomena. Others have adopted poststructural approaches, in which the communicative action itself constructs the "reality" of which it speaks. A recent edited book entitled *Equity in Discourse for Mathematics Education* (Herbel-Eisenmann et al. 2012) reflects this range of approaches and interpretations, combining detailed analyses of classroom interactions with concern for how these interactions and broader social practices affect the possibilities for participation in mathematics of students from different social groups.

In this article there is no space to provide a detailed review of the full range of approaches taken to discourse analysis. Instead, we provide a small number of contrasting cases, exemplifying the scope of discourse analytic methods and the problems in mathematics education that they may be used to address.

Critical Discourse Analysis

Critical discourse analysis (CDA) comprises a group of analytic approaches, all of which make strong analytic connections between forms of language use, social practices, and social structures. The label "critical" indicates a concern of the researchers to make use of the knowledge achieved through the analysis in order to enable critique and transformation of the social practices and/or structures. Research using CDA approaches thus tends to produce analyses that not only describe existing practices but also critique the ways these practices position students and/or teachers and the kinds of mathematics and mathematical identities that are valued and made possible.

CDA studies generally involve detailed analyses of texts, including oral and written texts produced and used by students and teachers in the classroom but also including texts such as the curriculum and policy documents that structure and regulate these educational practices and thus affect the interpretation of classroom texts. Within mathematics education, probably the most widely used type of CDA is based on the approach of Norman Fairclough (2003), using linguistic tools drawn from systemic functional linguistics (SFL). This approach has been used to investigate specific practices such as the assessment of student reports of mathematical investigation (Morgan 1998) or the use of "real-world problems" in an undergraduate mathematics course (Le Roux 2008). Research adopting a CDA approach may also use a range of other methods to address textual data, including corpus analysis of large data sets (e.g., Herbel-Eisenmann et al. 2010).

Whatever the linguistic tools used to describe the data, the interpretative stage of CDA involves considering how the features identified in the data function to construe the "reality" of the practice being studied and the social positionings and relations of the participants. As Fairlough argues, such interpretation requires explicit use of "insider knowledge" of the social practices studied (Fairclough 2003). This means that researchers in mathematics education need to bring knowledge of broader mathematics education practices and knowledge of mathematical practices to bear on their analyses. For example, Morgan's study of teachers' assessment practices is informed by an analysis of the constructs and values found in the associated curriculum documents, policy, and professional literature, while Le Roux draws on Sfard's (2008) characterization of mathematical discourse (discussed further below) to enable her analysis to address the nature of the mathematical activity involved in the use of real-world problems.

Poststructural Approaches

The approaches to discourse analysis discussed under the heading of postmodern or poststructural share with CDA approaches a concern with issues such as power and subjectivity that arise in considering relationships between individuals and social practices and structures. There are, however, both philosophical and methodological differences between the approaches. There is a range of philosophical positions associated with postmodern and poststructural thought; however, a shared foundation is a rejection of the notions of an objective world and of the fixed subjectivity of a rational knowing subject. These philosophical assumptions are shared by some but certainly not by all those employing CDA approaches, though there is a common interest in characterizing the key entities that play a role in a Discourse and the possibilities for individual subjectivities, identities, or positioning.

The major distinction drawn here between the approaches to discourse analysis discussed in this section and those identified under the heading CDA is methodological. While CDA involves close analysis of specific texts, usually employing analytical tools and methods drawn from linguistics, the starting point for postmodern/poststructural researchers tends to be at the level of the major functions of discourse. For example, Hardy (2004) uses the Foucauldian constructs of *power as production* and *normalization* as her analytical tools for interrogating a teacher training video produced as part of the English National Numeracy Strategy to demonstrate "effective teaching" of mathematics in a primary classroom. Rather than focus on detailed characteristics of the discourse of this video, Hardy uses these constructs to provide an alternative perspective on the data as a whole. This enables her to tell a story of what the Discourse of the National Numeracy Strategy achieves – how it produces assumptions about what is normal and what is desirable – a story that runs counter to the "common sense" stories about effective teaching.

A rather different approach is taken by Epstein et al. (2010), though again founded in Foucauldian theory. They first characterize the ways in which mathematics and mathematicians are represented in popular media – as hard, logical, and ultrarational but also as eccentric or even insane. Having identified different and in some cases apparently contradictory Discourses about mathematics, Epstein et al. then use these to analyze interviews with students, identifying how individual students deploy the various discursive resources in order to produce their own identities as mathematicians or as nonmathematicians and their relationships to mathematics as a field of study.

Mathematical Discourse, Thinking, and Learning

The main discourse analytic theories mentioned so far have their origins outside mathematics education, drawing on fields such as linguistics, ethnography, sociology, and philosophy. For mathematics education researchers, this raises the important theoretical and methodological problem of the extent to which the specifically mathematical aspects of the practices being studied may be captured and accounted for. In order to address this problem, an increasing number of researchers, including some of those working with CDA or other discourse analytic approaches, are turning to the work of Anna Sfard (2008). While Sfard draws on a number of sources, including Wittgenstein's notion of language game, her own theory of mathematical discourse has been developed within the field of mathematics education and is designed to address the problems arising in this field. Her communicative theory of cognition identifies thinking mathematically as participating in mathematical discursive practices, that is, as communicating (with oneself or with others) using the forms of discourse characteristic of mathematics. Sfard identifies four aspects of mathematical discourse that form the basis for her analytic method: specialized mathematical *vocabulary and syntax* (what may be considered the "language" of mathematics), *visual mediators* (nonlinguistic forms of communication such as algebraic notation, graphs, or diagrams), *routines* (well-defined repetitive patterns, e.g., routines for performing a calculation, solving an equation, or demonstrating the congruence of two triangles), and *endorsed narratives* (the sets of propositions accepted as true within a given mathematical community). Scrutinizing how these four aspects are manifested in discourse provides a means of describing mathematical thinking and hence allows one to address questions such as the following: How does children's thinking about a mathematical topic vary from that expected by their teacher or by an academic mathematical community? How does children's thinking develop (i.e., how does their use of a mathematical form of discourse change over time)? What kinds of mathematical thinking are expected of students taking an examination or using a textbook?

As may be seen from the research topics and questions illustrated in this article, discursive approaches can address a wide range of issues of concern within the field of mathematics education, bridging, as indicated in the title of Kieran et al.'s (2001) Special Issue of *Educational*

Studies in Mathematics, the individual and the social. While the various approaches share a basic assumption that language and social practices play a role in the ways that individuals make sense of mathematical activity, they differ in the ways they conceptualize this role (and, indeed, in how they conceptualize language, social practice, and mathematics). Hence they also differ in the research questions they pose and the methodological tools they employ. It can be argued that discourse analytic approaches allow us to see through *what is said* to reveal *what is achieved* by using language. The challenge for researchers and for the readers of research is to clarify how the theoretical and methodological tools employed enable this and to distinguish which kinds of actions and achievements are made visible by the different approaches.

Cross-References

▶ Discursive Approaches to Learning Mathematics
▶ Mathematical Language
▶ Poststructuralist and Psychoanalytic Approaches in Mathematics Education
▶ Sociological Approaches in Mathematics Education

References

Bernstein B (2000) Pedagogy, symbolic control and identity: theory, research and critique, rev edn. Rowman and Littlefield, Lanham

Epstein D, Mendick H, Moreau M-P (2010) Imagining the mathematician: young people talking about popular representations of maths. Discourse Stud Cult Politics Educ 31(1):45–60

Fairclough N (2003) Analysing discourse: textual analysis for social research. Routledge, London

Foucault M (1972) The archaeology of knowledge. Routledge, London

Gee JP (1996) Social linguistics and literacies: ideology in discourses, 2nd edn. Taylor & Francis, London

Hardy T (2004) "There's no hiding place": Foucault's notion of normalisation at work in a mathematics lesson. In: Walshaw M (ed) Mathematics education within the postmodern. Information Age, Greenwich, pp 103–119

Herbel-Eisenmann B, Wagner D, Cortes V (2010) Lexical bundle analysis in mathematics classroom discourse: the significance of stance. Educ Stud Math 75:23–42

Herbel-Eisenmann B, Choppin J, Wagner D, Pimm D (eds) (2012) Equity in discourse for mathematics education. Springer, Dordrecht

Howarth D (2000) Discourse. Open University Press, Buckingham

Kieran C, Forman EA, Sfard A (eds) (2001) Bridging the individual and the social: discursive approaches to research in mathematics education: a PME special issue. Educ Stud Math 46:1–298

Le Roux K (2008) A critical discourse analysis of a real-world problem in mathematics: looking for signs of change. Lang Educ 22(5):307–326

Lerman S (2000) The social turn in mathematics education research. In: Boaler J (ed) Multiple perspectives on mathematics teaching and learning. Ablex, Westport, pp 19–44

Morgan C (1998) Writing mathematically: the discourse of investigation. Falmer, London

O'Halloran KL (2005) Mathematical discourse: language, symbolism and visual images. Continuum, London

Ryve A (2011) Discourse research in mathematics education: a critical evaluation of 108 journal articles. J Res Math Educ 42(2):167–198

Sfard A (2008) Thinking as communicating: human development, the growth of discourses, and mathematizing. Cambridge University Press, Cambridge

Walkerdine V (1988) The mastery of reason: cognitive development and the production of rationality. Routledge, London

Wooffitt R (2005) Conversation analysis and discourse analysis: a comparative and critical introduction. Sage, London

D

Discrete Mathematics Teaching and Learning

Cécile Ouvrier-Buffet
UPEC and Laboratoire de Didactique André Revuz, Paris, France

Keywords

Discrete mathematics · Discrete · Continuous · Reasoning · Proof · Mathematical experience · Problem-solving

Definition

The teaching of "discrete mathematics" is not always clearly delimited in the curricula and can be diffuse. In fact, the meaning of "discrete mathematics teaching and learning" is twofold. Indeed, it includes the teaching and learning of discrete concepts (considered as defined objects inscribed in a mathematical theory), but it also includes skills regarding reasoning, modeling, and proving (such skills are specific to discrete mathematics or transversal to mathematics).

What Is Discrete Mathematics?

Discrete mathematics is a comparatively young branch of mathematics with no agreed-on definition (Maurer 1997): only in the last 30 years did it develop as a specific field in mathematics with new ways of reasoning and generating concepts. Nevertheless, the roots of discrete mathematics are older: some emblematic historical discrete problems are now well known, also in education where they are often introduced as enigma, such as the four color theorem (map coloring problem), the Königsberg's bridges (traveling problem), and other problems coming from the number theory, for instance.

There is no exact definition of *discrete mathematics*. The main idea is that *discrete mathematics* is the study of mathematical structures that are "discrete" in contrast with "continuous" ones. Discrete structures are configurations that can be characterized with a finite or *countable* set of relations. (A *countable* set is a set with the same cardinality (number of elements) as some subset of the set of natural numbers. The word "countable" was introduced by Georg Cantor.) And discrete objects are those that can be described by finite or countable elements. It is strongly connected to number theory, graph theory, combinatorics, cryptography, game theory, information theory, algorithmics, discrete probability, but also group theory, algebraic structures, topology, and geometry (discrete geometry and modeling of traditional geometry with discrete structures).

Furthermore, discrete mathematics represents a mathematical field that takes on growing importance in our society. For example, discrete mathematics brings with it the mathematical contents of computer science and deals with algorithms, cryptography, and automated theorem proving (with an underlying philosophical and mathematical question: is an automated proof a mathematical proof?).

The aims of discrete mathematics are to explore discrete structures, but also to give a specific modeling of continuous structures, as well as to bring the opportunity to consider mathematical objects in a new manner. Then new mathematical questions can emerge, as well as new ways of reasoning, which implies a challenge for mathematicians.

Some famous problems of discrete mathematics have inspired mathematics educators. That is the case of a combinatorial game: the game of Nim, played since ancient times with many variants. The regular game of Nim is between two players. It is played with three heaps of any number of objects. The two players alternatively take any number of objects from any single one of the heaps. The goal is to be the last one to take an object. Brousseau (1997) explicitly refers to the game theory to conceptualize the theory of didactical situations. The game of Nim is the background of the generic example of Brousseau's theory, "the Race to 20."

Why Teach and Learn Discrete Mathematics? New Context, Concepts, and Ways of Reasoning: A New Realm of Experience for the Classroom

To Integrate Discrete Mathematics into the School Curriculum: A Current Challenge

More and more fields of mathematics use results from discrete mathematics (topology, algebraic geometry, statistics, among others). Moreover, discrete mathematics is an active branch of contemporary mathematics. New needs for teaching are identified: they are linked to the evolution of the society and also other disciplines such as computer science, engineering, business, chemistry, biology, and economics, where discrete mathematics appears as a tool as well as an object. Then discrete mathematics should be an integral part of the school curriculum: the concepts and the ways of reasoning that should be taught in a specific field labeled "discrete mathematics" still should be more precisely identified. A dialog between mathematicians and mathematics educators can help for this delimitation.

However, the place of discrete mathematics in curricula is today very variable depending on the countries and on the levels. In a few countries, there has been a long tradition to introduce graph theory in the secondary level among other components of discrete mathematics. This place is strengthened and attested by the contents at the university level, in particular at the interface with computer science (Epp 2016; Ouvrier-Buffet et al. 2018). In other countries, only a very small number of discrete mathematics concepts are taught, especially those involved in the fields of combinatorics and number theory. Things are changing; the reader can refer to Rosenstein et al. (1997), DIMACS (2001) contributions, and the ICME monograph (Hart and Sandefur 2018) to go into details regarding the challenge of introducing discrete mathematics in curricula (especially the example of the NCTM standards [*National Council of Teachers of Mathematics*] which focuses on discrete mathematics as a field of teaching). The following arguments summarize the main ideas of these contributions, emphasizing the interests and the potential ways to implement discrete mathematics in the curricula:

Proof and abstraction are involved in discrete mathematics (for instance, in number theory, induction, etc.).

It allows an introduction to modeling and proving processes, but also to optimization and operational research, as well as experimental mathematics.

Problems are accessible and can be explored without an extensive background in school mathematics.

The results in discrete mathematics can be applied to real-world situations.

Discrete mathematics brings a specific work on algorithms and recursion.

The main problems in discrete mathematics are still unsolved in ongoing mathematical research: a challenge for pupils and students to be involved in a solving process close to the one of mathematicians and to promote cooperative learning (in a specific and suitable context: in particular, teachers should be trained to discrete problems and also to their teaching and management).

Benefits from Teaching and Learning Discrete Mathematics: Some Examples

Learning discrete mathematics clearly means learning new advantageous concepts but also new ways of reasoning, making room for a mathematical experience.

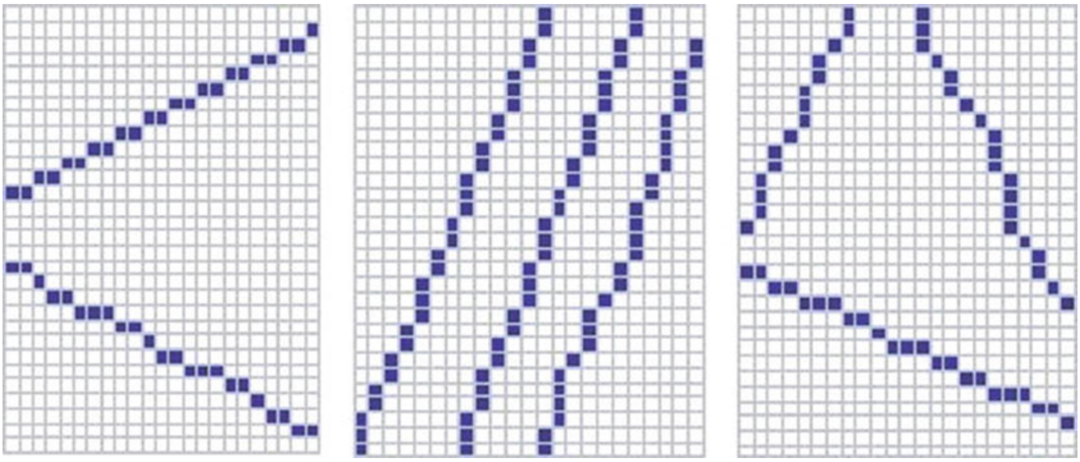
Many variants exist of the following famous problems that are developed below. Some of them are presented and analyzed, for instance, in Rosenstein et al. (1997), in Hart and Sandefur (2018), and on the website http://mathsamodeler.ujf-grenoble.fr/.

Accessible Problems and Concepts

Discrete concepts are easily graspable, applicable, accessible, and also neutral when not yet included in the curricula: this last argument implies that the way students deal with discrete concepts is quite new and different from the way they usually consider mathematics.

Traveling salesperson problem: the problem is to find the best route that a salesperson could take if he/she would begin at the home base, visit each customer, and return to the home base ("best" was defined as minimizing the total distance).

Map coloring problem (combinatorial optimization problem): a map coloring problem consists in discovering the minimum number of colors needed to *properly* color a map (or a graph). A map is *properly* colored if no two countries sharing a border have the same color. The proof of the minimum number of colors is also required. Similar coloring problems exist in graph theory. Such map and graph coloring problems are very useful to explore what discrete mathematical modeling is. Moreover, the proof of the five color theorem is accessible at the secondary and university levels. And the theorem of the four colors was proved with general-purpose theorem-proving software: such a proof brings epistemo-



Discrete Mathematics Teaching and Learning, Fig. 1 Are these lines straight lines?

logical debates in the mathematicians' community because of the use of a software.

Richness of Discrete Concepts: A Way to Deal with the Construction of Axiomatic Theory

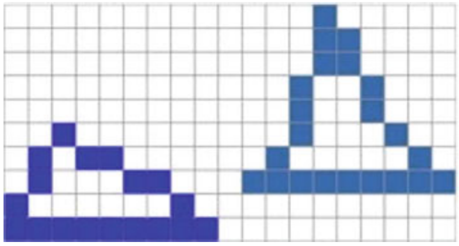
A certain amount of discrete objects can be defined in several ways, with different characterizations. The modeling of continuous concepts in the discrete case raises the problem of the construction of a mathematical consistent theory from an axiomatic point of view. It is illustrated with the following example of discrete geometry.

Discrete geometry: Example of discrete straight lines. Discrete straight lines form a concept accessible by its representation (e.g., perceptive and analytical aspects of geometry). It is noninstitutionalized (an institutionalized concept is a "curriculum" concept, i.e., a concept that has a place in the classic taught content). Delimiting what a straight line can be in a discrete context proves to be quite a challenge. Professional researchers in discrete geometry (both mathematicians and computer scientists) use several definitions, but the proof of the equivalence of these definitions remains worth considering. The complexity of the underlying axiomatization of discrete geometrical concepts is actually an open and interesting problem (it implies, for instance, the following questions: What is the intersection of two discrete straight lines? What does it mean to be parallel in the discrete case? etc.): the question of a "good" definition of a discrete straight line is currently an open and interesting problem. So are the questions of the definitions of other discrete geometrical concepts (Figs. 1, 2, and 3).

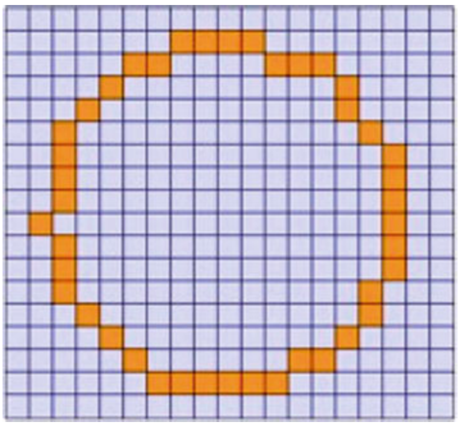
From a mathematical point of view, the discrete geometrical objects and more specifically the discrete straight lines can be approached in several ways: differential discrete analysis, the Bresenham algorithm, algorithms involving combinatorial analysis, several discretizations using algorithms which generate and study errors (Greene and Yao 1986; Freeman 1970; Pham 1986; Rosenfeld 1974), and the introduction by Reveillès (1991) of the arithmetical definition of a discrete straight line. For instance, the approach to the discretization of a real straight line by checking linearity conditions is directly related to number theoretical issues in the approximation of real numbers by rational numbers. These linearity conditions can be checked incrementally, leading to a decomposition of arbitrary strings into straight substrings as proved by Wu (1982). The ongoing mathematical problems in discrete geometry are intimately related to questions in other fields of mathematics and computer science. The construction and the manipulation of algorithms are important for this purpose.

Several Ways of Questioning, Proving, and Modeling

Besides, discrete mathematics arouses interest because it offers a new field for the learning and



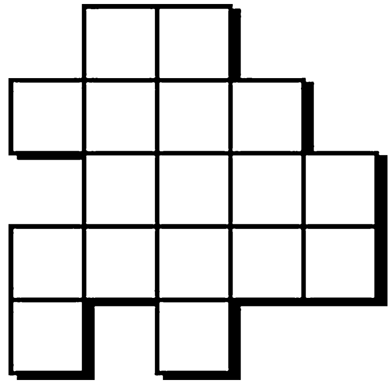
Discrete Mathematics Teaching and Learning, Fig. 2 Are these shapes triangles?



Discrete Mathematics Teaching and Learning, Fig. 3 Is it a circle?

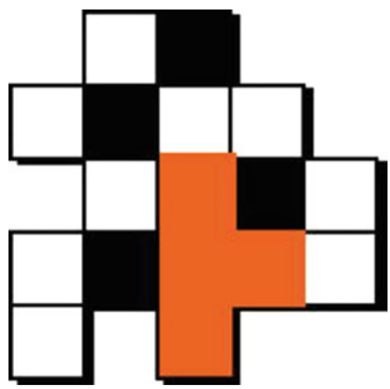
teaching of proofs (Grenier and Payan 1999; Heinze et al. 2004a, b; Hart and Sandefur 2018; http://mathsamodeler.ujf-grenoble.fr/). Some discrete problems fruitfully bring different ways to consider proof and proving processes. How can discrete mathematics contribute to make students acquire the fundamental skills involved in defining, modeling, and proving, at various levels of knowledge? It is still a fundamental question in mathematics education. The following example brings an opportunity to deal with an optimization problem which involves several kinds of reasoning. Besides, this problem is close to the contemporary research in discrete mathematics.

Hunting the beast. Your garden is a collection of adjacent squares (see Fig. 4), and a beast is itself a collection of squares (like the one drawn in Fig. 5). Your goal is to prevent a beast from entering your garden. To do this, you can buy traps. A trap is represented by a single black



Discrete Mathematics Teaching and Learning, Fig. 4 A garden

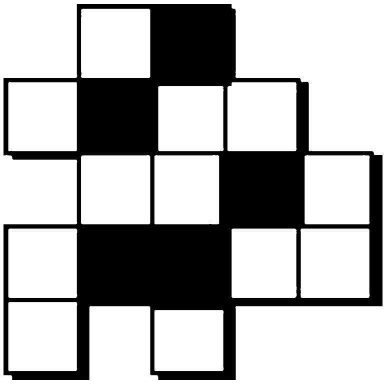
Discrete Mathematics Teaching and Learning, Fig. 5 A beast (a beast can be rotated or reversed)



Discrete Mathematics Teaching and Learning, Fig. 6 Not a solution

square that can be placed on any square of the garden. The question we ask is the following: What is the minimum number of traps you need to place so that no beast can land on your garden?

On Fig. 6, the disposition of the traps does not provide a solution to the problem, since a beast can be placed. On Fig. 7, a solution with five traps is suggested. Is it an optimal one for this configuration?



Discrete Mathematics Teaching and Learning, Fig. 7 A solution with five traps

In the literature, the problem *Hunting the beast* can be seen as a variation of the *Pentomino Exclusion Problem* introduced by Golomb (1994).

A Mathematical Experience

Discrete mathematics then brings the opportunity for students to be involved in a mathematical experience. Harel (2009) points out the following principle:

> The ultimate goal of instruction in mathematics is to help students develop ways of **understanding** and ways of **thinking** that are compatible with those practiced by contemporary mathematicians. (p. 91)

The "doing mathematics as a professional" component is clearly a new direction for the educational research in the problem-solving area, and discrete mathematics offers promising nonroutine potentialities to develop powerful heuristic processes, as underscored by Goldin (2009). Goldin (2018) also considers the affective dimension of studying discrete mathematics generally: he describes certain characteristics of discrete mathematics that can enable teachers to evoke student interest and engagement and develop students' powerful affect in relation to mathematics – emotions, attitudes, beliefs, and values.

Bearing in mind the aforesaid arguments, discrete mathematics provides a mathematical experience and is a field of experiments that questions concepts involved in other mathematical branches as well. Nevertheless, if the discrete problems are sometimes (and even often) easier to grasp than the continuous ones, the mathematics behind can be quite advanced. That is the reason why the didactic should analyze both the discrete mathematics for itself and the discrete mathematics helping the teaching of other concepts.

Interesting Perspectives for Research in Mathematics Education

Discrete mathematics is a relatively young science, still in progress with accessible and graspable concepts and ongoing questionings; hence the questions regarding the introduction of it in the curricula and in the classroom concern both mathematics educators and mathematicians.

Two separated but linked perspectives for the educational research emerge:

- The didactical study of teaching and learning discrete mathematics
- The didactical study of the teaching of concepts and skills (such as proof and modeling) with the help of discrete problems

Besides, discrete mathematics can be introduced either as a mathematical theory or as a set of tools to solve problems. The links between discrete mathematics as a tool and discrete mathematics as an object in teaching and learning should also be analyzed in depth, as well as the proof dimension involved in dealing with discrete concepts and structures. The didactic transposition of discrete concepts and ways of reasoning is still a current problem for mathematics education. It can raise the question of the development of models for teaching and learning discrete mathematics. Some epistemological models do exist (around transversal concepts such as implication, definition, and proof (see, for instance, Ouvrier-Buffet 2006) and specific contents such as the teaching of graph theory (see the work of Cartier 2008)), but the work is still in progress. Note that it involves the same questionings for mathematics education as the introduction of algorithmics in the curricula.

Furthermore, the introduction of discrete mathematics in the curricula clearly offers an opportunity to infuse new instructional techniques. In this perspective, the teacher training should be rethought at every level (primary, secondary, and university levels).

Cross-References

- ▶ Algorithmics
- ▶ Argumentation in Mathematics
- ▶ Inquiry-Based Mathematics Education
- ▶ Logic in Mathematics Education
- ▶ Mathematical Games in Learning and Teaching
- ▶ Mathematical Modelling and Applications in Education
- ▶ Mathematical Proof, Argumentation, and Reasoning
- ▶ Mathematics Teachers and Curricula
- ▶ Problem-Solving in Mathematics Education
- ▶ Word Problems in Mathematics Education

References

Brousseau G (1997) Theory of didactical situations in mathematics. Kluwer, Dordrecht

Cartier L (2008) Le graphe comme outil pour enseigner la preuve et la modélisation. Thèse Université Joseph Fourier, Grenoble. http://tel.archives-ouvertes.fr/tel-00416598/. Accessed 18 Jan 2013

DIMACS (2001) Center for discrete mathematics and theoretical computer science: educational program. http://dimacs.rutgers.edu/Education. Accessed 28 Sept 2018

Epp SS (2016) Discrete mathematics for computer science. In: Proceedings of the 13th international congress on mathematical education, Hamburg. http://condor.depaul.edu/sepp/ICME-13.Epp.pdf. Accessed 28 Sept 2018

Freeman H (1970) Boundary encoding and processing. In: Lipkin BS, Rosenfeld A (eds) Picture processing and psychopictorics. Academic, New York, pp 241–266

Goldin GA (2009) Problem solving heuristics, affect, and discrete mathematics: a representational discussion. In: Sriraman B, English L (eds) Theories of mathematics education: seeking new frontiers. Springer, Heidelberg, pp 241–250

Goldin GA (2018) Discrete mathematics and the affective dimension of mathematical learning and engagement. In: Hart E, Sandefur J (eds) Teaching and learning discrete mathematics worldwide: curriculum and research. ICME13 monograph. Springer, Cham, pp 53–65

Golomb SW (1994) Polyominoes – puzzles, patterns, problems and packings. Princeton Science Library, Princeton

Greene D, Yao F (1986) Finite-resolution computational geometry. In: 27th IEEE symposium on foundations of computer science, pp 143–152

Grenier D, Payan C (1999) Discrete mathematics in relation to learning and teaching proof and modelling. In: Schwank I (ed) Proceedings of CERME 1, vol 1. Forschungsinstitut für Mathematikdidaktik, Osnabrück, pp 143–155

Harel G (2009) Commentary on the theoretical, conceptual, and philosophical foundations for research in mathematics education. In: Sriraman B, English L (eds) Theories of mathematics education: seeking new frontiers. Springer, Heidelberg, pp 87–94

Hart E, Sandefur J (eds) (2018) Teaching and learning discrete mathematics worldwide: curriculum and research. ICME13 monograph. Springer, Cham

Heinze A, Anderson I, Reiss K (eds) (2004a) Discrete mathematics and proof in the high school. Zentralbl Didaktik Math 36(2):44–84. and 36(3): 82–116

Heinze A, Anderson I, Reiss K (2004b) Discrete mathematics and proof in the high school – introduction. Zentralbl Didaktik Math 36(2):44–45

Maurer SB (1997) What is discrete mathematics? The many answer. In: Rosenstein JG, Franzblau DS, Roberts FS (eds) Discrete mathematics in the schools. DIMACS series in discrete mathematics & theoretical computer science, vol 36. American Mathematical Society & NCTM, Providence, pp 121–132

Ouvrier-Buffet C (2006) Exploring mathematical definition construction processes. Educ Stud Math 63(3) 259–282

Ouvrier-Buffet C, Meyer A, Modeste S (2018) Discrete mathematics at university level – interfacing mathematics, computer science and arithmetic. In: Durand-Guerrier V, Hochmuth R, Goodchild S, Hogstad NM (eds) Proceedings of INDRUM 2018 – second conference of the international network for didactic research in university mathematics. University of Agder, Kristiansand, pp 255–264

Pham S (1986) Digital straight segments. Comput Vis Graph Image Process 36:10–30

Reveillès JP (1991) Géométrie discrète, calculs en nombres entiers et algorithmique. PhD thesis, Université Louis Pasteur, Strasbourg

Rosenfeld A (1974) Digital straight line segments. IEEE Trans Comput C-23:1264–1269

Rosenstein JG, Franzblau DS, Roberts FS (eds) (1997) Discrete mathematics in the schools. DIMACS series in discrete mathematics & theoretical computer science, vol 36. American Mathematical Society & NCTM, Providence

Website. http://mathsamodeler.ujf-grenoble.fr/. Accessed 28 Sept 2018

Wu LD (1982) On the chaincode of a line. IEEE Trans Pattern Anal Mach Intell 4:347–353

Discursive Approaches to Learning Mathematics

Anna Sfard
Department of Mathematics Education,
University of Haifa, Haifa, Israel

Keywords
Learning · Mathematics · Communication · Discourse · Language

Definition

Discursive approach to learning is a research framework grounded in the view that learning such subjects as mathematics, physics, or history is a communicational activity and should be studied as such. Learning scientists who adopt this approach treat discourse and its development as the primary object of exploration rather than as mere means to the study of something else (e.g., development of mental schemes). The term *discourse* is to be understood here as referring to a well-defined type of multimodal (not just verbal) communicational activity, which does not have to be audible or synchronous.

Background

Ever since human learning became an object of systematic study, researchers have been aware of its intimate relationship with language and, more generally, with the activity of communicating. The basic agreement on the importance of discourse notwithstanding, a range of widely differing opinions have been proposed regarding the way these two activities, learning and communicating, are related. At one end of the spectrum, there is the view that language-related activities play only the secondary role of means to learning; the other extreme belongs to those who look upon discourse as the object of learning. It is this latter position, the one that practically equates mathematic with a certain well-defined form of communicational activity, that can be said to fully reflect a discursive approach to learning.

Several interrelated developments in philosophy, sociology, and psychology combined together to produce this approach. It is probably the postmodern rejection of the notion of "absolute truth," the promise of which fuelled the positivist science, that put human studies on the path toward the "discursive turn." Rather than seeing human knowledge as originating in the nature itself, postmodern thinkers began picturing it as "a kind of discourse" (Lyotard 1979, p. 3) or as a collection of narratives gradually evolving in the "conversation of mankind" (Rorty 1979, p. 389).

Following this foundational overhaul, the interest in discourses began crossing disciplinary boundaries and established itself gradually as a unifying motif of all human sciences, from sociology to anthropology, to psychology, and so forth. Throughout human sciences, the discursivity – the fact that all human activities are either purely discursive or imbued with and shaped by discourses – has been recognized as a hallmark of humanity. Nowhere was this realization more evident than in the relatively young brand of psychology known as "discursive" (Edwards 2005) and defined as "one that takes language and other forms of communication as critical in the possibility of an individual becoming a human being" (Lerman 2001, p. 93). As evidenced by the steadily increasing number of studies dealing with interactions in mathematics classroom, the discursive turn has been taking place also in mathematics education research (Ryve 2011).

Foundations

For many discursively minded researchers, even if not for all, the shift to discourse means that some of those human activities that, so far, were considered as merely "mediated" or "helped" by concomitant discursive actions may now be rethought as being communicational in nature. For example, as an immediate entailment of viewing research as a communicational practice, one can now say that the research discipline known as mathematics is a type of discourse, and thus learning mathematics is a discursive activity as well.

Recognition of the discursive nature of mathematics and its learning, if followed all the way down to its inevitable entailments, inflicts a lethal blow to the famous "Cartesian split," the strict ontological divide between what is going on "inside" the human mind and what is happening "outside." Once thinking, mathematical or any other, is recognized as a discursive activity, mental phenomena lose their ontological distinctiveness and discourse becomes the superordinate category for the "cognitive" and the "communicational." This non-dualist position, which began establishing itself in learning sciences only quite recently, has been implicitly present already in Lev Vygotsky's denial of the separateness of human thought and speech and in Ludwig Wittgenstein's rejection of the idea of "pure thought," the amorphous entity supposed to preserve its identity through a variety of verbal and nonverbal expressions (Wittgenstein 1953).

In spite of the fact that the announcement of the ontological unity of thinking and communicating has been heralded by some writers as the beginning of the "second cognitive revolution" (Harré and Gillett 1995), non-dualism has not become, as yet, a general feature of discursive research. More often than not, discursivist researchers eschew explicit ontological commitments (Ryve 2011), whereas their occasional use of hybrid languages brings confusing messages about the nature of the objects of their study. One can therefore speak about weaker and stronger discursive approaches, with the adjective "strong" signaling the explicit adoption of non-dualist stance (Sfard 2008).

The ontological heterogeneity notwithstanding, all discursively oriented researchers seem to endorse Vygotsky's (1978) famous statement that uniquely human learning originates on the "social plane" rather than directly in the world. Consequently, they also view learning as a collective endeavor and recognize the need to always consider its broad social, historical, cultural, and situational context. Strong discursivists, in addition, are likely to claim that objects of discourses – *numbers* or *functions* in the case of mathematics and *conceptions* or *meanings* in the case of researcher's own discourse – grow out of communication rather than signifying any self-sustained entities preexisting the discourse about them. As a consequence, the researchers always keep in mind that any statement on the existence or nature of these entities is a matter of personal interpretation and must be presented as such. Moreover, since the protagonists of researchers' stories are themselves active storytellers, researchers must always inquire about the status of their own narratives vis-à-vis those offered by the participants of their study.

Strands

The current discursive research on learning at large and on mathematics learning in particular may be roughly divided into three main strands, according to perspectives adopted, aspects considered, and questions asked. The first two of these distinct lines of research are concerned with different features of the discourse under investigation and can thus be called intra-discursive or inward looking. The third one deals with the question of what happens between discourses or, more precisely, how inter-discursive relations impact learning.

The first intra-discursively oriented strand of research on mathematics learning focuses on learning-teaching interactions, whereas its main interest is in the impact of these interactions on the course and outcomes of learning. Today, when *inquiry learning*, *collaborative learning*, *computer-supported collaborative learning*, and other conversation-intensive pedagogies (also known as "dialogical") become increasingly popular, one of the main questions asked by researchers is that of what features of small group and whole-class interactions make these interactions conducive to high-quality learning. *Participation structure*, *mediation*, *scaffolding*, and *social norms* are among the most frequently used terms in which researchers formulate their responses. Whereas there is no doubt about theoretical and practical importance of this strand of research, some critics warn against the tendency of this kind of studies for being unhelpfully generic, which is what happens when findings regarding patterns of learning-teaching interactions are presented as if they were independent of their topic.

This criticism is no longer in force in the second intra-discursively oriented line of research on mathematics learning, which inquires about the development of mathematical discourse and thus looks on those of its features that make it into distinctly mathematical: the use of specialized mathematical words and visual mediators, specifically mathematical routines, and narratives about mathematical objects that the participants endorse as "true." Comparable in its aims to research conducted within the tradition of conceptual change, this relatively new type of study on learning is made distinct by its use of methods of discourse analysis, and this means, among others, its attention to contextual issues, its sensitivity to the inherent situatedness of learning, and its treatment of the discourse in its entirety as the unit of analysis, rather than restricting the focus to a single concept. Questions asked within this strand include queries about ways in which learners construct *mathematical objects*, develop *socio-mathematical norms*, engage in *argumentation*, or cope with uneasy transitions to *incommensurable discourses*. Methods of systemic functional linguistics (Halliday 2003) are often employed in this kind of study. One of the main tasks yet to be dealt with is to forge subject-specific methods of discourse analysis, tailored according to the distinct needs of the discourse under study. Another is to explore the possibility of improving school learning by overcoming its situatedness. Yet another regards the question of how mathematical learning occurring as if of itself while people are dealing with their daily affairs differs from the one that takes place in schools and results from teaching.

Finally, the inter-discursively oriented studies inquire about interactions between discourses and their impact on learning. This type of research is grounded in the recognition of the fact that one's participation in mathematics discourse may be supported or inhibited by other discourses. Of particular significance among these learning-shaping aspects of communication are those that pertain to specific cultural norms and values or to distinct ideologies. Studies belonging to this strand are often concerned with issues of *power*, *oppression*, *equity*, *social justice*, and *race*, whereas the majority of researchers whom this research brings together do not hesitate to openly admit their ideological involvement. The notion of *identity* is frequently used here as the conceptual device with which to describe the way cultural, political, and historical narratives impinge upon individual learning. Methods of critical discourse analysis (Fairclough 2010) are particularly useful in this kind of study.

Methods

As different as these three lines of research on learning may be in terms of their focus and goals, their methods have some important features in common. In all three cases, the basic type of data is the carefully transcribed communicational event. A number of widely shared principles guide the processes of collection, documentation, and analysis of such data. Above all, researchers need to keep in mind that different people may be using the same linguistic means differently and that in order to be able to interpret other person's communicational actions, the analysts have to alternate between being insiders and outsiders to their own discourse: they must sometimes look "through" the word to what they usually mean by it, and they also must be able to ignore the word's familiar use, trying to consider alternative interpretations. For the same reason, events under study have to be recorded and documented in their entirety, with transcriptions being as accurate and complete records of participants' verbal and nonverbal actions as possible. Finally, to be able to generalize their findings in a cogent way, researchers should try to support qualitative discourse analysis with quantitative data regarding relative frequencies of different discursive phenomena.

The admittedly demanding methods of discourse analysis, when at their best, allow the analyst to see what inevitably escapes one's attention in real-time conversations. The resulting picture of learning is characterized by high resolution: one can now see as different things or situations that, so far, seemed to be identical and is able to perceive as rational those discursive actions that in real-time exchange appeared as nonsensical.

Cross-References

- ▶ Argumentation in Mathematics
- ▶ Collaborative Learning in Mathematics Education
- ▶ Equity and Access in Mathematics Education
- ▶ Inquiry-Based Mathematics Education
- ▶ Mathematics Teacher Identity
- ▶ Scaffolding in Mathematics Education
- ▶ Sociomathematical Norms in Mathematics Education

References

Edwards D (2005) Discursive psychology. In: Sanders RE, Fitch KL (eds) Handbook of language and social interaction. Routledge, London, pp 257–273

Fairclough N (2010) Critical discourse analysis: the critical study of language, 2nd edn. Longman, Harlow

Halliday MAK (2003) On language and linguistics. Continuum, London

Harré R, Gillett G (1995) The discursive mind. Sage, Thousand Oaks

Lerman S (2001) Cultural, discursive psychology: a sociocultural approach to studying the teaching and learning of mathematics. Educ Stud Math 46:87–113

Lyotard J-F (1979) The postmodern condition: a report on knowledge. University of Minnesota Press, Minneapolis

Rorty R (1979) Philosophy and the mirror of nature. Princeton University Press, Princeton

Ryve A (2011) Discourse research in mathematics education: a critical evaluation of 108 journal articles. J Res Math Educ 42(2):167–198

Sfard A (2008) Thinking as communicating: human development, the growth of discourses, and mathematizing. Cambridge University Press, Cambridge

Vygotsky LS (1978) Mind in society: the development of higher psychological processes. Harvard University Press, Cambridge, MA

Wittgenstein L (1953/2003) Philosophical investigations: the German text, with a revised English translation (trans: Anscombe GEM), 3rd edn. Blackwell, Malden

Documentational Approach to Didactics

Luc Trouche[1], Ghislaine Gueudet[2] and Birgit Pepin[3]

[1]French Institute of Education, Ecole Normale Supérieure de Lyon, Lyon, France

[2]CREAD, ESPE de Bretagne, University of Brest, Rennes, France

[3]Eindhoven School of Education (ESoE), Technische Universiteit Eindhoven, Eindhoven, The Netherlands

D

Keywords

Curriculum materials · Digital resources · Documentational geneses · Operational invariants · Resource systems · Resources for teaching · Teachers' collective work · Teacher professional development

Introduction

Mathematics teachers interact with curriculum and other resources in their daily work, and their work with resources includes selecting, modifying, and creating new resources, in-class and out-of-class. This creative work is termed *teacher documentation work*, and its outcomes *teacher documentation*.

Typical curricular resources include text resources (e.g., textbooks, curricular guidelines, student worksheets), or digital curriculum resources (e.g., digital interactive textbooks). However, as there is now nearly unlimited access to resources on the web, teachers are often at a loss to choose the most didactically and qualitatively suitable resources for their mathematics teaching. Hence, the study of resources and mathematics teachers' interaction/work with those resources has become a prominent field of research (e.g., Pepin et al. 2013), not least because curriculum reforms in many countries go through the provision of reform oriented curriculum materials that are seen to help teachers enact the curriculum suitably and aligned with the reforms.

In theoretical terms, the work of teachers with curriculum resources has been studied from many angles and theoretical perspectives (e.g., Remillard 2005; Pepin et al. 2013), for example in the Anglo/American research literature through the notion of 'enacted curriculum'. In continental Europe, the notion of "*Didaktik*" is a common concept (e.g., Pepin et al. 2013). This entry describes, explains and illustrates the *Documentational Approach to Didactics* (DAD), which has its roots in French European Didactics.

Sources of the Approach

The documentational approach to didactics (DAD) has been introduced by Ghislaine Gueudet and Luc Trouche (2009) and has been developed further in joint work with Birgit Pepin (Gueudet et al. 2012). DAD is originally steeped in the French didactics tradition in mathematics (Trouche 2016), where concepts such as *didactical situation*, *institutional constraint*, and *scheme* are central. At the same time, it also leans on socio-cultural theory, including notions such as *mediation* (Vygotsky 1978) as a constitutive of each cognitive process. Moreover, the approach has also been developed due to the emerging digitalization of information and communication, which asks for new theoretical approaches.

The digitalization of information and communication and the development of Internet had indeed strong consequences: ease of quick access to many resources and of communication with many people. This necessitated a complete metamorphosis of thinking and acting, particularly in education: new balances between *static* and *dynamic* resources, between *using* and *designing* teaching resources, between *individual* and *collective* work (Pepin et al. 2017a). Taking into account these phenomena, DAD proposed a change of paradigm by analyzing teachers' work through the lens of "resources" for and in teaching: what they prepare for supporting their classroom practices and what is continuously renewed by/in these practices.

In addition to the French didactics tradition, the authors drew their inspiration from several main interrelated theoretical sources: the *field of technology use*, the field of *resources and curriculum design*, the field of *teacher professional learning/development*, the field of *information architecture*, and the field of *communities of practice*.

In the field of technology use, the central source for DAD was the *instrumental approach*. This theory has been developed by Rabardel (see Rabardel and Bourmaud 2003) in cognitive ergonomics and then integrated into mathematics didactics (Guin et al. 2005). It distinguishes between an *artifact*, available for a given user, and an *instrument*, which is developed by the user. Connected notions are those of *genesis*, *instrumentation*, and *instrumentalization* – these are also essential components of DAD (see section "The Documentational Approach to Didactics: A Holistic Approach to Teachers' Work"). The development of the instrumental approach corresponded to a period where teachers were facing the integration of new *singular* tools (a calculator, a computer algebra software, a dynamic geometry system...). It became clear that the instrumentatal approach was not sufficient, as teachers were often surrounded (via Internet) by a profusion and variety of *resources*.

This sensitivity to *resources* meets Adler's (2000) proposition of "think[ing] of a resource as the verb re-source, to source again or differently" (p. 207). Retaining this point of view, DAD took into consideration a wide spectrum of "resources" that have the potential to resource teacher activity (e.g., textbooks, digital resources, emails exchanged with colleagues, students' sheets), resources *speaking to the teacher* (Remillard 2005) and supporting her/his engagement in teaching.

This broad view on resources leads to a broad view on teacher professional learning. As Ball et al. (2005) stated in their study of "mathematical knowledge for teaching," teaching is not reduced to the work in class, but also includes planning, evaluating, writing assessments, discussing with parents, etc. DAD considers this: looking at teachers' work through their interactions with resources, and (following Cooney 1999) acknowledging that change of practice and change of professional knowledge or beliefs are connected (in a specific manner, as explained in Section 3).

Considering resources as the matter feeding teachers' work, a word was needed for naming what a teacher develops from these resources. The word *document* was retained: it had already been used in the field of information architecture (Salaün 2012) for designing 'something bearing an intention', and dedicated to a given usage in a given context.

Finally, the easiness to communicate via the Internet leads this approach to take into account the emergence of a spectrum of various forms of teachers' collective work: networks, online association, communities more, or less formal. Wenger's (1998) theory of *communities of practice,* and its concepts of *participation*, *negotiation*, *reification*, appeared as particularly fruitful for analyzing the design of teaching resources by collectives of teachers as a process of professional development.

Once described the sources of this theoretical approach, its structure and core concepts are presented in the following section.

The Documentational Approach to Didactics: A Holistic Approach to Teachers' Work

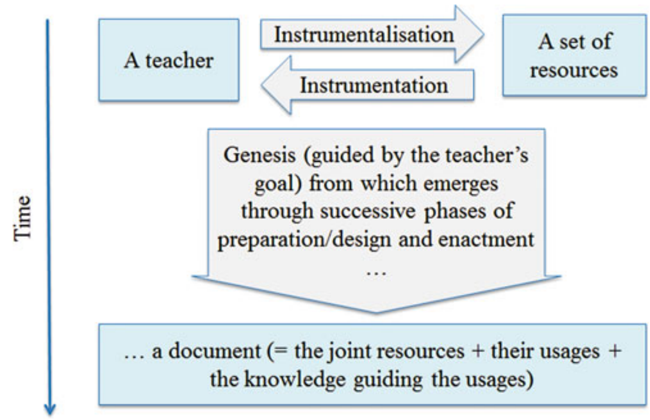
In this section, the 'ingredients' of the DAD, and the processes involved, are described. The following terms are defined: resources, documents, genesis, instrumentation, and instrumentalization.

In terms of "resource," we lean on a definition by Pepin and Gueudet (Entry ► Curriculum Resources and Textbooks in Mathematics Education) who refer to *mathematics curriculum resources* as all the resources (e.g., digital interactive, nondigital/traditional text) that are developed and used by teachers and pupils in their interaction with mathematics in/for teaching and learning, inside and outside the classroom. This also includes digital curriculum resources (Pepin et al. 2017a), and Pepin and Gueudet (Entry ► Curriculum Resources and Textbooks in Mathematics Education) make a distinction between digital curriculum resources and educational technology. They also distinguish between *material curriculum resources* (e.g., textbooks, digital curriculum resources, manipulatives, and calculators), *social resources* (e.g., a conversation on the web/forum), and *cognitive resources* (e.g., frameworks used to work with teachers). DAD has been mostly applied to teachers' work, but can also be used to study the work of teacher educators (e.g. Psycharis and Kalogeria online), or students' interaction with resources (e.g. Kock and Pepin 2018).

In terms of processes, during the interaction with a particular resource, or sets of resources, teachers develop their particular *schemes of usage* of these resource (see section "Deepening the Approach: Schemes and Systems" below). These are likely to be different for different teachers, although they may use the same resource, depending on, for example, their dispositions and knowledge. The outcome is the *document*, hence:

$$Resources + scheme\ of\ usage = document$$

The process of developing the document (including the teacher learning involved) has been coined *documentational genesis* (e.g., Gueudet and Trouche 2009). Pepin, Gueudet, and Trouche (2013) have provided theoretical perspectives on "re-sourcing teachers' work and interactions," and the documentational approach is particularly pertinent to viewing the "use" of resources as an interactive and potentially transformative process. This process works both ways: the affordances of the resource/s influence teachers' practice (that is the *instrumentation* process), as the teachers' dispositions and knowledge guide the choices and transformation processes between different resources (i.e., the *instrumentalization* process) (Fig. 1). Hence, the DAD emphasizes the dialectic nature of the teacher-resource interactions combining instrumentation and instrumentalization (Rabardel and Bourmaud 2003). These processes include the design, re-design, or "design-in-use" practices (where teachers change a document "in the moment" and according to their instructional needs).



Documentational Approach to Didactics, Fig. 1 A representation of a documentational genesis

The DAD proposes a model of the interactions between teachers and resources, which has implications for teachers' professional learning. Whilst there is an immensity of potentially suitable materials on the web, the web does not provide suitable support for relevant searching, which would be necessary if one wishes to search for particular (perhaps interactive) learning resources that combine with other (e.g., textbook) resources in their subtle epistemic or didactic features. In other words what is provided for teachers is often "a pile of bricks," without being given guidance on how these bricks might be put together to develop a coherent student's learning trajectory. Whether searching for tasks to supplement a given learning sequence, or planning learning paths through a flexible e-textbook, teachers will require professional support to help them develop *teacher design capacity* (Pepin et al. 2017b) – a mindfulness/sensitivity of mathematical and pedagogical aspects of learning resources and the flexibility to use them (see Window 1). This is in line with the DAD, and Wang (2018) regards teacher design capacity as part of teachers' *documentational expertise.*

Window 1 Resources at Secondary School: The Example of Vera's Documentation Work
Vera is one of the many mathematics school teachers working with Sésamath in France (Gueudet et al. 2013; Pepin et al. 2017b).

(continued)

Sésamath is an association of secondary school mathematics teachers in France, whose members have (since 2001) designed, and they freely offer, interactive e-textbooks on their website (http://www.sesamath.net/). Veras's documentation work for a new lesson is analyzed (Gueudet et al. 2013): it was the first time that Vera taught a grade 8 class on percentages, and she used a diversity of resources for this lesson, including Sésamath resources.

The analyzis focuses on a lesson cycle: her lesson preparation, the enactment of the lesson, the evaluation of students' understanding, and reflection. The choice of such a lesson cycle is in line with the ideas underpinning the DAD: the design was not restricted to the initial design of a given resource for teaching a particular content, but continued during the course of using the resource. The design work of Vera included, for example, the use of LaboMEP (a Sésamath tool) to propose different exercises to different students – this made her aware of the need to differentiate her teaching. LaboMEP also proposed variations of exercises with the same structure. Vera declared that it was a strong motivation for her to enhance her teaching, by mastering not only a set of familiar

(continued)

exercises but also those variations (with the same mathematical structure) related to a particular lesson.

These results are interpreted as evidencing an enhancement of her didactical flexibility, in other words the development of her design capacity when selecting, utilizing and transforming existing curricular resources effectively, designing/creating new materials, for the purpose of effective mathematics instruction.

Teaching is often regarded as design. This is in line with Brown (2009) who explains that the interpretation of teaching as design, and the notion of teachers as designers, is fitting with a range of cognitive theories that "emphasize the vital partnership that exists between individuals and the tools they use to accomplish their goals. . . . And it is not just the capacities of individuals that dictate human accomplishment, but also the affordances of the artifacts they use" (p. 19). Hence, Brown (2009) sees this relationship in the same way, as an interrelationship: that is, the activity of "designing" is not only dependent on the teacher's competence, but that it is an interrelationship between the teacher/s and the (curriculum) resources, the "teacher-tool relationship," that is at play here, and hence the affordances of the curriculum resources influence this relationship. This is in line with the DAD, emphasizing that any understanding of teacher as designer must include a conscious/deliberate act of designing, of creating "something new" (e.g., combining existing and novel elements) in order to reach a certain (didactical) aim. This is provided and supported by *schemes of usage*, as defined in the following section.

Deepening the Approach: Schemes and Systems

The concept of scheme (Vergnaud 1998) is central in the DAD. It is closely linked with the concept of class of situations that is in our context a set of professional situations corresponding to the same aim of the activity. For example, "managing the heterogeneity of the grade 8 class" is a class of situations for Vera, in the example given above. For a given class of situations, a subject (here a teacher) develops a stable organization of his/her activity, that is a scheme. A scheme has four components:

– The aim of the activity (the aim characterizing the class of situations)
– Rules of action, of retreiving information and of control
– Operational invariants, which are knowledge of two (associated) kinds: *theorem-in-action*, proposition considered as true, and *concept-in action*, a concept considered as relevant (see example below)
– Possibilities of inferences, of adaptation to the variety of situations

Over the course of his/her activity, the teacher can enrich his/her schemes, integrating new rules of actions, or she/he can develop new schemes: the scheme offers in fact a model for analyzing learning. In the DAD, the schemes that are considered are schemes of usage of a given resource (or set of resources). The resources and the scheme make up a document (as summarized by the equation provided in the previous section).

The set formed by all the resources used by the teacher is named his/her *resource system*. These resources are associated with schemes of usage, forming documents (the same resource can intervene in several documents). The documents developed by a teacher also form a system, called the *document system* of the teacher. Its structure follows the structure of the situations classes composing the professional activity of the teacher, according to the different aims of his/her activity.

When teachers share documentation work, for example, in a group preparing lessons collectively, they may also develop a shared resource system (Gueudet et al. 2012). Nevertheless, the different members of the group can develop different schemes for the same resource, resulting in different documents.

Window 2 presents a case of a resource at primary school, in order to illustrate/exemplify operational invariants, resource systems and document systems.

Window 2 Resources at Primary School: The Example of the Virtual Abacus

The virtual abacus (Fig. 2) is a free software developed in France by Sésamath, an association of mathematics teachers designing online resources (see Window 1).

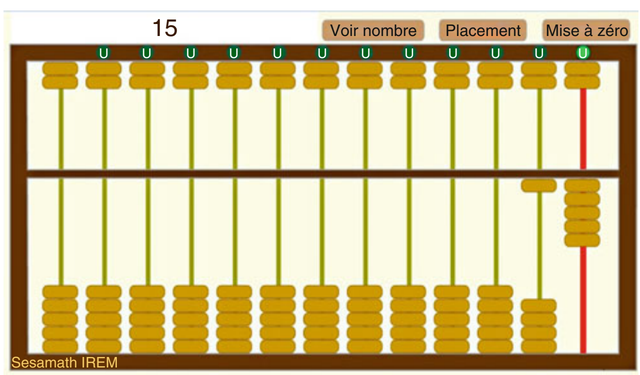
The Chinese abacus is separated in two parts by a central bar, called "the reading bar": only the beads on this bar are considered as "activated." There are two kinds of beads: 5-unit beads (two of them) and 1-unit beads (five of them). The Chinese abacus comprises 13 vertical rods. Each rod corresponds to a rank of the place-value system: units, tens, hundreds, etc. (from the right to the left). There are several possibilities to display the same number on the Chinese abacus: for example, 15 is represented on the abacus above using six beads (1-unit beads, one on the tenth rod and five on the unit rod); it can also be represented using only two beads, by replacing the five 1-unit beads on the right by a 5-units bead on the same rod.

Carlos is an experimented primary school teacher who has been "shadowed" over three years (Poisard et al. 2011). He has decided to use the abacus for his

(*continued*)

teaching of numbers with his grade 3 class. He has seen in his mathematics textbook an activity with the abacus, but did not want to use it in class before discovering the virtual abacus. He started with material abacus, the students manipulated them and formulated hypotheses on the way it works. Then they worked on the virtual abacus and wrote instructions for its use. After that Carlos proposed exercises: inscribing a given number on the abacus, reading a number inscribed on the abacus. For the final assessment, such exercises were given on paper, to avoid a strategy of trial and error possible on the software.

Carlos developed several documents incorporating the virtual abacus and other associated resources (Poisard et al. 2011). For the aim "Discover how the abacus works," he used both material abaci and the virtual abacus and asked the students to compose posters. It was important for him to let his students discover by themselves the principles of the abacus. This corresponds to an operational invariant: a theorem-in-action like "the students must discover by themselves a far as possible the new tools they meet." It is also linked to the associated concept-in-action: "self-discovery," and both were developed by Carlos before meeting the virtual abacus. Another operational invariant intervened in his choices: "it is important for students in grade 3 to manipulate material resources." The scheme for the aim



Documentational Approach to Didactics, Fig. 2 The virtual abacus "discover how the abacus works" comprises these operational invariants and associated rules of actions: "propose material abaci to the students to afford manipulation" and "propose the virtual abacus to the students to allow them to check which number is displayed."

Along his work with the abacus, Carlos observed that for the exercise: "display a given number on the abacus," if the students use the virtual abacus, they develop trial and error strategies using the button "display number." Hence, he decided to do a final assessment on paper. He developed a new document, for the aim: "teach the students how to display a number on the abacus." This document incorporates the virtual abacus, but also abaci drawn on paper, and an operational invariant like "on the virtual abacus students can use trial and error strategies."

Carlos is an experienced primary school teacher. For the teaching of numbers in grade 3, he had for many years developed a (sub-)system of documents. Some of these documents correspond to aims directly linked with the abacus, like "discover how the abacus works," "teach the students how to display a number on the abacus." In other documents, the abacus did not appear in the aim, but was nevertheless used for this aim. For example, for the aim "teach the principles of the base ten place-value system," he used the abacus to evidence the "grouping and exchanging" principles (like grouping two 5-unit beads on a rod and exchanging it with a 1-unit bead on the next rod). Other resources in his resource system intervened in these documents, like posters written by the students. Some of these resources have been decisive in his choice to use the abacus: the textbook in particular, which offered him a first opportunity to meet a possible use of the abacus in class.

The whole document system of a teacher comprises many subsystems. It can be described at different levels, ranging from a very general view of the activity to a very specific focus on a given mathematical content. It is possible to consider, for example, that a teacher develops a single document for the aim "assess the students' skills." For research in mathematics education, the more specific levels, taking into account the mathematics content (the aim could be then "assess the students' skills on percentages in grade 8"), are more informative about teachers' interactions with resources and their consequences. The DAD claims that analyzing teachers' documentation work requires specific methodology, which is the purpose of the following section.

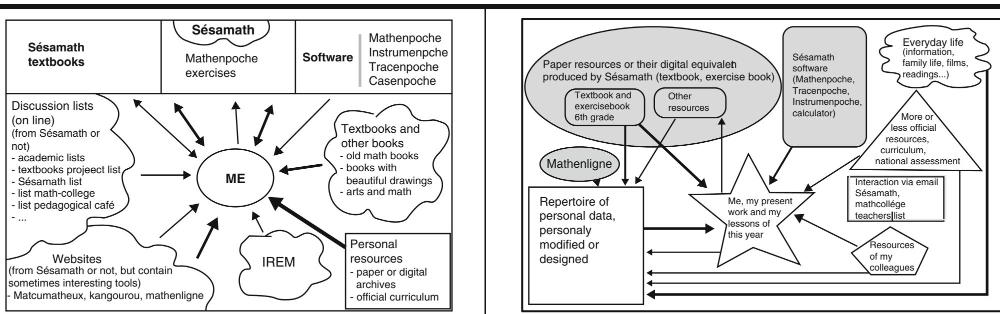
D

Reflexive Investigation: A Developing Methodological Construct

This section presents the research design typically linked to the DAD; the principles grounding this design; then describes one tool illustrating these principles; finally presents some issues that the reflective investigation methodology is facing. Analyzing teachers' activity through their documentation work requires to take into account the followings: the variety of resources feeding, and produced by, this work; the variety of interactions (collective, institutional as well as social) influencing this work; and the time for developing documentational geneses. These epistemological considerations lead us (Gueudet et al. 2012, p. 27) to develop a specific methodology, that we named *reflective investigation of teachers documentation work*. We present in this section the principles grounding this design; then we describe one tool incarnating these principles; finally we present some issues that the reflective investigation methodology is facing.

This methodology gives a major role to teachers themselves, and it is underpinned by five main principles:

- The principle of *broad collection of the material resources* used and produced in the documentation work, throughout the follow-up.
- The principle of *long-term follow-up*. Geneses are ongoing processes and schemes develop over long periods of time,
- The principle of *in- and out-of-class follow-up*. The classroom is an important place where the teaching elaborated is implemented, bringing adaptations, revisions, and improvisations. However, an important part of teachers' work takes place beyond the students' presence – at



At the top of this representation, appear Sésamath resources: textbooks, exercises (Mathenpoche) and software. Different arrows allow distinguishing different types of activity: thick for preparing lessons, medium for preparing exercises, thin for preparing activities. Most of the arrows concern activities (i.e. problem solving, open ended questions, which constitute the heart of Pierre's teaching).

The reflective follow-up over one year makes Pierre more aware (his words) of his colleagues as part of "the sources of his resources."

Pierre also distances himself from Sésamath resources (his words), appearing no more as his own horizon: through his new SRRS, appear new inspiring resources (films, readings, . . .).

Pierre is a French mathematics middle school teacher strongly involved in collective documentation work, particularly in Sésamath, a teacher association designing and making freely available, at a large scale, online textbooks and tools (see Window 1). He was asked to draw his RSSR in February 2009 (left drawing), then in February 2010 (right drawing)

Documentational Approach to Didactics, Fig. 3 A teacher's SRRS (Gueudet et al. 2012, p. 314 and 318)

school, at home, in teacher development programs, etc.

- The principle of *reflective follow-up* of the documentation work.
- The principle of *permanent confronting the teacher' views on her documentation work, and the materiality of this work* (materiality coming, for example, from the collection of material resources; from the teacher's practices in her classrooms).

The active involvement of the teacher is a practical necessity, as she/he is the one having access to the teacher's documentation work (beyond the direct observation of the researcher). It also yields a reflective stance leading the teacher to an introspective attitude, sometimes making visible what could be hidden resources, or hidden links within his/her resource system. This long and close follow-up of teachers' documentation work needs to make clear, as far as possible, what the researcher is asking for, and for which purpose. This necessity leads Sabra (2016) to propose the notion of *methodological contract*, linking a teacher and a researcher following his/her documentation work.

Based on these principles, selected data collection strategies and tools were designed, adapted to the various contexts and research questions. For example, a tool seemingly fruitful is the *schematic representation of a teacher's resource system* (SRRS, see Fig. 3). The teacher is asked to draw a map of her resources, evidencing the resources s/he had identified and appropriated, from which repositories, and for which purpose (e.g. Pepin et al. 2016).

Since the beginning of DAD, this tool has been developed in several directions:

- Hammoud (2012), working in chemistry education, proposed an approach for analyzing SRRS as mind maps; she used also these schematic representations for asking teachers to describe their interactions with colleagues, or within different collectives.
- Rocha (2018) renamed SRRS as "Reflective mapping of teacher resource system" (RMRS), for two reasons: underlining the role of reflectivity; denoting, with the word "mapping" a process of progressive exploration of an unknown (for the researcher, but also in some points for the teacher herself) territory. Rocha (2018), following the fifth principle (see above) confront also permanently RMRS, made by the teacher, and what she names "Inferred mapping of teacher resource system" made by the researcher herself.

Beyond this specific tool, new methodological developments of the reflective investigation have occurred, to access, as close as possible, the "true" teacher's documentation work at the time, when it happens. It means observing a teacher during episodes of interacting with resources, not only in public episodes, as in classrooms, both also during more intimate episodes. For example: preparing a progression for the year; preparing a lesson; revising it. A video-recorded follow-up of these successive episodes has been experienced (Bellemain and Trouche 2016), giving access to gestures and words of a teacher in the course of her documentation work, allowing to infer elements of schemes. This kind of follow-up raises some difficulties, among them the following:

- Under which "natural" conditions could a teacher work alone with resources, and at the same time describe the rationale of her activity? Wang (2018) introduces the notion of teacher's *documentation-working mate*, meaning a teacher regularly sharing the documentation work of the targeted teacher. The follow-up of the pair of teachers working together gave access to their mutual explanations and then to a part of the teacher's knowledge guiding her documentation work.
- How is it possible to store (for analysis purposes) the heterogeneous and numerous data resulting from the follow-up of teachers' documentation work? This issue is addressed by the development of *webdocuments* (Bellemain and Trouche 2016), storing videos as well as resources intervening, or produced by teachers' documentation work, for subsequent analysis and sharing of both the data and their analyses within a research community.

Analyzing teachers' documentation work is a complex process. The five principles of the reflective investigation methodology provide guidelines for methodological choices. Researchers using these principles need to make motivated choices for limiting the abundance of data: choices of critical moments for teachers' documentation work (see the notion of *documentational incidents*, Sabra 2016) or critical resources for a teacher's resource system (see the notion of *pivotal resource*, Gueudet 2017). This work is in progress, the methodological and conceptual needs being strongly interrelated.

Perspectives for Further Evolutions

D

The documentational approach to didactics is a recent theoretical framework in mathematics education. While the concepts presented in sections "Sources of the Approach" and "The Documentational Approach to Didactics: A Holistic Approach to Teachers' Work": resources, documents, documentational geneses, are now well established, more elaborated concepts like resource system and document systems are still evolving. If we consider, for example, secondary school mathematics teachers, can we observe different types of structures for their document systems, linked with some professional profiles of the teachers? Do some groups of teachers (communities of practice in particular) share collective document systems?

Moreover, the fields of application of the DAD have evolved over the years. In terms of school level, they now range from preschool (Besnier and Gueudet 2016) to university (Gueudet 2017) and also consider the work of teacher educators with resources (Psycharis and Kalogeria online). In terms of disciplines, the documentational approach has been used in experimental sciences like physics and chemistry (Hammoud 2012) and also in language education (Quéré 2017). What are the specificities of teachers' documentation work and documentation systems in these new contexts?

The use of DAD in various social and cultural contexts (e.g., Brazil, China, Lebanon, Norway, Senegal. . .) also opens up questions arising from the different naming systems used by teachers in their daily documentation work. This diversity is the result of historical, social, and cultural context in which teachers' work takes place. Research investigating these questions could lead to a better understanding to a more nuanced view of the nature of teachers' interactions with resources and to a deepening of the DAD related concepts.

In a recent study, the work of students with resources has been investigated (e.g., Kock and Pepin 2018). Other studies (e.g., Gueudet and Pepin 2018) have suggested possible links with other theories. The theory of didactical situations (TDS), for example, introduces the concept of *milieu* which includes all the objects with which the individual student interacts in a mathematical situation. These objects can be considered as resources. What are the consequences of such theoretical links?

As the range of teaching and learning phenomena studied with this approach increases, the concepts and methods also develop. This is evident in numerous conference presentations, publications, and specialized conferences (e.g., Re(s)sources 2018 International Conference https://resources-2018.sciencesconf.org/, Gitirana et al. 2018; Trouche et al. 2019) using DAD. These studies and activities constitute milestones in the ongoing development of a living theoretical framework.

Cross-References

▶ Activity Theory in Mathematics Education
▶ Communities of Practice in Mathematics Teacher Education
▶ Curriculum Resources and Textbooks in Mathematics Education
▶ Education of Mathematics Teacher Educators
▶ Frameworks for Conceptualizing Mathematics Teacher Knowledge
▶ Information and Communication Technology (ICT) Affordances in Mathematics Education
▶ Instrumentalization in Mathematics Education
▶ Instrumentation in Mathematics Education
▶ Lesson Study in Mathematics Education
▶ Mathematics Teacher as Learner
▶ Mathematics Teacher Education Organization, Curriculum, and Outcomes
▶ Mathematics Teacher Educator as Learner
▶ Mathematics Teachers and Curricula
▶ Pedagogical Content Knowledge Within "Mathematical Knowledge for Teaching"
▶ Professional Learning Communities in Mathematics Education
▶ Reflective Practitioner in Mathematics Education
▶ Teacher Beliefs, Attitudes, and Self-Efficacy in Mathematics Education
▶ Teaching Practices in Digital Environments
▶ Technology and Curricula in Mathematics Education
▶ Subject Matter Knowledge Within "Mathematical Knowledge for Teaching"

References

Adler J (2000) Conceptualising resources as a theme for teacher education. J Math Teach Educ 3:205–224

Ball DL, Hill HC, Bass H (2005) Knowing mathematics for teaching. Who knows mathematics well enough to teach third grade, and how can we decide? Am Educ 30(3):14–17, 20–22, 43–46

Bellemain F, Trouche L (2016) Comprendre le travail des professeurs avec les ressources de leur enseignement, un questionnement didactique et informatique, conférence invitée au premier symposium latino-américain de didactique des mathématiques, Bonito, 3 novembre. https://drive.google.com/file/d/0B6OphkgfrkD3ZFRtTDJ2anRfSWM/view

Besnier S, Gueudet G (2016) Usages de ressources numériques pour l'enseignement des mathématiques en maternelle: orchestrations et documents. Perspect Educação Matemática 9(21):978–1003. Repéré à http://seer.ufms.br/index.php/pedmat/article/view/2215/2279

Brown MW (2009) The teacher-tool relationship: theorizing the design and use of curriculum materials. In: Remillard JT, Herbel-Eisenmann BA, Lloyd GM (eds) Mathematics teachers at work: connecting curriculum materials and classroom instruction. Routledge, New York, pp 17–36

Cooney TJ (1999) Conceptualizing teachers' ways of knowing. Educ Stud Math 38:163–187

Gitirana V, Miyakawa T, Rafalska M, Soury-Lavergne S, Trouche L (eds) (2018) Proceedings of the Re(s)sources 2018 international conference. ENS de Lyon, retrieved at https://hal.archives-ouvertes.fr/hal-01764563

Gueudet G (2017) University teachers' resources systems and documents. Int J Res Undergrad Math Educ 3(1):198–224

Gueudet G, Pepin B (2018) Didactic contract at the beginning of university: a focus on resources and their use. Int J Res Undergrad Math Educ 4(1):56–73

Gueudet G, Trouche L (2009) Towards new documentation systems for mathematics teachers? Educ Stud Math 71(3):199–218

Gueudet G, Pepin B, Trouche L (eds) (2012) From text to 'lived' resources: mathematics curriculum materials and teacher development. Springer, New York

Gueudet G, Pepin B, Trouche L (2013) Collective work with resources: an essential dimension for teacher documentation. ZDM 45(7):1003–1016

Guin D, Ruthven K, Trouche L (eds) (2005) The didactical challenge of symbolic calculators: turning a

computational device into a mathematical instrument. Springer, New York
Hammoud R (2012) Le travail collectif des professeurs en chimie comme levier pour la mise en œuvre de démarches d'investigation et le développement des connaissances professionnelles. Contribution au développement de l'approche documentaire du didactique. PhD, Lyon 1 University. https://tel.archives-ouvertes.fr/tel-00762964
Kock Z-J, Pepin B (2018) Student use of resources in calculus and linear algebra. In: INDRUM conference, Kristiansand, Apr 2018
Pepin B, Gueudet G, Trouche L (2013) Re-sourcing teachers' work and interactions: a collective perspective on resources, their use and transformations. ZDM 45(7):929–944
Pepin B, Choppin J, Ruthven K, Sinclair N (2017a) Digital curriculum resources in mathematics education: foundations for change. ZDM 49(5):645–661
Pepin B, Gueudet G, Trouche L (2017b) Refining teacher design capacity: mathematics teachers' interactions with digital curriculum resources. ZDM 49(5): 799–812, https://rdcu.be/tmXb
Poisard C, Bueno-Ravel L, Gueudet G (2011) Comprendre l'intégration de ressources technologiques en mathématiques par des professeurs des écoles. Rech Didactique Mathématiques 31(2):151–189
Psycharis G, Kalogeria E (online) Studying the process of becoming a teacher educator in technology-enhanced mathematics. J Math Teach Educ. https://link.springer.com/article/10.1007/s10857-017-9371-5
Quéré N (2017) Collective design and sharing of open educational resources and teachers' professional development: a study of the French CARTOUN platform. Communication présentée au colloque international European Association for Computer Assisted Language Learning (EuroCALL). Southampton, 23–26 août
Rabardel P, Bourmaud G (2003) From computer to instrument system: a developmental perspective. In: Rabardel P, Waern Y (eds) Special issue "From computer artifact to mediated activity", part 1: organisational issues. Interact Comput 15(5):665–691
Remillard JT (2005) Examining key concepts in research on teachers' use of mathematics curricula. Rev Educ Res 75(2):211–246
Rocha KDM (2018) Uses of online resources and documentational trajectories: the case of Sésamath. In: Fan L, Trouche L, Rezat S, Qi C, Visnovska J (eds) Research on mathematics textbooks and teachers' resources: advances and issues. Springer, Cham, pp 235–258
Sabra H (2016) L'étude des rapports entre documentations individuelle et collective: incidents, connaissances et ressources mathématiques. Rech Didactique Mathématiques 36(1):49–95
Salaün J-M (2012) Vu, lu, su. Les architectes de l'information face à l'oligopole du Web. La Découverte, Paris
Trouche L (2016) Didactics of mathematics: concepts, roots, interactions and dynamics from France. In: Monaghan J, Trouche L, Borwein JM (eds) Tools and mathematics, instruments for learning. Springer, New York, pp 219–256
Trouche L, Gueudet G, Pepin B (eds) (2019) Resources in mathematics teacher professional activity. Springer, Cham
Vergnaud G (1998) Toward a cognitive theory of practice. In: Sierpinska A, Kilpatrick J (eds) Mathematics education as a research domain: a search for identity. Kluwer, Dordrecht, pp 227–241
Vygotsky LS (1978) Thought and language. MIT Press, Cambridge. (Original work published 1934)
Wang C (2018) Mathematics teachers' expertise in resources work and its development in collectives. A French and a Chinese cases. In: Fan L, Trouche L, Rezat S, Qi C, Visnovska J (eds) Research on mathematics textbooks and teachers' resources: advances and issues. Springer, Cham, pp 193–213
Wenger E (1998) Communities of practice. Learning, meaning, identity. Cambridge University Press, New York

Down Syndrome, Special Needs, and Mathematics Learning

Sophie Brigstocke
Department of Psychology, University of York, York, UK

Keywords
Genetic disorder · Mathematics difficulties · Number difficulties · Cognitive impairment

Characteristics

Down syndrome is a genetic disorder which has serious consequences for cognitive development. Most children with Down syndrome show mild to moderate cognitive impairments with language skills typically being more severely impaired than nonverbal abilities (Næss et al. 2011). Children with Down syndrome are frequently reported to have problems with short-term and working memory. While a relatively large number of studies have investigated the language and reading skills (Hulme et al. 2011) of children with Down syndrome, much less is known about the development of number skills in this group.

Early case studies and studies using highly selected samples have reported some relatively high levels of arithmetic achievement in individuals with Down syndrome. However, for the majority of individuals with Down syndrome, simple single digit calculations and even counting represent a significant challenge (Gelman and Cohen 1988). Carr (1988) reported that more than half of her sample of 41 individuals aged 21 years could only recognize numbers and count on the Vernon's arithmetic-mathematics test. Buckley and Sacks (1987) surveyed 90 secondary school-age children with Down syndrome in the and found that only 18% could count beyond 20 and only half of the sample could solve simple addition problems.

Studies conducted on larger samples consistently report low arithmetic achievement in individuals with Down syndrome relative to other scholastic skills such as reading accuracy (Hulme et al. 2011; Buckley and Sacks 1987; Carr 1988). Age equivalents on standardized number tests are typically reported to lag age equivalent reading scores by around 2 years in children with Down syndrome (e.g., Carr 1988).

Arithmetic performance is reported to improve with chronological age in children with Down syndrome, but this varies widely within IQ levels and is not true for all children (e.g., Carr 1988). It seems highly plausible that a relationship might exist between IQ level and arithmetic performance level, but thus far, there is no consensus in the literature. Education has a positive influence on arithmetic performance as might be expected, and individuals in mainstream school are reported to achieve higher levels of mathematical attainment compared to special school (e.g., Carr 1988).

Individual differences in response to intervention are primarily determined by quality and quantity of teaching (Nye et al. 2005). In the UK, Jo Nye has written a book on adapting Numicon for use with children with Down syndrome, "Teaching Number Skills to Children with Down Syndrome Using the Numicon Foundation Kit." In the USA, DeAnna Horstmeier has written a book titled "Teaching Math to People with Down Syndrome and Other Hands-On Learners: Basic Survival Skills." More research is needed to determine the origin of the difficulties that individuals with Down syndrome before a theory driven intervention program can be designed.

Cross-References

- ▶ 22q11.2 Deletion Syndrome, Special Needs, and Mathematics Learning
- ▶ Autism, Special Needs, and Mathematics Learning
- ▶ Deaf Children, Special Needs, and Mathematics Learning
- ▶ Inclusive Mathematics Classrooms
- ▶ Language Disorders, Special Needs and Mathematics Learning
- ▶ Learning Difficulties, Special Needs, and Mathematics Learning

References

Brigstocke S, Hulme C, Nye J (2008) Number and arithmetic skills in children with down syndrome. Down Syndr Res Pract http://www.down-syndrome.org/reviews/2070/reviews-2070.pdf

Buckley S, Sacks B (1987) The adolescent with down syndrome: life for the teenager and for the family. Portsmouth Polytechnic, Portsmouth

Carr J (1988) Six weeks to twenty-one years old: a longitudinal study of children with down's syndrome and their families. J Child Psychol Psychiatry 29(4):407–431

Gelman R, Cohen M (1988) Qualitative differences in the way Down Syndrome and normal children solve a novel counting problem. In: Nadel L (ed) The psychobiology of Down syndrome. MIT Press/Bradford Books, Cambridge, pp 51–99

Horstmeier D (2004) Teaching math to people with Down syndrome and other hands on learners. Woodbine House, Bethesda

Hulme C, Goetz K, Brigstocke, S, Lervag, A, Snowling M (2011) The development of reading and language skills in children with Down syndrome. Dev Sci 15(3):320-329

Irwin K (1989) The school achievement of children with Down's syndrome. N Z Med J 102:11–13

Næss KB, Lyster SH, Hulme C, Melby-Lervåg M (2011) Language and verbal short-term memory skills in children with Down syndrome: a meta-analytic review. J Res Dev Disabil 32(6):2225–2234

Nye J (2006) Teaching number skills to children with Down syndrome using the Numicon Foundation Kit. Down Syndrome Education International, Portsmouth

Nye J, Buckley S, Bird G (2005) Evaluating the Numicon system as a tool for teaching number skills to children with Down syndrome. Down Syndr News Update 5(1):2–13. http://www.down-syndrome.org/updates/352/